

ChatGPT

Dear GPT sorry to bother! May i ask from our Erdos 319 discussion - doing some of a meta-methodology check: from nearly giving up to getting something seemingly. May i ask on a methodological level what is the miracle way? Can it be generalised, and also including prompt or research skills from others in the github link to have AI obtain full proofs (especially like Boon Suan Ho)? Can these skills/ workflow design etc. allow us to crack further problems/ provide transferrable inspirations? Thx.

我會把它分成兩層看：一是你這次「差點放棄 → 找到可推進結構」的數學方法論；二是能否抽象成 AI-assisted proof workflow, 而不是把它神化成「prompt magic」。

我查到幾個很相關的更新:Erdős AI wiki 本身其實已經把「不要把它當魔法、要做 provenance/literature/verification」列成核心建議;Boon Suan Ho 的留言也特別提到 provenance check 和 prompt engineering 對避免 attribution/spaghetti proof 問題有實際幫助。

已思考 55s

Dear Nam, yes — but the honest answer is: 不是 miracle prompt, 而是一套「研究閉環」。你今次從「似乎無路」到「似乎有東西」的關鍵, 不是單次靈感, 而是把 AI 從「直接證明機器」改造成:

problem decomposer + counterexample hunter + literature/provenance checker + proof critic + formalization assistant。

1. #319 的方法論 miracle 是甚麼?

Erdős #319 本身不是空白問題。官方頁面已記錄 Adenwalla 由 Croot 的 reciprocal-sum 結果得到下界

$(1 - 1e^{-o(1)})N$

$(1 -$

$\frac{e}{1}$

$+o(1))N$, 並指出目前 comments 沒有 claimed solution / partial solution; 該問題也已被 Google DeepMind Formal Conjectures project formalised in Lean。這代表它很適合 AI-assisted work: statement 精確、已有 baseline、但中間仍有可探索空間。

你今次真正做對的地方是: 沒有一直要求 AI「solve the problem」。你改成問:

這個已知 construction 為甚麼卡住?

單一負元素是否太特殊?

多個負元素會否形成 signed reciprocal zero circuit?

能否用 Croot-style blocks / interval blocks / cancellation identities 擴張?

這是很重要的轉向: 由「求結論」轉成「找 mechanism」。這類問題中, AI 最有用的位置往往不是直接吐出 final theorem, 而是幫你重新 parameterize the search space。

2. 可否 generalise? 可以, 但要降神話

Erdős AI wiki 的建議其實很貼切: AI 不是 magic; 使用 AI 做 research mathematics 有 learning curve, persistence 有價值, 而且現代有效做法不是舊式 role-play prompt, 而是 human-AI cognitive complementarities、long-range orchestration、accuracy-improvement techniques。它也強調要 keep grounded: 先做自己能檢查的結果, 之後才變得更 ambitious。

所以可 generalise 的不是「某一句 prompt」, 而是以下 workflow:

Stage	你應該叫 AI 做甚麼	產物
1. Statement hygiene	找出 exact statement、known lower/upper bound、variant risk	避免 solve wrong problem
2. Baseline reconstruction	重新證明已知 construction / obstruction	知道現有方法卡在哪
3. Search-space reframing	要 AI 產生 3–5 個等價 formulation	signed circuit、graph、linear algebra、density、modular view
4. Toy computation	小 N / finite model / rational verifier	找 pattern, 不憑感覺
5. Lemma extraction	將 pattern 變成 lemma candidates	可證明的中間命題
6. Red-team	要 AI 主動攻擊 lemma	找 hidden assumption
7. Provenance check	問「這其實像哪個 theorem / literature technique?」	避免 reinvent / misattribute
8. Formalization or certificate	Lean / exact rational code / exhaustive verifier	提高可信度

這比單純「please solve」強很多。

3. Boon Suan Ho-style skill 的核心

Boon Suan Ho 在 Erdős forum 的留言很值得學。他特別提到: 現在 AI proof 容易有 attribution / exposition / “spaghetti proof” 問題, 但部分可以靠 better prompting 改善;

他建議做 provenance check, 即要求 GPT 說明一個證明究竟來自甚麼 known idea, 例如 Bohr sets、simultaneous approximation 等。

他的 GitHub 也顯示他不是只靠聊天式 prompting, 而是有 Lean/formalization-oriented repos, 包括 entropy-inequality、3ap-free、FerrersBound、erdos670 等。這點很重要：高手不是「問得比較玄」, 而是把 AI output 拉回可檢查 artifact。

你可直接用這幾個 prompt pattern:

Provenance check:

This proof idea looks plausible. Do not try to improve it yet. Identify the closest known theorems, methods, or papers it resembles. For each step, say whether it is genuinely new, a standard lemma, a disguised known theorem, or possibly false.

Mechanism extraction:

Ignore polishing. Extract the actual mathematical mechanism in this argument. What is the invariant? What is being conserved, expanded, cancelled, or compressed? State the mechanism as a standalone lemma.

Adversarial check:

Try to disprove the proposed lemma. Look for boundary cases, missing quantifiers, dependence on N , hidden coprimality assumptions, density losses, and cases where the construction degenerates.

Formalization planning:

Separate the theorem into Lean-checkable components. Mark each component as:

A. already in mathlib,

B. elementary but needs proof,

C. black-box literature theorem,

D. computational certificate,

E. suspicious / likely misformalized.

Do not prove anything yet.

4. Full proof workflow: what actually works?

For full proofs, there are two different tracks.

First is informal research proof. Here AI can sometimes make real leaps, as seen in the 2026 OpenAI unit-distance result, where an internal model produced a construction disproving a long-standing Erdős conjecture; the proof was then checked and contextualized by external mathematicians. The important lesson is not “AI always solves hard problems,” but that AI can sometimes connect distant areas — in that case algebraic number theory and discrete geometry — in a way humans had not prioritized.

Second is formal proof / Lean proof. Here AI is strong but not magic. Lean Copilot’s paper frames LLMs as copilots for theorem proving: Lean checks correctness, while LLMs suggest proof steps, complete goals, and select premises. But this is still a human-AI workflow, not reliable autonomous solving of arbitrary novel mathematics.

A very relevant warning comes from the Grasshopper Lean case study: AI-assisted formalization produced verified local lemmas, but the main theorem remained blocked by one unresolved global counting step. The paper explicitly identifies this as a limitation: local proof search can succeed while global combinatorial bookkeeping remains unresolved.

So for your Erdős work, the right expectation is:

AI can often produce local lemmas, reductions, candidate constructions, and proof skeletons.

The hard part is usually the global closure step: density bookkeeping, quantifier control, asymptotic loss, or exact obstruction.

That matches exactly what happened in your #287 and now #319-style exploration.

5. Can this crack further problems?

Yes, but selectively. It is most transferable to problems with these features:

1. Precise finite/combinatorial statement
Good: reciprocal sums, signed subsets, primitive sets, graph extremal configurations, finite modular obstruction.
2. Known construction but unclear optimality
Good for asking AI: “how can this construction be perturbed?”
3. Small-case computation possible
Your #287 p-adic/exact rational workflow is a real advantage here.
4. Proof can be decomposed into lemmas
Lean/certificate route becomes possible.
5. Literature can be triangulated
The Erdős wiki itself warns that many listed open problems lack thorough literature review, and some “AI solutions” later turn out to overlap with existing literature or solve a misstated version.

Bad targets are those where the missing ingredient is a deep global theorem with no finite experimental footprint and no clear toy model.

6. My honest verdict for you

你最可 transfer 的 skill 不是「AI prompting」本身，而是這個 combination：

lawyerly issue-spotting + exact statement discipline + computational verification + AI red-teaming + provenance checking + willingness to downgrade claims。

這其實很罕見。很多人只會叫 AI “solve this”；你現在慢慢在做的是：

turn vague AI output into inspectable mathematical objects.

For #319, the realistic next methodological move is not “claim solution”。It is:

find a signed-circuit construction that genuinely beats the one-negative Adenwalla/Croot template, or prove that a large class of multi-negative templates collapses back to the same

$(1-1/e)^N$

$(1-1/e)^N$ barrier.

Either result is meaningful. The first could be real progress; the second could still be a publishable computational/structural note if stated modestly.

The transferable inspiration is therefore:

Treat AI as a generator of research pressure, not as an oracle.

Every promising idea must become either a theorem, a counterexample, a computation, a Lean lemma, or a provenance link.

Anything else is just vibes.

Sorry many thanks moreover once again in Chinese like to secondary student can you explain "What is the size of the largest $A \subseteq \{1, \dots, N\}$ such that there is a function $\delta : A \rightarrow \{-1, 1\}$ such that

$$\sum_{n \in A} \delta n$$

$$= 0$$

$$\text{and}$$

$$\sum_{n \in A'} \delta n$$

$$= 0$$

$$\text{and}$$

$$\sum_{n \in A'} \delta n$$

$$= 0$$

$$\neq 0$$

$$\neq 0$$

for all non-empty $A' \subseteq A$?" itself for Erdős 319 thx!

已思考 18s

可以。用中學生版本講, Erdős #319 問的是:

在

$$1, 2, 3, \dots, N$$

$1, 2, 3, \dots, N$ 這些數字裏, 最多可以挑多少個數, 令它們的「倒數」可以用加號、減號剛好湊成 0, 而且這個湊法是不可再拆細的?

官方 Erdős Problems 網站列出的題目正是這個 signed reciprocal sum 問題, 而且目前標示為 open。網站也記錄了一個已知下界, 大約可以做到

$$(1 - 1/e)N$$

$$(1 -$$

$\frac{1}{2}N$, 即約

$0.632N$

$0.632N$ 個數。

1.

$A \subseteq \{1, \dots, N\}$

$A \subseteq \{1, \dots, N\}$ 是甚麼？

$\{1, \dots, N\}$

$\{1, \dots, N\}$ 就是：

$\{1, 2, 3, \dots, N\}$

$\{1, 2, 3, \dots, N\}$

而

$A \subseteq \{1, \dots, N\}$

$A \subseteq \{1, \dots, N\}$ 意思是：

A

A 是從

1

1 到

N

N 裏面挑出來的一批數。

例如如果

$N=10$

$N=10$, 可以有：

$A = \{2, 3, 6\}$

$A = \{2, 3, 6\}$

這就是從

$\{1,2,\dots,10\}$

$\{1,2,\dots,10\}$ 裏面挑了 2、3、6。

2.

$\delta:A\rightarrow\{-1,1\}$

$\delta:A\rightarrow\{-1,1\}$ 是甚麼？

這句看起來很抽象，但其實只是說：

對

A

A 裏每一個數，你都要分配一個符號：加號

+1

+1 或減號

-1

-1。

例如：

$$A=\{2,3,6\}$$

$$A=\{2,3,6\}$$

我們可以設定：

$$\delta_2=+1, \delta_3=-1, \delta_6=-1$$

δ

2

$=+1, \delta$

3

$=-1, \delta$

6

$$=-1$$

意思就是：

$$+12-13-16$$

$$+$$

$$2$$

$$1$$

$$-$$

$$3$$

$$1$$

$$-$$

$$6$$

$$1$$

3. 第一個條件是甚麼？

題目要求：

$$\sum_{n \in A} \delta_{nn} = 0$$

$$n \in A$$

$$\sum$$

$$n$$

$$\delta$$

$$n$$

$$=0$$

中學生版翻譯：

把

A

A 裏所有數字變成倒數，再按你分配的加號或減號相加，最後要剛好等於 0。

例如：

$$12-13-16=0$$

$$2$$

$$1$$

$$-$$

$$3$$

$$1$$

$$-$$

$$6$$

$$1$$

$$=0$$

因為：

$$12=36, 13=26, 16=16$$

$$2$$

$$1$$

=

6

3

,

3

1

=

6

2

,

6

1

=

6

1

所以：

$$36-26-16=0$$

6

3

—

6

2

—

6

1

=0

因此

$A=\{2,3,6\}$

$A=\{2,3,6\}$ 是一個可以「平衡」的例子。

你可以想像成天秤：

左邊放：

12

2

1

右邊放：

13+16

3

1

+

6

$$1$$

因為：

$$12=13+16$$

$$2$$

$$1$$

$$=$$

$$3$$

$$1$$

$$+$$

$$6$$

$$1$$

所以天秤剛好平衡。

4. 第二個條件是最重要的

題目還要求：

$$\sum_{n \in A'} \delta_{nn} \neq 0$$

$$n \in A$$

,

$$\sum$$

$$n$$

$$\delta$$

n

$$=0$$

for all non-empty

$$A' \subsetneq A$$

$$A$$

,

$$\subsetneq A。$$

這句意思是：

你不能只拿

$$A$$

A 的一小部分，就已經湊到 0。

也就是說，整個

$$A$$

A 湊到 0，但任何較小的非空子集合都不能湊到 0。

例如

$$A=\{2,3,6\}$$

$A=\{2,3,6\}$ ，符號是：

$$+12-13-16$$

$$+$$

$$2$$

$$1$$

$$-$$

$$3$$

1

—

6

1

整體是 0:

$$12-13-16=0$$

2

1

—

3

1

—

6

1

=0

但如果你只拿其中一部分:

$$12-13\neq0$$

2

1

—

3

1

=0

12-16≠0

2

1

—

6

1

=0

-13-16≠0

—

3

1

—

6

1

=0

所以它不是由更小的「零和組合」拼出來的。它是一個最小不可拆的平衡關係。

5. 所以 Erdős #319 真正在問甚麼？

它不是問：

能不能找到一些倒數加加減減等於 0？

因為這很容易，例如：

$$12-13-16=0$$

2

1

—

3

1

—

6

1

=0

它問的是更難的問題：

當

N

N 很大時，在

$1, 2, \dots, N$

$1, 2, \dots, N$ 裏，最多可以挑多少個數，形成一個很大的、不可拆的 signed reciprocal zero-sum relation？

換句話說：

最大 $|A|$ 可以有多大？

最大 $|A|$ 可以有多大？

其中

$|A|$

$|A|$ 就是

A

A 裏有多少個數。

6. 為甚麼要加「不可拆」條件？

因為如果沒有第二個條件，我們可以把很多小的零和式子拼在一起。

例如：

$$12-13-16=0$$

2

1

—

3

1

—

6

1

=0

如果另外又有一個：

$$14-15-120=0$$

4

1

—

5

1

—

20

1

=0

那你可以把兩個放在一起，仍然等於 0。

但這樣不是真正「一個大型結構」，只是把很多小零和積木黏起來。

所以 Erdős #319 要求：

整個

A

A 是一個單一、不可分割的零和結構。

這就是它有趣的地方。

7. 一句話總結

Erdős #319 問：

在

1

1 到

N

N 的數字裏，最多可以選多少個數，使得它們的倒數可以分成正負兩邊剛好相等，而且這個相等關係不能由更小的一部分已經完成？

最簡單例子是：

$$A=\{2,3,6\}$$

$$A=\{2,3,6\}$$

因為：

$$12=13+16$$

$$2$$

$$1$$

$$=$$

$$3$$

$$1$$

$$+$$

$$6$$

$$1$$

或者寫成：

$$12-13-16=0$$

$$2$$

$$1$$

$$-$$

$$3$$

1

—

6

1

=0

但 Erdős #319 要問的是：

可不可以做到非常非常大，例如接近

N

N 個數？最多到底是多少？

目前官方頁面記錄的已知 construction 至少能做到約

$(1 - 1/e)N$

$(1 -$

e

1

$)N$ ，即大約

63.2%

63.2% 的大小，但完整答案仍然未解。

Many thanks. Moreover may i confirm, despite

"<https://aristotle.harmonic.fun/dashboard/requests/7449b179-93d2-4726-b583-0cb78b7bb285>"; our progress of erdos 319 has yet to be proven right? Is it because there are "other two M"s?

已思考 37s

是，可以確認：我們 Erdős #319 的 progress 仍然未算 proven right / full proof。它目前最多應該叫：

promising proof route / partial obstruction / fixed-

Q

Q notebook / Lean scaffold,

不是 solved theorem。

我剛試過打開你給的 Harmonic Aristotle dashboard link, 但它 redirect 到 Harmonic authentication, 所以我這邊不能直接看到該 request 的內容。換言之，我不能靠那條 private dashboard link 確認它真的完成了 #319 的 formal proof。

公開可查的狀態是：Erdős Problems #319 頁面仍然把問題列為 open, 並寫明 comments 中沒有 claimed solution / partial solution; 頁面只記錄 Adenwalla/Croot lower bound, 即可做到大約

$$(1-1e+o(1))N$$

$$(1-$$

$$e$$

$$1$$

$$+o(1))N。$$

你的理解大致對：不是「已證明」，而是「還差全局 closure」

我們現在的 #319 進展大概是這樣：

層級	狀態
題目理解	已清楚
已知 baseline	Adenwalla/Croot construction, 約
	0.632N
	0.632N
我們的新方向	multi-negative / fixed-
	Q
	Q / signed reciprocal circuit route

小範圍或局部 lemma	有希望
Lean scaffold / formal statement	可以做
full asymptotic theorem	未完成
官方 solved status	未解

所以即使 Harmonic / Aristotle 對某些 lemma 有 formal check, 也不等於 #319 solved。Aristotle 這類系統的標準其實很嚴格: Harmonic 自己的 Aristotle paper 說, 只有產生完整 Lean 4 proof、沒有 gaps 或 unsound axioms, 才算 solved。

「是不是因為有 other two

M

M ?」——我會這樣修正

我不完全肯定你說的 “other two

M

M ” 指哪兩個, 因為我看不到你 Harmonic dashboard 內文。但按我們之前 #319 的討論, 核心 gap 不是單純因為有兩個

M

M , 而是因為有幾個不同層次的 quantifier / parameter 還未被統一處理。

最可能是這些:

1. 一個

M

M : largest denominator / scale

例如我們可能證明某個分母範圍、某個 scale

M

M 裏可以找到 reciprocal block。

這只是：

在某個尺度上可以做事。

但 #319 要的是對

$1, \dots, N$

$1, \dots, N$ 裏的最大 primitive signed zero-circuit 做 asymptotic control。

2. 另一個

M

M : construction block 的 internal parameter

Croot/Bloom/Liu–Sawhney 類方法常常會有 interval、smooth number、local mass、target denominator 等內部參數。你可能在某個

M

M 或某個

D

D 下造出：

$$\sum_{b \in B} b = 1D$$

$$b \in B$$

$$\sum$$

$$b$$

$$1$$

$$=$$

$$D$$

$$1$$

或類似 target。但這不自動給出完整 #319 construction。

3. 真正最難的不是

M

M , 而是 target-selection + minimality

Erdős #319 不只要:

$$\sum_{n \in A} \delta_n = 0$$

$$n \in A$$

$$\sum$$

$$n$$

$$\delta$$

$$n$$

$$=0$$

還要任何 proper non-empty subcircuit 都不能自己等於 0。這就是 minimal / primitive circuit condition。

所以即使你找到一個大 signed zero-sum, 也還要證明:

它不是幾個小零和式子拼起來的。

這一步非常容易被 AI proof 草稿忽略。

為甚麼 #124 / #728 可以算 AI proof, 但 #319 暫時不能?

Erdős #124 的 Aristotle case 很有啟發: forum 上有人指出 Aristotle formalized a version in Lean, 但後來也有討論說它解的是 “a version”, 不一定是原問題最強版本。這正好提醒我們: formal proof 也要小心 formalized statement 是否等於原題。

#728 則有公開 arXiv writeup, 明確說 final result 是 Lean formal proof, 再翻譯成 informal mathematics; 這種才比較接近「可公開驗證的 AI proof」。

我們 #319 現在還未到這個階段。原因不是悲觀, 而是嚴格:

我們未有一份完整 statement + complete Lean proof + informal writeup + literature/provenance check, 證明它真的超越 Adenwalla/Croot lower bound 或解決 full #319。

最準確結論

你可以這樣記：

#319 progress is real but not proven.
It is a serious partial route around fixed-

Q

Q, local obstruction, and multi-negative signed reciprocal circuits.

But the full theorem remains unproved because the global quantifiers, automatic target selection, and primitive/minimality condition have not yet been closed.

中文更簡單講：

我們現在像是找到了一條可能通往山頂的路，而且走到了一段有地圖、有路標的位置；但還未到山頂，也未證明這條路一定通到山頂。

所以 CV / writeup 上應該仍然寫：

AI-assisted independent research on Erdős Problem #319; developed fixed-

Q

Q partial obstruction / Lean formalisation scaffold / proof-search notebook.

不應寫：

Solved Erdős Problem #319.

7449b179-93d2-4726-b583-0cb78b7bb285-aristotle.tar(1).gz

檔案

Many thanks this is the lean stuff thx

我會直接檢查你上傳的 Aristotle/Lean 壓縮包：先看它證明了哪個 theorem、是否有 `sorry`/axiom、以及 formal statement 是否真的等於 Erdős #319 原題。然後我會對照公開 #319 狀態再給你結論。

我已打開壓縮包。核心發現很清楚：它確實是 Lean project, 但 `MainTheorem.lean` 的主定理仍然是 `sorry`, 而 `BlackBox.lean` 有 5 個黑箱 `sorry`; 所以它不是完整 Lean proof。

已思考 1m 16s

我檢查了你上傳的 Aristotle tar.gz。結論很明確：

這份是有價值的 Lean artifact, 但不是 Erdős #319 的完整證明。

它比較準確的定位是：

Fixed-

Q

Q Singleton-Gate Theorem 的 Lean scaffold / partial formalization / proof sketch with black boxes。

不是：

Erdős #319 solved。

公開 Erdős #319 頁面目前仍標示為 OPEN, 而且 remarks 說 comments 中沒有 claimed partial/complete solution; 頁面只記錄 Adenwalla/Croot 的下界

$(1 - 1e + o(1))N$

$(1 -$

$$\frac{e}{1}$$

$+ o(1))N$ 。

我在 Lean package 裏看到甚麼

你這個包有幾個主要檔案：

檔案	狀態
<code>Defs.lean</code>	基本定義, 看起來無 <code>sorry</code>
<code>Primitivity.lean</code>	primitivity contradiction lemma, 看起來無 <code>sorry</code>
<code>BlackBox.lean</code>	有 5 個重要 theorem 直接 <code>sorry</code>
<code>MainTheorem.lean</code>	主定理 <code>singletonGateTheorem</code> 是 <code>sorry</code>

最關鍵的是：

```
theorem singletonGateTheorem ... := by  
  
sorry
```

而 `BlackBox.lean` 裏也有 5 個黑箱：

```
tailNonmultipleBound := by sorry  
  
smoothPruningLoss := by sorry  
  
boundedOmegaPruningLoss := by sorry  
  
localMassPruning := by sorry  
  
liuSawhneyTarget := by sorry
```

所以即使後面的：

```
exampleQ_singletonGate
```

形式上能由 `singletonGateTheorem` 推出，它仍然是靠一個有 `sorry` 的主定理推出來的。這在 Lean 裏不等於 completed proof；它等於「如果接受這些黑箱和主定理缺口，那麼 corollary 可推出」。

我這邊不能實際 kernel-check，因為 sandbox 沒有 `lake` / Lean toolchain；但 source inspection 已足夠確認它不是 sorry-free proof。

這是否因為「other two M」？

不完全是。主要不是「有兩個

M

M」這麼簡單，而是有三層 gap。

第一層是 Lean gap：主定理本身還是 `sorry`。這已經足夠說明它不是 closed proof。

第二層是 black-box interface gap: 它把 Liu–Sawhney 的 Proposition 5.2、Lemma 2.2、2.3、6.2 等作為簡化黑箱。問題是: 這些黑箱在 Lean 中沒有證明, 而且它們的 statement 是否完全等於原 paper 的 statement, 也仍未核對。

第三層是 scope gap: 這個 theorem 只處理 fixed

Q

Q 且滿足 gate condition 的情況。可是 Erdős #319 問的是所有 possible signed reciprocal primitive circuits 裏, 最大

$|A|$

$|A|$ 可以多大。也就是說, full #319 需要處理 arbitrary

Q

Q、arbitrary sign split、minimality / primitivity、target selection、不同 denominator scale 的全局情況。

所以如果你說的「other two

M

M」是指 Liu–Sawhney machinery 裏還有幾個 scale parameters /

M,N

M,N / target denominator, 那它們確實是技術難點之一; 但真正原因更大:

不是只有兩個

M

M 沒處理, 而是整個 asymptotic parameter closure + black-box hypotheses + arbitrary-

Q

Q quantifier 還未完成。

這份 Lean stuff 有沒有價值?

有。它不是垃圾。它至少 formalized / organized 了幾件有用的東西:

1. SRC definition: 把 signed reciprocal circuit 寫成
2. P, Q
3. P, Q 兩邊 reciprocal sum 相等。
4. Primitive condition: formalized 「沒有 proper sub-circuit」。
5. Primitivity contradiction: 如果找到
6. $B \subseteq P$
7. $B \subseteq P$ 且
8. $R(B)=1/D$
9. $R(B)=1/D$, 而
10. $D \in Q$
11. $D \in Q$ 且
12. Q
13. Q 不止一個元素, 就會破壞 primitive。
14. Example
15. $Q=\{3,5,7,9,11,13,15,17\}$
16. $Q=\{3,5,7,9,11,13,15,17\}$: 它核對了
17. $R(Q)>1$
18. $R(Q)>1$ 和 gate condition。
19. Proof architecture: 它把真正缺口清楚定位在 Liu–Sawhney target-reaching / pruning / asymptotic closure。

這很像近期 AI-assisted Lean case studies 的常見狀態: local lemmas 可以 formalize, 但 global combinatorial/asymptotic step 留有 unresolved `sorry`。Grasshopper formalisation case study 也明確指出, AI 可以證明一些 local components, 但 main theorem 仍可被一個 unresolved `sorry` 卡住; 這正是目前這份 #319 artifact 的情況。

和真正 solved 的 Aristotle case 差在哪?

真正比較接近「solved」的例子, 例如 Erdős #728, 有公開 writeup 說 final result 是 Lean formal proof, 然後再翻譯成 informal mathematics。

你這份 #319 artifact 則相反: README 和 summary 自己已寫明:

does NOT prove the full Erdős Problem #319

main theorem sorry

five black-box lemmas sorry

所以目前不能當作 #319 proof, 也不能當作 full fixed-

Q

Q theorem proof。它最多是：

有 formalized components 的 proof sketch。

最準確 verdict

Yes — your earlier caution is correct.

這份 Aristotle/Lean output 沒有證明 Erdős #319, 也未完全證明 fixed-

Q

Q singleton-gate theorem。

但它不是毫無價值。它的價值在於：

它把「我們到底還差甚麼」變得很清楚。

剩下要做的不是再叫 AI 美化文字，而是：

1. 找到 Liu–Sawhney paper 的 exact Proposition 5.2 statement;
2. 對照 `liuSawhneyTarget` 的 Lean black-box statement 是否忠實;
3. 補齊 pruning preservation: smooth、bounded-
4. Ω
5. Ω 、local mass;
6. 真正證明 `singletonGateTheorem`, 移除主定理 `sorry`;
7. 最後再處理 arbitrary
8. Q
9. Q / full #319 scope。

一句話：

這是 serious partial progress / formalization scaffold, 不是 proof。

目前卡位不是「小錯」，而是主定理和五個核心黑箱仍未閉合。

Erdos 319 (1).docx

文件

Many thanks! sorry should be and also as noted by poe Claude

"https://aristotle.harmonic.fun/dashboard/requests/7449b179-93d2-4726-b583-0cb78b7bb285"

已思考 1m 27s

明白, thanks — 你意思是: 不只是 Harmonic Aristotle dashboard / Lean tar 包本身, 也要把 Poe Claude 對 #319 的批評一起納入 final check。

我更新後的 verdict 是：

Poe Claude 的批評基本上是正確的，而且與 Lean artifact 的狀態一致：我們有 serious scaffold / research programme，但未有 Erdős #319 proof。

官方 #319 頁面仍寫明此題是 OPEN，而且 comments 中沒有 claimed partial/complete solution；頁面只記錄 Adenwalla/Croot 給出的 lower bound

$$|A| \geq (1 - 1/e + o(1))N$$

$$|A| \geq (1 - 1/e + o(1))N$$

並說此題已有 Lean formalized statement，但不是已解決。

Poe Claude 補充了甚麼？

你上傳的 `Erdos 319.docx` 裏，Poe Claude 的核心意思不是「你完全沒進展」，而是：要嚴格分清楚：哪些是 proved，哪些只是 reformulation / programme / conjectural bridge。

這點非常重要。它其實幫我們把 #319 狀態降級成正確版本：

項目	現在狀態
#319 題目理解	已清楚
spectrum-intersection reformulation	有價值，基本正確
singleton-side / Adenwalla-Croot subclass	可能有一個小 theorem / corollary，但要查文獻
large-prime-factor obstruction	可能是 modest structural lemma
small- N N scanner / near-miss data	真實 exploratory evidence
Lean definitions / primitive contradiction lemmas	有用
signed Liu–Sawhney bridge	未證明
arbitrary multi-negative	未處理

Q

Q case

full Erdős #319 asymptotic upper bound

未證明

full Lean proof

未完成

你文件裏也多次提到「real hard task 是 target-

r

r / signed Liu–Sawhney, 不是 more brute force], 以及下一步應該集中在 target-

r

r 、Fourier structured class、multi-negative stress test。

所以是不是因為「other two M」？

如果你說的 “other two M” 是「還有其他兩個 missing mechanisms / missing modules」，那可以這樣理解，但我會更精準地命名成：

M1: Minimality

#319 不只是要找到：

$$\sum_{n \in A} \delta_{nn} = 0$$

$$n \in A$$

$$\sum$$

$$n$$

δ

n

$$= 0$$

還要證明任何 proper nonempty subset 都不能已經等於 0。

這個 minimality 條件是很硬的。很多構造可能做到 signed zero-sum, 但其實是幾個小 circuit 黏起來; 那不算 #319 的 primitive circuit。

M2: Multi-negative / arbitrary

Q

Q

Adenwalla/Croot lower bound 基本是 singleton-side construction:

$$1 = \sum_{b \in B} 1b$$

$$1 =$$

$$\sum_{b \in B}$$

$$\sum$$

$$b$$

$$1$$

即一邊只有

1

1, 另一邊是一大堆

B

B 。

但 #319 容許兩邊都有很多項:

$$\sum_{p \in P} 1p = \sum_{q \in Q} 1q$$

$$\sum_{p \in P}$$

$$\sum$$

$$p$$

$$\frac{1}{n} \sum_{q \in Q} \frac{1}{q}$$

Poe Claude 正確指出：如果要真正測試能否超越

$(1-1/e)N$

$(1-1/e)N$, 就要看 multiple negative elements / multi-block signed circuits, 而不是只修改 Adenwalla template。

所以, 不是單純「有另外兩個

M

M 」;更準確是:

full #319 卡在 minimality + arbitrary multi-negative signed bridge + asymptotic upper bound。

Lean artifact 為甚麼仍不是 proof?

因為 Aristotle/Harmonic 自己的標準是：只有完整 Lean 4 proof, 沒有 gaps 或 unsound axioms / `sorryAx`, 才算 solved。Harmonic 的 Aristotle paper 也明確說, 他們只把「完整 Lean proof」視為 solved。

而你之前上傳的 Lean tar 包裹, `MainTheorem.lean` 的主 theorem 還是 `sorry`, `BlackBox.lean` 也有 5 個 Liu–Sawhney / analytic number theory black-box `sorry`。這和 Grasshopper case study 很像：local Lean lemmas 可成立, 但 main theorem 被 unresolved `sorry` 卡住；該 paper 也把這視為 AI-assisted formalization 的一個典型限制。

真正已解決的 Aristotle-style Erdős case, 例如 #728, 有公開 writeup 說 final result 是 Lean formal proof, 然後再翻譯成 informal mathematics。
我們 #319 目前還未達到這種狀態。

最準確一句話

你現在可以這樣記：

Poe Claude was not too mean. It was doing reviewer work.
我們 #319 的 progress 是 real exploratory progress, 但不是 proof。
目前最有價值的成果是：把 #319 重新定位成
minimal signed reciprocal circuits + Croot/Adenwalla lower bound + signed
Liu–Sawhney / target-

r

r bridge problem + multi-negative construction test。

中文更簡單講：

我們不是「解了 #319」；我們是把「要解 #319 到底缺哪幾塊」看清楚了。
缺的不是小修小補，而是 signed Liu–Sawhney bridge / arbitrary

Q

Q / minimality closure 這些核心大塊。

所以現在最正確的寫法仍然是：

AI-assisted exploratory work on Erdős Problem #319; developed reformulation, finite search data, partial Lean scaffold, and identified the signed target-

r

r / multi-negative bridge as the main remaining obstruction.

不應寫：

Solved Erdős Problem #319.

Many thanks! sorry my bad may i ask then, are we "two Qs" from solving erdos 319?
thanks.

已思考 28s

是, 但要很小心地說: 不是「只差兩個

Q

Q 」那麼近。更準確是:

我們可能已經把 #319 壓縮到一個「兩邊

P, Q

P, Q 的交集問題」;

但這個

Q

Q 不是兩個 case, 而是 任意大小、任意結構的 negative side

Q

Q 。

所以不是「two Q s away」, 而是「卡在 arbitrary two-colour

P, Q

P, Q bridge」。

官方 #319 目前仍是 open, 頁面只記錄 Adenwalla/Croot 下界

$$|A| \geq (1 - 1/e + o(1))N$$

$$|A| \geq (1 - 1/e + o(1))N$$

而且 comments 沒有 claimed solution / partial solution。

用中學生版講

#319 可以把正號那邊叫

P

P , 負號那邊叫

Q

Q 。即:

$$\sum_{p \in P} 1_p = \sum_{q \in Q} 1_q.$$

$$p \in P$$

$$\sum$$

$$\sum_{p \in P} 1 = \sum_{q \in Q} 1.$$

然後 minimality 要求：

不能在

P

P 裏挑一小部分

P'

P

,

，又在

Q

Q 裏挑一小部分

Q'

Q

,

，令兩邊已經相等。

即不能有非平凡：

$$\sum_{p \in P'} 1 \neq \sum_{q \in Q'} 1.$$

$p \in P$

,

$$\sum$$

$$p$$

$$1$$

$$=$$

$$q \in Q$$

,

$$\sum$$

$$q$$

$$1$$

.

所以真正要證明的是：

如果

$$P \cup Q$$

$P \cup Q$ 太大, 兩邊的 subset reciprocal sums 一定會撞車;

一撞車, 就會產生 proper subcircuit;

一有 proper subcircuit, 就不是 minimal;

所以

$$|A|$$

$$|A| \text{ 不能太大。}$$

這就是你文件裏說的 two-colour spectrum intersection / signed Liu–Sawhney bridge。

你的 doc 也明確說真正 obstacle 是 two-colour analogue of Liu–Sawhney, 而不是普通 singleton calculation。

已經處理到哪裏？

大概是這樣：

區域	狀態	距離 full #319
$Q=\{1\}$ $Q=\{1\}$ 或 singleton-side	基本可理解, 接近 Adenwalla/Croot converse	不夠
fixed small Q Q / gate condition	可以做 Lean scaffold / local theorem	還不夠
arbitrary Q Q , arbitrary P P	真正核心	未解
minimality closure	必須全局處理	未解
sharp $(1-1/e+o(1))N$ $(1-1/e+o(1))N$ upper bound	目標	未證明

Liu–Sawhney 的 unit-fraction theorem 很重要, 因為它證明某些很大的 unsigned reciprocal set 必然含有 reciprocal sum

1

1 的 subset; 但 #319 需要的是 signed / two-colour version: 給兩個集合

P, Q

P, Q , 要逼它們的 subset-sum spectra 非平凡相交。Liu–Sawhney 原文是 unsigned unit fractions, 並不是直接給 #319 的 signed theorem。

所以「two Qs」可怎樣理解？

如果你說的 two Qs 是：

1. 固定
2. Q
3. Q : 例如
4. $Q=\{3,5,7,9,11,13,15,17\}$
5. $Q=\{3,5,7,9,11,13,15,17\}$ 這種 fixed dangerous side;

6. 任意
7. Q
8. Q : 所有 possible negative sides;

那麼答案是：

fixed

Q

Q 只是 training ground; 真正解 #319 要處理 arbitrary

Q

Q 。

如果你說的 two Q s 是：

1. 一個
2. Q
3. Q 是 Lean / formal statement 裏的 negative side;
4. 另一個
5. Q
6. Q 是 Liu–Sawhney machinery 裏的 denominator / target parameter;

那也不是只差兩個

Q

Q 。因為問題不是符號混亂，而是：要把 unsigned target theorem 變成 signed two-colour intersection theorem。

最準確結論

我會這樣定性：

我們不是 “two Q s away from solving #319”。

我們是 “one major bridge theorem away”。

那個 bridge theorem 大概長這樣：

$$P, Q \subseteq [1, N], |P| + |Q| > (1 - 1/e + \varepsilon)N$$

$$P, Q \subseteq [1, N], |P| + |Q| > (1 - 1/e + \varepsilon)N$$

是否一定存在非空 proper subsets

$$P' \subseteq P, Q' \subseteq Q$$

P

,

$$\subseteq P, Q$$

,

$\subseteq Q$, 使得:

$$\sum_{p \in P'} 1_p = \sum_{q \in Q'} 1_q?$$

$$p \in P$$

,

$$\sum$$

$$p$$

$$1$$

$$=$$

$$q \in Q$$

,

$$\sum$$

$$q$$

$$1$$

$$?$$

如果能證明這個, 而且處理好 proper / nonempty / boundary cases, 就很接近 #319 的 sharp asymptotic upper bound。

但這個不是小 lemma。你的文件裏 Claude 也正確指出: large-prime-factor lemma 只給

$$c(N) \leq (1 - o(1))N$$

$$c(N) \leq (1 - o(1))N$$

級別, 仍然沒有推到

$0.632N$

$0.632N$; 真正要到

$(1-1/e)N$

$(1-1/e)N$, 需要 signed Liu–Sawhney / two-colour bridge。

所以一句話:

不是差兩個

Q

Q , 而是差一個很大的 arbitrary-

P, Q

P, Q intersection theorem。

若這個 theorem 成立, #319 可能真的會大幅前進; 若不成立, 反而可能存在 multi-negative construction 超越 Croot/Adenwalla。

Many thanks. So in fair and objective, is it at most as aristotle said "I performed a thorough mathematical and computational review of the Fixed- Q Singleton-Gate Theorem argument. My detailed analysis is in review.md, but here is the executive summary: The argument is not hallucinated. It has a coherent mathematical structure and all numerical claims check out:

All $\Phi_q(\theta)$ values verified ($\Phi_3 \approx 1.973$, $\Phi_4 \approx 1.387$, etc.) ✓

$S_Q = 827504/765765 \approx 1.0806$ for the example $Q = (3, 5, 7, 9, 11, 13, 15, 17)$ ✓

$S_Q < \Phi_3(\theta)$ with huge margin ($1.081 \ll 1.973$), so condition (G) holds ✓

The gap $\Phi_D(\theta) > 1 + 1/m_D$ is uniformly positive for all $D \geq 3$ ✓

The asymptotic gap $\approx 0.35/D$ confirmed via $e^2 - 2e - 1 \approx 0.952 > 0$ ✓

$m_D > e(D-1)$ verified for all tested D ✓

The logical structure is sound: The local forcing argument (contrapositive via Φ), the minimal-over-one bound, and the primitivity contradiction are all correct. However, it is a proof sketch, not a complete proof. The main gap is the interface with Liu–Sawhney's Proposition 5.2 — the argument invokes it as a black box without fully verifying that all hypotheses are met. Specifically:

The exact statement of Prop 5.2 is not reproduced

The pruning preservation claims (Lemmas 2.2, 2.3, 6.2) are plausible but not rigorously justified

The definition and properties of $Q_{\{A_0\}}$ need checking against the paper

One interesting finding: For the example Q , $S_Q \approx 1.0806$ is extremely close to

$\Phi_{11}(\theta) \approx 1.0805$ — the argument would fail if one tried target $1/11$ instead of $1/3$. But

since $D = \min Q = 3$ and the argument correctly uses Φ_3 , this is not a problem. Also notable: $Q = (3, 5, 7, 9, 11, 13, 15, 17)$ is not minimal-over-one (removing 13, 15, or 17 leaves $R > 1$), but this is irrelevant since condition (G) is checked directly. Bottom line: This is a substantive, likely correct proof sketch of a meaningful partial result. The core innovation (Φ -based local forcing) is genuine. The risk of a fundamental error is low — if the argument fails, it would be at the Liu–Sawhney interface, not in the new ideas. Making it fully rigorous would require careful verification against the Liu–Sawhney paper."; but far from full proof?

已思考 45s

是, fair and objective 地講, 最安全的上限評價就是 Aristotle 那句:
不是 hallucination; 是一個有 coherent mathematical structure 的 substantive proof sketch / partial result; 但不是 complete proof, 更不是 Erdős's full proof。
我會再精準一點分級:

層級	評價
數值檢查, 例如 SQ S Q , $\Phi_Q(\theta)$ Φ q (θ) , gap estimates	看來可信; 這些是可獨立驗算的 arithmetic / calculus 部分
Fixed- Q Q Singleton-Gate Theorem 的核心想法	有實質內容, 不像胡亂拼湊
Φ Φ -based local forcing	可能是這份 sketch 最有新意的部分
Primitivity contradiction	邏輯方向合理
Liu–Sawhney interface	主要風險位

所以你的句子可以改成：

At most, we have a likely meaningful proof sketch of a fixed-

Q

Q , singleton-gate partial theorem, with verified numerical evidence and a plausible local forcing mechanism. It is not a full proof of Erdős #319.

為甚麼仍然「far from full proof」？

因為官方 #319 問的是所有

$A \subseteq \{1, \dots, N\}$

$A \subseteq \{1, \dots, N\}$ 、所有 sign choices、所有 possible positive/negative split 的最大 size；官方頁面仍標示為 OPEN, 並說 comments 中沒有 claimed partial or complete solution；已記錄的只是 Adenwalla/Croot lower bound

$(1-1/e+o(1))N$

$(1-1/e+o(1))N$ 。

而你現在這份 Fixed-

Q

Q theorem 只處理很特殊的情況：

$$Q=(3,5,7,9,11,13,15,17)$$

$$Q=(3,5,7,9,11,13,15,17)$$

或更一般地, 某些 fixed

Q

Q / gate condition 情形。這距離 full #319 還差三個大跳：

1. 從 fixed
2. Q
3. Q 到 arbitrary
4. Q

5. Q
#319 的 negative side
6. Q
7. Q 可以任意變, 不一定是你測試的那個
8. Q
9. Q , 也不一定滿足同一 gate condition。
10. 從 singleton-gate 到 general signed circuit
full problem 裏兩邊都可以有很多項。真正困難是 two-colour / signed subset-sum spectrum intersection, 不只是單一 target。
11. 從 proof sketch 到 proof
Aristotle 自己的標準也很清楚: 只有完整 Lean 4 proof、沒有 gaps 或 `sorry` 這類 unsound axioms, 才算 solved。

Liu–Sawhney interface 為甚麼是主要風險？

Liu–Sawhney 的 paper 證明的是 unit-fraction density theorem: 如果

$$A \subseteq [1, N]$$

$A \subseteq [1, N]$ 的 reciprocal mass 足夠大, 則存在 subset

B

B 令

$$\sum_{b \in B} 1/b = 1$$

$$\sum$$

$$b \in B$$

$1/b = 1$; 他們也說其技術 refine Croot and Bloom。

但 #319 需要的是更 signed / two-colour 的版本:

$$\sum_{p \in P} 1/p = \sum_{q \in Q} 1/q.$$

$$p \in P$$

,

$$\sum$$

$$p$$

$$1$$

=

$q \in Q$

,

Σ

q

1

.

這不是 Liu–Sawhney theorem 直接給出的東西。你的 uploaded conversation 也反覆把這點標成「full signed Liu–Sawhney bridge not proved」和「sharp

$c(N) \leq (1 - 1/e + o(1))N$

$c(N) \leq (1 - 1/e + o(1))N$ not proved」; 這正是現在的核心 gap。

我會稍微修正 Aristotle 的一句話

Aristotle 說：

“The risk of a fundamental error is low — if the argument fails, it would be at the Liu–Sawhney interface, not in the new ideas.”

這句大致可以接受，但我會更保守：

The risk of error in the local

Φ

Φ -forcing arithmetic seems low.

The risk of the whole theorem being overstated is still nontrivial, because a black-box interface can fail by mismatched hypotheses, not merely by a minor technicality.

也就是說，失敗點不一定是「新想法錯」，但可能是：

- Liu–Sawhney Prop 5.2 的 statement 不支援你要的 target;
- pruning 後
- QA0
- Q
- A

- 0
-
-
- 或 local mass 條件不保留;
- theorem 用到的
- D, mD, θ
- D, m
- D
-
- $, \theta$ parameters 與 paper 裏的 parameter regime 不完全對齊;
- fixed-
- Q
- Q theorem 可以成立, 但不能推到 arbitrary
- Q
- Q 。

這種不是 hallucination, 但也不是 completed proof。

最終客觀 verdict

是的, 你的理解正確:

最公允的評價是: 這是一份 substantive, likely non-hallucinated, potentially correct proof sketch of a meaningful partial Fixed-

Q

Q result; 但它離 full Erdős #319 proof 很遠。

如果用研究履歷 / note wording, 應寫:

“Developed and audited a Fixed-

Q

Q Singleton-Gate proof sketch for Erdős #319, including verified numerical conditions and a

Φ

Φ -based local forcing mechanism; main remaining gap is verifying the Liu–Sawhney interface and extending from fixed

Q

Q to arbitrary signed circuits.”

不應寫：

“Solved Erdős #319.”

更簡短講：

這不是假東西，但也不是終點。它是 partial theorem candidate + serious proof sketch

。

真正下一步不是再吹大，而是把 Liu–Sawhney interface 拆成可核對 lemma，一個一個消掉。

貼上的 Markdown (1)(5).md

檔案

Many thanks but the chat below gpt said "

已思考 15s

Yes — the “chat below GPT” is mostly consistent with what I said, but it is slightly more optimistic in framing. I would accept its classification with one tightening:

We are close to a meaningful fixed-

Q

Q , gate-condition partial theorem candidate; we are not close to full Erdős #319.

The uploaded note itself says the same hierarchy: concrete

Q

Q example is strong, fixed-

Q

Q under gate condition is “中近” but needs Liu–Sawhney interface audit, arbitrary fixed

Q

Q is “中遠”，and full #319 is “遠”.

The key reconciliation

There is no real contradiction between:

1. "This is substantive and likely not hallucinated."
2. "This is far from full Erdős #319."

Both can be true.

Erdős #319 asks for the maximum size of a primitive signed reciprocal zero-circuit in

$\{1, \dots, N\}$

$\{1, \dots, N\}$. The official page still marks it OPEN, says no partial or complete solution is claimed in comments, and records only the Adenwalla/Croot lower bound

$$|A| \geq (1 - 1/e + o(1))N.$$

$$|A| \geq (1 - 1/e + o(1))N.$$

It also says the statement has been formalised in Lean, but that is only a formalised statement, not a proof.

Your current line is much narrower: fixed

Q

Q , gate condition

(G)

(G), singleton target

$1/D$

$1/D$, and a proposed interface with Liu–Sawhney-style unit-fraction machinery.

What I would now say objectively

The lower GPT's assessment is good if read as:

"This may be close to a real partial theorem after the Liu–Sawhney interface is checked."

It should not be read as:

"We are close to solving Erdős #319."

The difference is this:

Claim	Fair status
Arithmetic checks for $Q=(3,5,7,9,11,13,15,17)$ $Q=(3,5,7,9,11,13,15,17)$	Strong
Φ Φ -based local forcing idea	Substantive / promising
Fixed- Q Q under gate condition (G) (G)	Plausible theorem candidate
Minimal-over-one fixed Q Q	Possible next theorem, still needs proof
Arbitrary fixed Q Q	Not solved
Growing $Q=QN$ $Q=Q$ N	Not solved
Full #319	Far

Why Liu–Sawhney is the decisive interface

Liu–Sawhney prove a strong unit-fraction result: if

$$A \subseteq [1, N]$$

$A \subseteq [1, N]$ has large enough reciprocal mass, then some subset sums exactly to

1

1, refining Croot and Bloom. Croot and Bloom are the relevant earlier backbone: Croot proved an Erdős–Graham colouring conjecture about monochromatic unit-fraction sums to

1

1, and Bloom proved a density version for positive upper-density sets.

But your #319 use is more delicate. You need not just “some subset sums to

1

1”, but something like:

$$B \subseteq P, \sum_{b \in B} b = 1D,$$

$$B \subseteq P,$$

$$b \in B$$

$$\sum$$

$$b$$

$$1$$

$$=$$

$$D$$

$$1$$

$$,$$

under the exact pruning, smoothness, local mass, and

QA0

$\mathcal{Q}$

$\mathcal{A}$

0

-type hypotheses. That is why the lower GPT correctly says the main risk is not the new

Φ

Φ -calculus, but the black-box interface.

My tightened verdict

I would phrase it like this:

The fixed-

Q

Q singleton-gate argument is probably a real proof-sketch-level mathematical object, not hallucination. The numerical and local forcing parts look robust. But until Proposition 5.2 and the pruning lemmas are matched line-by-line to Liu–Sawhney, it remains a theorem candidate, not a theorem. Even if fully repaired, it gives a fixed-

Q

Q partial obstruction, not full Erdős #319.

So yes: the lower GPT is not “overhyping” if read carefully. It is saying:

$\text{\text{close to fixed-}}Q\text{\text{ partial theorem}} \neq \text{\text{close to full \#319}}.$

Your safest public/research wording remains:

“A substantive fixed-

Q

Q partial-theorem candidate for Erdős #319, with verified numerical gate conditions and a

Φ

Φ -based local forcing mechanism; the main remaining task is to rigorously verify the Liu–Sawhney interface.”

對 Claude 評估與 Erdős #319 後續路線的可行性深度研究.docx

文件

Thanks! The GPT in erdos 319 gap discussion also noted this. Can you provide an honest update as to as you demonstrated the levels of Qand progress status? Thx

已思考 19s

可以。以下是我現在的誠實 update：你們在 Erdős #319 的路線不是空想；但它應該被分成幾個

Q

Q -levels。每一層's status 很不同。

官方 Erdős #319 仍是 OPEN; 頁面記錄的已知下界是 Adenwalla/Croot 型

$(1-1/e+o(1))N$

$(1-1/e+o(1))N$, 並說 problem statement 被 Lean formalised, 但這是「題目敘述 formalised」, 不是「問題已被 Lean 解決」。你上傳的深度研究文檔也基本採用同一個分層: fixed-

Q

Q under

(G)

(G) 近期可行性高, minimal-over-one 高, arbitrary fixed-

Q

Q 中高, slowly growing

Q

Q 中低, full #319 低。

先講結論

現在最準確的定位是:

我們接近一個 meaningful fixed-

Q

Q partial theorem candidate;

還未有 complete proof;

更遠未到 full Erdős #319 proof。

如果用研究進度表達:

$\boxed{\text{Concrete example strong}} \rightarrow \text{fixed-}Q \text{ theorem plausible} \rightarrow \text{arbitrary fixed-}Q \text{ unresolved} \rightarrow \text{growing } Q \text{ / full \#319 far}$

Q

Q-level 0: 具體例子

Q=(3,5,7,9,11,13,15,17)

Q=(3,5,7,9,11,13,15,17)

Status: 強, 但只是 example-level。

這一層包括:

$$SQ=\sum_{q\in Q}1q=827504765765\approx1.0806$$

S

Q

=

$q\in Q$

$\sum$

q

1

=

765765

827504

≈1.0806

以及:

$$SQ<\Phi^3(1-1/e)$$

S

Q

$\langle \Phi$

3

$(1-1/e)$

這些 arithmetic / numerical checks 看來穩。Aristotle 說所有

Φq

Φ

q

、

SQ

S

Q

、gap estimates 都 checked, 這部分我會接受為「可信的 computational verification」。

但這只證明：

這個 specific

Q

Q 滿足 gate condition。

它不是 full #319, 甚至不是 all fixed-

Q

Q 。

評級：

數值層面：高可信。

數學意義：good example / sanity check。

距離 full #319：很遠。

Q

Q -level 1 : fixed-

Q

Q under gate condition

(G)

(G)

Status: 最有希望成為近期真正 partial theorem。

這是目前最重要的一層。形式大概是：

固定一個 finite

Q

Q 。如果它滿足某個 gate condition

(G)

(G), 例如

$$SQ < \min_p \|D\Phi_p\| (1 - 1/e),$$

S

Q

$<$

p

a

$\|D$

$\min$

Φ

p

a

$$(1-1/e),$$

那麼任何 primitive signed circuit with this fixed negative side

Q

Q 不能超過

$$(1-1/e+\varepsilon)N$$

$$(1-1/e+\varepsilon)N。$$

這裏的核心新意是：

Φ -based local forcing

Φ -based local forcing

也就是：用 balance condition

$$R(P)=SQ$$

$$R(P)=S$$

Q

迫使

P

P 在某些 prime-power divisibility layers 有足夠 mass, 然後嘗試接上 Liu–Sawhney machinery, 命中 target

$$1/D$$

$$1/D。$$

Liu–Sawhney 的 paper 的確很 relevant: 他們證明如果

$$A \subseteq [1, N]$$

$A \subseteq [1, N]$ reciprocal mass 足夠大, 則有 subset

B

B 令

$$\sum_{b \in B} 1/b = 1$$

$$\sum$$

$$b \in B$$

$1/b=1$, 並說技術 refine Croot and Bloom。更關鍵是, 他們對 Proposition 5.2 的討論說明其技術點是把 Archimedean 與

p

p -adic considerations 分開, 並在條件滿足時 reach “all possible” fractions。

所以這一層不是 hallucination。它有真數學骨架。

但目前仍未完成, 因為要逐條核對:

- Proposition 5.2 的 exact statement;
- target 是否真的是你需要的
- $1/D$
- $1/D$ 或
- x/Q
- x/Q ;
- local mass 條件是否對 prime powers 而不只是 primes;
- pruning 後 protected prime powers 是否仍保留;
- smoothness / bounded-
- Ω
- $\Omega /$
- QA
- Q
- A
-
- notation 是否完全吻合。

評級:

概念可信度: 中高。

近期可成 note: 高。

是否已 proven: 未。

最大 gap: Liu–Sawhney interface audit。

Q

Q -level 1.5: minimal-over-one fixed

Q

Q

Status: 自然、重要、可能是下一個里程碑。

Minimal-over-one 意思是：

$$SQ > 1,$$

S

Q

$$> 1,$$

但移走任何一個重要元素後可能不再超過 1, 或者

Q

Q 是剛好跨過 1 的 minimal core。

為甚麼重要？因為在 primitive signed circuit 裏，如果

Q

Q 本身有很多多餘元素，它可能不是真正 hard core。真正難的 negative side 很可能是 minimal-over-one 類型。

你們現在想證的東西大概是：

$$SQ \leq 1 + 1mD$$

S

Q

$$\leq 1 +$$

m

D

1

以及：

$$1 + \frac{1}{m} D < \Phi D (1 - 1/e).$$

$1 +$

m

D

1

$< \Phi$

D

$$(1 - 1/e).$$

如果這兩個能對所有 relevant

$D \geq 3$

$D \geq 3$ 寫成嚴格 proof, 就可以說：

all fixed minimal-over-one

Q

Q satisfy gate condition

(G)

(G)。

這會比「某個 example

Q

Q 」自然很多。

評級：

數學自然性：高。

近期可行性: 高。

主要困難: technical inequality + finite small-

D

D checking。

不是 conceptual wall。

這也是為甚麼我會說: minimal-over-one 不是支線, 而是 fixed-

Q

Q 路線的自然升級。

Q

Q -level 2: arbitrary fixed

Q

Q

Status: 中遠; 真正開始困難。

這裏

Q

Q 還是固定, 不隨

N

N 增長, 但不一定滿足 gate condition

(G)

(G)。

問題是: 如果用

$D = \min Q$

$D=\min Q$ 和 target

$1/D$

$1/D$ 失敗, 不代表 fixed-

Q

Q method 沒救。可能只是 target 選錯。

你需要找 proper subset:

$$Q'\subsetneq Q$$

Q

,

$$\subsetneq Q$$

令:

$$r=R(Q')=\sum_{q\in Q'}1_q$$

$r=R(Q$

,

$$)=$$

$q\in Q$

,

$$\sum$$

$$q$$

$$1$$

的 denominator prime powers 都是 forceable 的。這就是你們說的 target-selection problem。

這一步可能需要 bad-prime quotient / finite abelian group subset-sum / zero-sum theory。你上傳文檔也把這條路評成 arbitrary fixed-

Q

Q 最值得押注的方向: 把 bad prime powers 做成有限 abelian group, 然後問是否存在

Q'

Q

,

使 target denominator 避開壞 prime powers。

評級:

可計算探索: 高。

理論 proof: 中等偏難。

是否接近 solved: 否。

最值得做: small fixed-

Q

Q exhaustive search + MITM target-selection experiment。

Q

Q -level 3: slowly growing

$Q=QN$

$Q=Q$

N

Status: 中低; fixed-

Q

Q 技術開始失效。

這裏

Q

Q 不再固定, 而是隨

N

N 慢慢變大。

為甚麼難很多？

因為 fixed-

Q

Q proof 依賴 constants:

$$c_{Q,\epsilon} > 0$$

c

Q, ϵ

$$> 0$$

只要

Q

Q 固定, 這些 constants 不會消失。但如果

$Q = Q_N$

$Q = Q$

N

變大, local mass margin、protected pruning margin、gate gap 都可能趨近 0。

也就是說:

fixed-

Q

Q 可以靠「每個 protected prime-power 有固定正 mass」;

growing-

Q

Q 則可能沒有 uniform positive margin。

這一層要開始做 uniform estimates, 不能只靠 single

Q

Q arithmetic。

評級:

概念可見: 有。

近期 proof: 不太近。

主要風險: constants collapse。

建議: 先做 toy regimes, 例如

$$|Q| = O(\log \log \log N)$$

$$|Q| = O(\log \log \log N),$$

$$SQ = O(1)$$

S

Q

$$= O(1),$$

$$\max Q = N o(1)$$

$$\max Q = N$$

$$o(1)$$

。

Q

Q -level 4: full Erdős #319

Status: 遠。

Full #319 不是固定

Q

Q , 也不是只選 target

1/D

1/D。它允許：

$$P, Q \subseteq \{1, \dots, N\}$$

$$P, Q \subseteq \{1, \dots, N\}$$

兩邊都大、都隨

N

N 變、都可以有複雜 structure, 並要求 primitive:

$$\sum_{p \in P} 1_p = \sum_{q \in Q} 1_q$$

$$p \in P$$

$$\sum$$

$$p$$

$$1$$

$$=$$

$$q \in Q$$

$$\sum$$

$$q$$

$$1$$

但不能有 proper subcircuit。

真正 full problem 可能要證明：

$$\Sigma(P) \cap \Sigma(Q) \neq \{0, SQ\}$$

$$\Sigma(P) \cap \Sigma(Q)$$

$=\{0,S$

Q

$\}$

也就是兩邊 reciprocal subset-sum spectra 必然有非平凡交集。

這比 fixed target hitting 難很多。Liu–Sawhney 的 machinery 可能提供 reachability intervals, 但 full #319 還需要證明兩邊 intervals / denominator lattices /

p

p -adic structures 必然 intersect。這是新的 two-sided structure theorem, 不是現成 theorem 直接給出的。

評級:

目前距離:遠。

需要工具:two-sided subset-sum spectrum,

p

p -adic top-layer cancellation, entropy/Fourier/additive combinatorics, possibly containers later。

是否可由 current fixed-

Q

Q note 直接推出:否。

Bloom formalisation 對我們的啟示

Bloom 的 density conjecture paper 確實有完整 Lean formalisation; paper 寫明主結果 proofs 已由 Bloom 與 Bhavik Mehta 用 Lean 完整形式化, 並檢查到 axiomatic level。但這反而提醒我們: formalised Bloom theorem

$\neq$

= formalised Liu–Sawhney Proposition 5.2

$\neq$

= formalised #319 fixed-

Q

Q theorem。

Bloom appendix 還說，即使人類覺得「顯然」的 asymptotic estimates, 在 Lean 裏也需要很長 formal proof。所以你們現在不能說「Aristotle checked numerics, 所以 theorem solved」。更準確是：

Aristotle helped audit local numerics and structure;

Lean package is useful scaffold;

full proof still needs human-readable interface audit before formalisation.

Honest progress dashboard

Level	Object	Current status	Confidence	Next action
Q0	Specific $Q=(3,5,7,9,11,13,15,17)$ $Q=(3,5,7,9,11,13,15,17)$	numerics/gate check strong	高	write clean example
Q1	fixed- Q Q under (G) (G)	plausible theorem candidate	中高	Liu–Sawhney line-by-line audit
Q1.5	minimal-over-one fixed Q Q	natural next class	中高	prove harmonic/gate inequality
Q2	arbitrary fixed Q Q	unresolved target-selection	中	bad-prime quotient search
Q3	slowly growing Q Q	constants may collapse	中低	toy regimes only
Q4	full #319	far	低	two-sided spectrum theory

最終誠實判斷

你現在不是在「接近 full Erdős #319 proof」。

但也不是在做 meaningless toy work。

最準確是：

You have a serious fixed- Q partial-theorem programme.

You have a serious fixed- Q partial-theorem programme.

如果成功完成 Q_1 ，它已經是值得給專家看的 partial result。

如果再完成 $Q_{1.5}$ ，它會變成一個自然 class 的 obstruction theorem。

如果 Q_2 有進展，才開始真正接近「fixed negative side general theory」。

只有 Q_3/Q_4 被打開，才談得上 full #319。

所以我會把整體 update 寫成一句：

目前最成熟的是 fixed-

Q

Q under gate condition；最現實的 breakthrough 是把它變成一篇短 note；full #319 仍遠，主要卡在 arbitrary/growing

Q

Q and two-sided spectrum intersection。

Many hanks! However may I confirm, so essentially are the only missing mathematically the remaining being since Arbitrary P ? Many thanks!

已思考 20s

不是。如果只講你現在的 Stage 1 fixed-

Q

Q theorem，主要缺口可以說是「能否對 arbitrary

P

P 成功套上 Liu-Sawhney machinery」。但如果講 full Erdős #319，缺口絕對不只是 arbitrary

P

P 。

官方 #319 仍列為 OPEN, 題目是問所有

$$A \subseteq \{1, \dots, N\}$$

$A \subseteq \{1, \dots, N\}$ 和所有 sign choices 的最大 size, 不是固定某個

Q

Q 的特殊情況。

最精準分法

把 signed circuit 寫成：

$$P = \{n: \delta n = +1\}, Q = \{n: \delta n = -1\}$$

$$P = \{n: \delta$$

n

$$= +1\}, Q = \{n: \delta$$

n

$$= -1\}$$

那條等式就是：

$$\sum_{p \in P} 1_p = \sum_{q \in Q} 1_q.$$

$$p \in P$$

$$\sum$$

$$p$$

$$1$$

=

$$q\in Q$$

$$\sum$$

$$q$$

$$1$$

.

你現在 fixed-

Q

Q 路線是：

固定

Q

Q , 然後證明任何很大的

P

P 都會被迫含有某個 subset

$B\subseteq P$

$B\subseteq P$, 例如

$$\sum_{b\in B}1b=1D,$$

$$b\in B$$

$$\sum$$

$$b$$

$$1$$

=

$$D$$

,

於是

$B \cup \{D\}$

$B \cup \{D\}$ 形成 proper subcircuit, 破壞 primitivity。

所以在這個 fixed-

Q

Q under gate condition 層級, 對, 核心缺口很接近:

arbitrary

P

P 是否真的可由 Liu–Sawhney Proposition 5.2 / pruning machinery 處理。

Liu–Sawhney 的 unit-fraction paper 確實是 relevant backbone, 因為它研究如何從 dense reciprocal set 中 force exact reciprocal subset sums; 但你的用法要逐條核對 target、local mass、smoothness、pruning、prime-power hypotheses。

但 full #319 還有其他缺口

full #319 不只是 fixed

Q

Q +arbitrary

P

P 。它要處理:

缺口	是不是已處理?	意思
arbitrary		
P		
P for fixed good	未完全	Liu–Sawhney interface audit
Q		
Q		
arbitrary fixed		
Q	未處理	Q

Q	Q 不一定滿足 gate condition, 要 target-selection
growing $Q=QN$ $Q=Q$ N	fixed- Q 未處理 Q constants 可能隨 N N 消失
arbitrary P, Q P, Q both large	需要 two-sided subset-sum spectrum 未處理 intersection
full primitivity closure	要保證任何 large circuit 必有 proper subcircuit 未完全

你上傳的研究路線文檔也正是這樣分層: fixed-

Q

Q under gate condition 近期可行性高, minimal-over-one 高, arbitrary fixed-

Q

Q 中高, slowly growing

Q

Q 中低, 而 full #319 仍低 / 遠。

一句話回答

Stage 1: 可以說主要缺 arbitrary

P

P 的 rigorous handling。

也就是: 給定 fixed

Q

Q , 能否對所有 sufficiently large

P

P force target subset。

Full #319: 不可以說只缺 arbitrary

P

P 。

因為 full problem 還要處理 arbitrary

Q

Q 、growing

Q

Q 、以及

P,Q

P, Q 兩邊的 subset-sum spectra 必然非平凡相交。

最準確講法是：

目前最接近完成的是 fixed-

Q

Q theorem; 它的主要數學缺口是 arbitrary

P

P via Liu–Sawhney interface。

但完整 Erdős #319 的主要缺口是 arbitrary/growing

Q

Q plus two-sided

P,Q

P, Q spectrum intersection, 不只是

P

P 。

MAny thanks! May i then ask, so does that mean even if we prove arbitrary Q, there is still growing Q obstructing, before full proof? Thanks.

已思考 7s

Yes — exactly, with one important nuance.

If you prove arbitrary fixed

Q

Q , that would be a serious step, but it still would not automatically prove full Erdős #319, because full #319 also allows

Q

Q to grow with

N

N .

Officially, Erdős #319 is still listed as open, with the known Adenwalla/Croot lower bound

$$|A| \geq (1 - 1/e + o(1))N,$$

$$|A| \geq (1 - 1/e + o(1))N,$$

and the site treats the Lean item as formalisation of the problem statement, not a solved proof.

The hierarchy is:

1. Fixed good

Q

Q

You fix one negative side, e.g.

$$Q = (3, 5, 7, 9, 11, 13, 15, 17).$$

$$Q = (3, 5, 7, 9, 11, 13, 15, 17).$$

Then ask: for this particular

Q

Q , can any very large

P

P survive primitivity?

This is your current strongest route.

2. Arbitrary fixed

Q

Q

You prove the result for every fixed finite

Q

Q .

That means

Q

Q can be anything like:

$\{3,5,7\}, \{4,6,11,25\}, \{10,17,19,23,41\},$

$\{3,5,7\}, \{4,6,11,25\}, \{10,17,19,23,41\},$

but once chosen, it stays fixed as

$N \rightarrow \infty$

$N \rightarrow \infty$.

This would be much stronger than the current fixed-

Q

Q gate theorem.

3. Growing

$$Q=QN$$

$$Q=Q$$

$$N$$

Now

$$Q$$

Q is allowed to change and grow with

$$N$$

N . For example:

$$|QN| \rightarrow \infty$$

$$|Q$$

$$N$$

$$| \rightarrow \infty$$

as

$$N \rightarrow \infty$$

$$N \rightarrow \infty.$$

This is genuinely harder, because all the fixed-

$$Q$$

Q constants may collapse. In fixed-

$$Q$$

Q , you can say: "this prime-power layer has some positive mass

$$cQ > 0$$

$$c$$

$$Q$$

>0 .” But if

$$Q=QN$$

$$Q=Q$$

N

grows, that mass might become:

$$cQN \rightarrow 0.$$

c

Q

N

$$\rightarrow 0.$$

Then protected pruning, local forcing, and target-selection arguments may stop working uniformly.

4. Full #319

Full #319 allows both sides to vary:

$$P=PN, Q=QN,$$

$$P=P$$

N

$$,Q=Q$$

N

,

with both potentially large and structured. Then the problem is no longer just “hit a target from

P

P.” It becomes a two-sided subset-sum spectrum problem:

$$\Sigma(P) \cap \Sigma(Q) \neq \{0, SQ\}.$$

$$\Sigma(P) \cap \Sigma(Q)$$

$$= \{0, S$$

Q

}.

This is the true full-problem difficulty.

So the answer is:

Yes. Even if arbitrary fixed

Q

Q is proved, growing

Q

Q remains a separate obstruction before full #319.

But arbitrary fixed

Q

Q would still be a very meaningful milestone. It would say:

No fixed negative-side pattern can beat the

$(1-1/e)$

$(1-1/e)$ -type barrier.

That would strongly suggest any counterexample or improvement must use a negative side

QN

Q

N

whose complexity grows with

N

N.

Clean status table

Stage	What it proves	Would it solve #319?
Specific Q $\mathcal{Q}$	One example family blocked	No
Fixed Q $\mathcal{Q}$ under gate condition	A class of fixed Q $\mathcal{Q}$'s blocked	No
Minimal-over-one fixed Q $\mathcal{Q}$	Natural fixed Q $\mathcal{Q}$ cores blocked	No
Arbitrary fixed Q $\mathcal{Q}$	Every fixed finite Q $\mathcal{Q}$ blocked	Still no
Slowly growing QN $\mathcal{Q}$ N	Some growing negative sides blocked	Not necessarily
Full arbitrary PN,QN P v $\mathcal{Q}$ N	All primitive circuits controlled	Yes

So the honest sentence is:

Arbitrary fixed

Q

$\mathcal{Q}$ would be a major partial theorem, but full #319 still needs uniform control over growing

QN

Q

N

and ultimately a two-sided

P, Q

P, Q spectrum-intersection theorem.

Many thanks! So can we now push for the proof of arbitrary P with the methods mentioned by Claude yesterday? Many thanks! For " Two very important things first, before I answer the substantive question.

Critical clarification about Lean verification

You wrote "the proof was later verified in Lean in Aristotle." I need to flag this carefully: the GPT document mentions that "primitive contradiction and gate condition are Lean-checked in the Aristotle scaffold." That is much weaker than "the proof was verified in Lean."

What was likely Lean-checked: the elementary parts — that $S_Q < \Phi_D(\theta)$ holds for your specific Q , that primitive failure follows IF you have B with $R(B) = 1/D$, etc. These are the easy bookkeeping steps.

What was almost certainly NOT Lean-checked: the actual Liu-Sawhney machinery (Proposition 5.2, Lemmas 2.2, 2.3, 6.2). Lean formalizations of Croot's Fourier analytic methods, Bloom's smoothing arguments, or Liu-Sawhney's Proposition 5.2 do not exist as far as I am aware. Bloom mentions Lean verification of his density result with Bhavik Mehta, but that is for Bloom's theorem itself, not for an application to #319.

So "Lean-verified in Aristotle" likely means the scaffold around the proof was checked, not the actual specialist content. Be careful about how you describe this when posting.

Now the substantive question

1. Real potentials in GPT's deep map

****Genuinely promising:****

The ****bad-prime quotient target selection**** idea is the strongest new direction. The observation that target selection for arbitrary fixed Q reduces to a zero-sum problem in a finite abelian group G_B is correct and connects to the well-developed Davenport constant theory. This is a real bridge that GPT did not over-sell.

The ****reachability interval corollary**** of Proposition 5.2 is plausible and would be useful. Liu-Sawhney's own description that their proposition can reach "all possible fractions" does suggest an interval theorem might be extractable. This is worth pursuing.

****Plausible but speculative:****

The p -adic layer peeling idea has real precedent in zero-sum theory but is far from a proof technique for #319.

The two-sided spectrum intersection framing is correct as a reformulation but does not contain new tools.

****Probably not productive:****

The entropy and absorption methods are interesting in principle but very far from any concrete proof path for #319.

The toric/Graver and Chevalley-Waring routes are mathematically elegant but, as GPT itself acknowledges, do not naturally see the $1-1/e$ constant.

2. What GPT missed or under-estimated

****One genuine omission worth flagging:****

GPT under-estimates how much the ****balance constraint $R(P) = R(Q)$ **** is doing in the proof and how much more it could do. The singleton-gate theorem uses balance once — to force local mass at $q \parallel D$ via the contradiction $R(P) > S_Q$. But the same balance constraint applied modulo every prime power p^a gives congruence conditions on the p -adic structure of P . GPT's section 5 mentions this briefly under "p-adic layer peeling" but treats it as a separate speculative route. Actually, integrating balance-derived p -adic constraints into the bad-prime quotient framework might be more powerful than either alone.

****One under-estimation:****

The minimal-over-one Q case is more important than GPT presents it. GPT treats Stage 2 of the roadmap (fixed minimal-over-one Q) as a step toward Stage 3 (arbitrary fixed Q). But minimal-over-one Q is the natural primitive structure — any non-minimal-over-one Q can be reduced because removing an element keeps $S_Q > 1$ and gives a forbidden target. So if you can prove minimal-over-one rigorously (which the current theorem does, conditional on Liu-Sawhney), you have actually handled the structurally distinct case. Stage 3 may be more accessible than it appears.

****The honest verdict:****

There are real potentials, primarily in the bad-prime quotient target selection direction. But they are research projects of months, not weeks. The current fixed- Q singleton-gate theorem, properly verified, is the real near-term deliverable. Post that, get specialist feedback, and only then decide whether to invest in the more speculative directions GPT mapped.

The advice remains: write the note, post it, wait for feedback. Do not start Stage 3 before Stage 1 has been validated by the community. " Thanks!

可以推，但要把目標縮準：先推「固定

Q

Q 、任意

P

P 」的 theorem，不要一開始推 arbitrary

Q

Q 或 full #319。我會先核對公開來源中的 #319 狀態與 Liu–Sawhney/Bloom 背景，再給你一個可執行 proof programme。

已思考 53s

Yes — we can now push the “arbitrary

P

P ” proof, but only in the precise Stage-1 sense:

Fixed

Q

Q , arbitrary large

P

P , under the gate condition

(G)

(G).

Not arbitrary

Q

Q , not growing

Q

Q , not full #319 yet.

Officially, #319 remains open; the page records the Adenwalla/Croot lower bound

$(1-1/e+o(1))N$

$(1-1/e+o(1))N$, says no partial or complete solutions are claimed in comments, and notes only that the problem statement has been formalised in Lean. Liu–Sawhney is the correct external machinery to audit because their paper proves strong unit-fraction reachability results and says their techniques refine Croot and Bloom. Bloom’s paper, by contrast, proves a positive-density unit-fraction theorem and has an appendix co-written with Bhavik Mehta, but that does not automatically formalise Liu–Sawhney or your #319 argument.

The exact theorem we should now try to prove

The target should be something like:

Fixed- Q Arbitrary- P Gate Theorem.

Fixed- Q Arbitrary- P Gate Theorem.

Let

Q

Q be a fixed finite negative side, let

$$SQ = \sum_{q \in Q} 1q,$$

S

$\mathcal{Q}$

$=$

$q \in \mathcal{Q}$

Σ

q

1

,

and suppose

$\mathcal{Q}$

$\mathcal{Q}$ satisfies the gate condition

(G)

(G), roughly:

$$SQ < \min p a \| D \Phi p a (1 - 1/e),$$

S

$\mathcal{Q}$

$<$

p

a

$\| D$

min

Φ

p

a

$$(1-1/e),$$

for the chosen target denominator

$$D \in Q$$

$D \in Q$. Then for every

$$\varepsilon > 0$$

$\varepsilon > 0$, for all sufficiently large

$$N$$

N , there is no primitive signed reciprocal circuit

$$\sum_{p \in P} 1/p = \sum_{q \in Q} 1/q$$

$$p \in P$$

$$\sum$$

$$p$$

$$1$$

$$=$$

$$q \in Q$$

$$\sum$$

$$q$$

$$1$$

with

$$|P| > (1-1/e+\varepsilon)N$$

$$|P| > (1-1/e+\varepsilon)N$$

and

Q

Q fixed.

Here “arbitrary

P

P means:

P

P has no special assumed structure; it is any sufficiently large positive side satisfying the balance equation

$$R(P)=R(Q)$$

$$R(P)=R(Q).$$

That is the right next proof target.

What parts are already strong?

The strongest current pieces are:

1. The
2. Φ
3. Φ -based local forcing idea
4. If
5. P
6. P is too large but lacks enough mass in the prime-power layers needed for
7. D
8. D , then the reciprocal balance
9. $R(P)=SQ$
10. $R(P)=S$
11. Q
12. should be impossible. This is where
13. Φ
14. Φ enters.
15. The primitivity contradiction
16. If Liu–Sawhney machinery produces
16. $B \subseteq P, \sum_{b \in B} 1/b = 1/D,$

$$17. B \subseteq P,$$

$$18. b \in B$$

$$19. \sum$$

$$20.$$

$$21. b$$

$$22. 1$$

$$23.$$

$$24. =$$

$$25. D$$

$$26. 1$$

$$27.$$

$$28.,$$

$$29. \text{and}$$

$$30. D \in Q$$

$$31. D \in Q, \text{ then}$$

$$32. B \cup \{D\}$$

$$33. B \cup \{D\} \text{ gives a smaller zero-sum circuit, unless it is the whole circuit. For}$$

$$34. |Q| > 1$$

$$35. |Q| > 1, \text{ this is usually safely proper.}$$

$$36. \text{The fixed-}$$

$$37. Q$$

$$38. Q \text{ advantage}$$

$$\text{Since}$$

$$39. Q$$

$$40. Q \text{ is fixed, all local forcing margins are fixed constants. This is why pruning preservation is plausible. Your uploaded roadmap also identifies fixed-}$$

$$41. Q$$

$$42. Q \text{ under}$$

$$43. (G)$$

$$44. (G) \text{ as the highest-feasibility route, with the Liu–Sawhney interface as the main risk.}$$

What remains to prove for arbitrary

P

P?

The proof should be reduced to four lemmas.

Lemma 1: Local forcing from balance

For each protected prime power

$p_a \parallel D$

p

a

$\parallel D$, prove:

$$R(P_{p_a}) \geq p_{p_a}, Q, \varepsilon > 0,$$

$R(P$

p

a

$) \geq \rho$

p

a

$, Q, \varepsilon$

$> 0,$

where

P_{p_a}

P

p

a

is the part of

P

P lying in the required

p_a

p

a

-divisibility layer.

Meaning in plain English:

Any sufficiently large

P

P balanced against this

Q

Q must contain enough denominator mass in the prime-power layers needed to build target

$1/D$

$1/D$.

This is the part where Claude is right: the balance condition

$R(P)=R(Q)$

$R(P)=R(Q)$ should not be used only once. It should be extracted prime-power by prime-power.

Lemma 2: Protected pruning

Show that after applying the smoothness / bounded-

Ω

Ω / local pruning needed by Liu–Sawhney, those protected layers still survive:

$$R(P_{\text{pruned}}) \geq 12 \rho_{P, Q, \epsilon}.$$

$R(P$

p

a

pruned

) $\geq$

2

1

ρ

p

a

$,Q,\varepsilon$

This is where fixed

Q

Q matters. Since

$\rho_{p,a,Q,\varepsilon}$

ρ

p

a

$,Q,\varepsilon$

is a fixed positive constant, we can choose the pruning loss smaller than it.

This would fail or become much harder for growing

Q_N

Q

N

, because the margin might tend to 0.

Lemma 3: Liu–Sawhney reachability

Use the exact Liu–Sawhney proposition to prove:

$\exists B \subseteq P_{\text{pruned}}$ such that $R(B) = 1/D$.

$\exists B \subseteq P$

pruned

such that $R(B) =$

D

This is the main audit point. We must check whether their Proposition 5.2 really gives this target under exactly our hypotheses.

This is not a cosmetic detail. If the target in Liu–Sawhney is

$$x/QA$$

$$x/Q$$

$$A$$

, or requires a specific denominator lattice, or requires local mass in a slightly different sense, the theorem statement may need to be weakened.

Lemma 4: Primitivity contradiction

Once

$$B \subseteq P$$

$$B \subseteq P \text{ with}$$

$$R(B)=1/D$$

$$R(B)=1/D \text{ exists, and}$$

$$D \in Q$$

$$D \in Q, \text{ we get:}$$

$$R(B)-1D=0.$$

$$R(B)-$$

$$D$$

$$1$$

$$=0.$$

Thus

$$B \cup \{D\}$$

$B \cup \{D\}$ is a non-empty zero-sum subcircuit. If it is proper, the original circuit is not primitive.

This is the cleanest part.

Should we use the “bad-prime quotient” now?

Not for Stage 1.

Bad-prime quotient target selection is probably for arbitrary fixed

Q

Q , where target

$$1/D$$

$1/D$ may be the wrong target. For the current fixed-

Q

Q gate theorem, the cleaner route is:

Φ -forcing $\Rightarrow$ protected prime powers $\Rightarrow$ Liu–Sawhney target $1/D \Rightarrow$ primitivity contradiction.

Φ -forcing $\Rightarrow$ protected prime powers $\Rightarrow$ Liu–Sawhney target $1/D \Rightarrow$ primitivity contradiction.

So yes, Claude is right that bad-prime quotient is a real direction, but I would not mix it into the first proof unless the

$$1/D$$

$1/D$ target fails.

What I recommend doing now

Push Stage 1 in this order:

1. Write the exact theorem statement.
Include all hypotheses: fixed
2. Q
3. $\mathcal{Q}$,
4. $D \in Q$
5. $D \in \mathcal{Q}$,
6. $|Q| > 1$
7. $|\mathcal{Q}| > 1$, gate condition,
8. $P \cap Q = \emptyset$
9. $P \cap \mathcal{Q} = \emptyset$,
10. $R(P) = SQ$
11. $R(P) = S$
12. $\mathcal{Q}$
- 13.
- 14.,
15. $|P| > (1 - 1/e + \varepsilon)N$
16. $|P| > (1 - 1/e + \varepsilon)N$.
17. Write Lemma 1 fully.
This is your original contribution:
18. Φ
19. Φ -forcing from the balance equation.
20. Create a Liu–Sawhney parameter dictionary.
Match your notation to theirs line by line: their
21. $A, QA, S, K, M, N, \Gamma$
22. $A, \mathcal{Q}$
23. A
- 24.
25. S, K, M, N, Γ , local mass conditions, target fraction, smoothness assumptions.
26. Prove pruning preservation only for fixed
27. Q
28. $\mathcal{Q}$.
Do not attempt growing
29. Q
30. $\mathcal{Q}$. Fixed
31. Q
32. $\mathcal{Q}$ is where constants stay positive.

33. Only then draft the note.

The note should be titled something like:

A fixed-negative-side obstruction related to Erdős Problem #319
not “Solution to Erdős #319.”

Honest verdict

Yes, we can now push the arbitrary-

P

P proof.

But the exact honest framing is:

Push fixed- Q , arbitrary- P under gate condition.

Push fixed- Q , arbitrary- P under gate condition.

The make-or-break point is:

Does Liu–Sawhney Proposition 5.2 really produce the target $1/D$ after our pruning?

Does Liu–Sawhney Proposition 5.2 really produce the target $1/D$ after our pruning?

If yes, Stage 1 can become a genuine partial theorem.

If no, the

Φ

Φ -forcing idea may still be real, but the theorem statement must be repaired.

Many thx may i confirm - did we find proposition 5.2 from a book? is it more likely than not correct?

已思考 38s

Yes — we found Proposition 5.2 in Liu–Sawhney’s arXiv paper, not from a book.

The source is:

Yang P. Liu and Mehtaab Sawhney, “On further questions regarding unit fractions,”
arXiv:2404.07113.

The paper states that it proves a strong unit-fraction theorem and that its techniques refine Croot and Bloom.

What Proposition 5.2 actually says, roughly

Proposition 5.2 is not just a vague citation. It is explicitly stated in the paper. In simplified language, it says:

If

$$A \subseteq [M, N]$$

$A \subseteq [M, N]$ is smooth enough, has bounded number of prime factors, has enough local mass in every prime-power layer

$$q \in Q_A$$

$$q \in Q$$

A

, and the target fraction

$$x/Q$$

x/Q lies in the right Archimedean range relative to

$$R(A)$$

$R(A)$, then there exists

$$B \subseteq A$$

$B \subseteq A$ such that

$$R(B) = \sum_{b \in B} 1/b = x/Q.$$

$$R(B) =$$

$$\sum_{b \in B}$$

$$\sum$$

$$b$$

$$1$$

=

Q

x

.

The formal statement includes conditions like

$$A \subseteq [M, N]$$

$A \subseteq [M, N]$, every

$$n \in A$$

$n \in A$ being

S

S -smooth,

$$\max_{n \in A} \Omega(n) \leq 5 \log \log N$$

$\max$

$$n \in A$$

$\Omega(n) \leq 5 \log \log N$, and a local mass condition

$$\min_{q \in Q} A_q R(A_q) \geq \eta$$

$\min$

$$q \in Q$$

A

$$q R(A$$

q

$) \geq \eta$. Then, if

$$R(A)$$

$R(A)$ is in the required interval around

$$x/Q$$

x/Q , it concludes that some

$$B \subseteq A$$

$B \subseteq A$ satisfies

$$R(B)=x/Q$$

$$R(B)=x/Q.$$

The authors themselves describe the importance of Proposition 5.2 as separating “Archimedean” and “

p

p-adic” considerations, and say it lets one reach “all possible” fractions under the right conditions.

Is Proposition 5.2 more likely than not correct?

Yes — as a theorem inside the Liu–Sawhney paper, it is more likely than not correct. I would treat it as a serious modern research result, not as an AI-invented claim or random internet statement.

But there are two separate questions:

Question	My honest assessment
Is Proposition 5.2 itself likely correct?	Yes, likely.
Does it automatically prove our fixed- Q Q Erdős #319 partial theorem?	No, not automatically.
Is our intended application plausible?	Yes, but still needs line-by-line interface audit.

The main risk is not that Liu–Sawhney’s Proposition 5.2 is fake. The main risk is that our

$$P,Q,D,\eta,S,K,M,N,QA,x/Q$$

$$P,Q,D,\eta,S,K,M,N,Q$$

A

,*x*/*Q* do not exactly satisfy its hypotheses after pruning.

The exact risk

For our #319 fixed-

Q

Q argument, we want to use Proposition 5.2 to get:

$$B \subseteq P, R(B) = 1D.$$

$$B \subseteq P, R(B) =$$

$$D$$

$$1$$

.

To do that, we need to match their target:

$$xQ$$

$$Q$$

$$x$$

with our desired target:

$$1D.$$

$$D$$

$$1$$

.

So we need:

$$xQ = 1D,$$

$$Q$$

$$x$$

$$=$$

D

1

,

which means

Q

Q must be divisible by

D

D , and

$x=Q/D$

$x=Q/D$ must be an integer. That is plausible if their

Q

Q is the lcm-like denominator generated by

QA

Q

A

, but it must be checked, not assumed.

Then we must also verify:

1. P
2. P can be pruned to an
3. $A \subseteq [M, N]$
4. $A \subseteq [M, N]$;
5. the pruned
6. A
7. A remains
8. S
9. S -smooth;
10. $\Omega(n) \leq 5 \log \log N$
11. $\Omega(n) \leq 5 \log \log N$;
12. every required
13. $q \in QA$

14. $q \in Q$
15. A
- 16.
17. has local mass
18. $qR(Aq) \geq \eta$
19. $qR(A$
20. q
- 21.
22. $) \geq \eta;$
23. $R(A)$
24. $R(A)$ lies in the required interval relative to
25. x/Q
26. $x/Q;$
27. the proposition's
28. QA
29. Q
30. A
- 31.
32. includes the prime-power factors needed for
33. D
34. D .

That is why we keep calling this the Liu–Sawhney interface.

Important distinction

Officially, Erdős #319 is still listed as OPEN, with no partial or complete solutions claimed in comments. The page records only the Adenwalla/Croot lower bound and says the problem statement, not a solution, has been formalised in Lean.

So the correct sentence is:

We found a real Proposition 5.2 in a serious arXiv paper, and it is more likely than not correct. But our application to #319 remains unproved until the hypotheses are checked exactly.

Bottom line

No, it was not from a book. It was from Liu–Sawhney's research paper.

Yes, Proposition 5.2 itself is likely correct.

But for us, the key question is narrower:

Can our arbitrary

P

P , after

Φ

Φ -forcing and pruning, be made to satisfy exactly the hypotheses of Proposition 5.2 with target

$1/D$

$1/D$?

That is the next serious proof task.

Many thanks. Does this answer your question: Yes, it is fixable. Let me go through the six conditions GPT identified one by one with the precise status of each.

Condition by condition

**1. P can be pruned to $A \subseteq [M, N]$ **

Already handled. We set $A_1 = P \cap [\lambda N, N]$ at the start of the proof. The Liu-Sawhney parameter requires $M \leq N/10^4$. Choose $\lambda = 1/10^5$. Then $A_1 \subseteq [\lambda N, N]$ satisfies $M = \lambda N \leq N/10^4$. This condition is met by construction.

2. The pruned A remains S -smooth

After applying Liu-Sawhney Lemma 2.3, the remaining elements are by definition S -smooth for some S in the parameter range. Their condition requires $S \geq N^{0.99}$. For our $\lambda N \geq N^{0.99}$ to hold, we need $N^{0.01} \geq 1/\lambda$, which holds for $N \geq \lambda^{-100}$. For $\lambda = 1/10^5$, that means $N \geq 10^{500}$. Large but finite — this is a legitimate "for sufficiently large N " condition, exactly what an asymptotic theorem requires.

**3. $\Omega(n) \leq 5 \log \log N$ **

Satisfied by definition after applying Lemma 2.2. No issue here.

**4. Local mass $qR(A_q) \geq \eta$ for every $q \in Q_A$ **

For the primes dividing D , our Φ -forcing argument gives $q \cdot R((A_1)_q) \geq c_q/2$ which is a positive constant, far above the threshold $\eta \cdot \xi_0 / (2 \log \log N) \rightarrow 0$. So these primes survive all pruning. For all other prime powers in Q_A , Lemma 6.2 supplies the local mass directly. This condition is met.

****5. $R(A)$ in the required Archimedean window****

Lower bound: $R(A_0) \geq 1/D + \Delta/4 > (1 + 1/\log N)/D$ for large N . ✓

Upper bound: $R(A_0) \leq R(P) = S_Q$ which is a fixed constant much less than $(\log N)/D$ for large N . ✓

Both bounds are satisfied for sufficiently large N .

****6. $Q_{\{A_0\}}$ includes the prime-power factors for D ****

Since our Φ -forcing guarantees positive local mass at D , multiples of D remain in A_0 after all pruning. Therefore $D \mid Q_{\{A_0\}}$, making $x = Q_{\{A_0\}}/D$ an integer, and the target $x/Q_{\{A_0\}} = 1/D$ is valid.

The bottom line

All six conditions are satisfiable. None represents a conceptual obstruction. The only thing needed is for someone to sit with the Liu-Sawhney paper and write out the explicit parameter table — choosing $\lambda, S, K, M, \eta, \xi$ explicitly as functions of N and verifying the chain $N^{0.99} \leq S \leq K \leq M \leq N/10^4$ holds. This is technical bookkeeping, not new mathematics.

The threshold N_0 for "sufficiently large N " may be very large — potentially $N \geq 10^{500}$ or similar — but that is standard for theorems of this type and does not affect the validity of the asymptotic statement.

So the correct characterization of the current state is: the proof is complete in structure, all new mathematics is verified, and the remaining step is a technical parameter verification that is likely routine for a specialist in the area. Post the note.

已思考 2m 1s

Yes — this substantially answers the right question, but I would slightly downgrade the final conclusion.

My honest update:

This makes the Liu–Sawhney interface look fixable and probably not conceptually blocked. But it is not yet safe to say “the proof is complete except routine bookkeeping.” It is safer to say: “the proof structure is now plausible; the remaining work is a serious but probably manageable parameter-verification and pruning-preservation audit.”

I checked Proposition 5.2 again. It really does have the shape we need: if

$$A \subseteq [M, N]$$

$$A \subseteq [M, N] \text{ is}$$

S

S -smooth, has

$$\Omega(n) \leq 5 \log \log N$$

$$\Omega(n) \leq 5 \log \log N, \text{ satisfies a local-mass condition}$$

$$\min_{q \in Q} A_q R(A_q) \geq \eta$$

$\min$

$$q \in Q$$

A

$$q R(A$$

q

$$) \geq \eta, \text{ and}$$

$$R(A)$$

$$R(A) \text{ lies in the interval}$$

$$[(1+1/\log N)xQ, (\log N)xQ],$$

$$[(1+1/\log N)$$

$$Q$$

$$x$$

$$, (\log N)$$

Q

x

$]$,

then there exists

$$B \subseteq A$$

$B \subseteq A$ with

$$R(B) = x/Q$$

$$R(B) = x/Q.$$

Condition-by-condition verdict

1.

P

P can be restricted to

$$A1 = P \cap [\lambda N, N]$$

A

1

$$= P \cap [\lambda N, N]$$

Mostly yes. This is a valid starting move. But for a theorem with arbitrary

$$\varepsilon > 0$$

$\varepsilon > 0$, I would not hard-code

$$\lambda = 10^{-5}$$

$$\lambda = 10$$

-5

. Better choose

$$\lambda = \min(10^{-5}, \varepsilon/100)$$

$$\lambda = \min(10$$

$$-5$$

$$, \varepsilon/100)$$

or something similar. Otherwise, if

$$\varepsilon < 10^{-5}$$

$$\varepsilon < 10$$

$$-5$$

, truncating below

$$\lambda N$$

λN may lose more than the excess density you are trying to exploit.

So: fixable, but choose

$$\lambda = \lambda(\varepsilon, Q)$$

$$\lambda = \lambda(\varepsilon, Q), \text{ not one universal fixed number.}$$

2. Smoothness after pruning

Yes, broadly correct. Liu–Sawhney use Lemma 2.3 to control integers with large prime-power divisors, and their later proof explicitly removes non-

S

S -smooth numbers while showing the reciprocal-mass loss is small.

But the statement should not only check

$$N^{0.99} \leq S \leq M.$$

$$N$$

$$0.99$$

$$\leq S \leq M.$$

Proposition 5.2 also requires the full chain

$$N^{0.99} \leq S \leq K \leq M \leq N/10^4$$

N

0.99

$$\leq S \leq K \leq M \leq N/10$$

4

and the stronger inequalities involving

$$S, K, M, \eta, \Gamma$$

S, K, M, η, Γ . So this condition is not just “smoothness exists”; it must be part of a full parameter table.

3.

$$\Omega(n) \leq 5 \log \log N$$

$$\Omega(n) \leq 5 \log \log N$$

Yes. Liu–Sawhney Lemma 2.2 gives that the set of

$$n \leq N$$

$n \leq N$ with

$$\Omega(n) > 5 \log \log N$$

$\Omega(n) > 5 \log \log N$ is small, and the paper later removes such

n

n with small reciprocal-mass loss.

This part is relatively routine.

4. Local mass

$$qR(Aq) \geq \eta$$

$$qR(A$$

q

$$) \geq \eta$$

Plausible, but this is one of the two most important places to write carefully.

There are two subcases:

For protected prime powers dividing

D

D , your

Φ

Φ -forcing must prove local mass inside the tail

$A_1 = P \cap [\lambda N, N]$

A

1

$= P \cap [\lambda N, N]$, not merely somewhere in

P

P . If it only proves mass in all of

P

P , small denominators below

λN

λN could carry too much of the mass.

For other

$q \in Q_A$

$q \in Q$

A

, Lemma 6.2 does exactly the right kind of pruning: it removes low-local-mass layers and returns

$$A' \subseteq A$$

$$A$$

,

$$\subseteq A \text{ such that for all}$$

$$q \in Q_{A'}$$

$$q \in Q$$

$$A$$

,

,

$$qR(Aq') \geq \eta \xi^{2 \log \log N}$$

$$qR(A$$

$$q$$

,

$$) \geq$$

$$2 \log \log N$$

$$\eta \xi$$

in the paper's notation.

But to apply Proposition 5.2, you should rename the parameter carefully. For example:

$$\eta_{LS} := \eta \xi^{2 \log \log N}.$$

η

LS

$:=$

$2\log\log N$

η

0

ζ

.

Then check:

$\eta\text{LS}\geq 1\log N.$

η

LS

$\geq$

$\log N$

1

.

This is true for large

N

N , if

η_0,ξ

η

0

, ξ are fixed positive constants, but it should be written explicitly.

5. Archimedean window for

$$R(A)$$

$$R(A)$$

Likely yes, but this is the other crucial proof obligation.

To hit target

$$1/D$$

$1/D$, we set

$$x/QA=1/D.$$

$$x/Q$$

$$A$$

$$=1/D.$$

Proposition 5.2 requires

$$R(A)\in[(1+1/\log N)1D, (\log N)1D].$$

$$R(A)\in[(1+1/\log N)$$

$$D$$

$$1$$

$$,(\log N)$$

$$D$$

$$1$$

].

The upper bound is easy because

$$R(A) \leq R(P) = SQ = OQ(1)$$

$$R(A) \leq R(P) = S$$

$$Q$$

$$= O$$

$$Q$$

(1), while

$$(\log N)/D \rightarrow \infty$$

$$(\log N)/D \rightarrow \infty.$$

The lower bound is the important one. Your text says:

$$R(A_0) \geq 1/D + \Delta/4.$$

$$R(A$$

$$0$$

$$) \geq 1/D + \Delta/4.$$

If that has genuinely been proved after tail restriction, smooth pruning,

$$\Omega$$

Ω -pruning, and Lemma 6.2 pruning, then yes, the lower window condition is satisfied for large

$$N$$

$$N.$$

But this line should be its own lemma:

Archimedean Mass Preservation Lemma: after all pruning,

$$R(A_0) > 1/D$$

$$R(A$$

$$0$$

$$) > 1/D \text{ by a fixed margin.}$$

Without that lemma written cleanly, the proof is still not closed.

6.

$D \mid QA_0$

$D \mid Q$

A

0

Likely yes, but phrase it in prime-power language.

In Liu–Sawhney,

QA

Q

A

is built from prime-power factors appearing in elements of

A

A , and

Q

Q is essentially the lcm generated by those prime powers; the paper describes

QA

Q

A

as the set of prime-power factors dividing at least one element of

A

A , with

Q

Q the least common multiple of those integers.

So to make

$$x=QA_0/D$$

$$x=Q$$

A

0

$/D$

an integer, you need every prime-power factor

$p_a \parallel D$

p

a

$\parallel D$ to appear in

QA_0

Q

A

0

. You do not necessarily need many multiples of the whole composite

D

D ; it is enough that the required prime-power layers survive.

If your

Φ

Φ -forcing gives positive local mass for every

$p_a \parallel D$

p

a

// D , and pruning preserves those layers, then this condition is fine.

The important correction

Your quoted response says:

“The only thing needed is ... verifying the chain

$$N^{0.99} \leq S \leq K \leq M \leq N/10^4$$

N

0.99

$$\leq S \leq K \leq M \leq N/10$$

4

.”

That is not enough.

Proposition 5.2 also requires the quantitative inequalities:

$$S \leq \min(M^2CN, \eta MK^2N^2(\log N)^3),$$

$$S \leq \min($$

$$CN$$

M

2

,

N

2

$(\log N)$

3

ηMK

2

),

$$K \leq M \exp(-(\log N)^{1-\delta}),$$

$$K \leq M \exp(-(\log N)^{1-\delta})$$

$1-\delta$

),

and

$$C \leq \Gamma^2 (\log N)^{2\varepsilon} (\log(N/M) + (\log N)^{1-\delta}).$$

$$C \leq$$

$$(\log N)$$

2ε

$$(\log(N/M) + (\log N)^{1-\delta})$$

$1-\delta$

)

Γ

2

.

These are explicitly part of the proposition.

So the remaining audit is not merely the simple chain. It is the full parameter inequality table.

Bottom line

Does this answer the question?

Yes, it strongly suggests the Liu–Sawhney interface is fixable.

Does it prove the theorem already?

Not yet.

Is it more likely than not that Stage 1 can be made rigorous?

I would now say yes, more likely than not, provided the

Φ

Φ -forcing and pruning-preservation lemmas are exactly as stated.

The safest characterization is:

The proof is complete in architecture, but not yet complete as mathematics. The remaining work is a nontrivial parameter audit plus two preservation lemmas: protected local mass and Archimedean mass after pruning.

I would prepare the note, but phrase it as:

"A fixed-

Q

Q partial obstruction related to Erdős #319; Liu–Sawhney interface checked modulo explicit parameter verification."

Not yet:

"Proof complete."

Many thanks! Do you agree "I cannot close that gap. I want to be honest with you about why, because this matters.

The remaining work GPT identifies is a real mathematical task that requires sitting with the Liu-Sawhney paper open, working through Proposition 5.2's full parameter table, and writing two preservation lemmas with all the explicit inequalities verified. Specifically the chain $S \leq \min(M^2/(CN), \eta MK^2/(N^2(\log N)^3))$, $K \leq M \cdot \exp(-(\log N)^{1-\delta})$, and $C \leq \Gamma^2/((\log N)^{2\epsilon} \cdot (\log(N/M) + (\log N)^{1-\delta}))$ all have to be satisfied simultaneously with explicit choices of every parameter.

This is several hours of careful manual verification by someone with the paper in hand. It is not something I can do in a chat exchange. If I tried to produce the

parameter table here, I would be guessing at exact values rather than verifying them, and that would be exactly the kind of false confidence that has caused trouble throughout this conversation. Earlier I made an error claiming "the cascade closes automatically" that needed correction. I do not want to repeat that mistake at the verification stage where it matters most.

The honest path forward is the one we already identified. Post the note with the parameter audit listed as the remaining technical step. A specialist reading it can either verify the parameters in an afternoon, or tell you precisely what additional inequality fails and needs work. That feedback is worth more than any further AI exchange we have here, because the question is now narrowly technical rather than conceptual.

What you have is genuinely valuable as it stands: a proof complete in architecture, with all the new mathematical ideas verified and the remaining work clearly identified as parameter bookkeeping. That is a real contribution. Trying to push further in this chat without the paper open in front of someone who knows the area will produce more elaborate framing without actually closing anything.

Write the note with the honest framing GPT suggested at the end: "a fixed-Q partial obstruction related to Erdős #319, Liu-Sawhney interface checked modulo explicit parameter verification." Post it. Let the specialists do the verification step. That is the right next action and the only one that will actually move things forward.

Many thanks! I see - so it is like "this is a known step" but not like we can pull that step out of thin air. Thx!

8:24 PM

Exactly. You have put it perfectly.

The parameter verification is a known type of step — specialists do exactly this kind of audit routinely when applying someone else's technical machinery to a new problem. It is not mysterious or conceptually hard. But it requires the actual paper open, the actual inequalities in front of you, and careful manual checking of each constraint. It cannot be conjured from reasoning alone.

A useful analogy: it is like confirming that a bridge design satisfies a building code. The bridge design is real and sound. The building code is real and published. Whether the specific design meets every clause of the code is a genuine engineering audit — you cannot just assert it does without checking each clause.

In your case:

The bridge design = the proof architecture (new, correct, verified by Aristotle)

The building code = Proposition 5.2's parameter table (real, published, in the Liu-Sawhney paper)

The audit = what a specialist can do with the paper open in an afternoon

The contribution is real. The remaining step is real but known in character. That is an honest and good place to be.

Now write the note and post it.

"

已思考 36s

是, 我基本同意這段判斷, 而且我覺得它比前面一些較樂觀的說法更成熟。

最準確地說:

這不是 conceptual gap; 但仍是 real mathematical verification gap。

它是「已知類型的技術審核」, 不是「可以憑空略過的步驟」。

我同意的部分

我同意這句核心判斷:

The proof is complete in architecture, but not complete as a checked proof.

原因是 Liu-Sawhney Proposition 5.2 的確不是一句簡單「dense set gives target」; 它

有完整 parameter table, 包括:

$$S \leq \min(M^2 CN, \eta M K^2 N^2 (\log N)^3),$$

$$S \leq \min($$

$$CN$$

$$M$$

$$2$$

,

$$N$$

$$2$$

$$(\log N)$$

3

$$\eta MK$$

2

),

$$K\leq M\exp(-(\log N)^{1-\delta}),$$

$$K\leq M\exp(-(\log N)$$

$^{1-\delta}$

),

以及

$$C\leq \Gamma 2(\log N)^{2\varepsilon}(\log(N/M)+(\log N)^{1-\delta}).$$

$$C\leq$$

$$(\log N)$$

$^{2\varepsilon}$

$$(\log(N/M)+(\log N)$$

$^{1-\delta}$

)

Γ

2

.

它還要求

$$A\subseteq [M,N]$$

$$A\subseteq [M,N]、所有$$

$$n\in A$$

$$n\in A \text{ 是}$$

S

S -smooth、

$$\Omega(n) \leq 5 \log \log N$$

$\Omega(n) \leq 5 \log \log N$ 、local mass 條件、以及 Archimedean window；最後才推出存在

$$B \subseteq A$$

$B \subseteq A$ 使

$$R(B) = x/Q$$

$R(B) = x/Q$ 。這些條件是 paper 裏明確寫出的，不是 GPT 自己想像出來的。

所以，Claude 說「不能憑 chat 交換就宣稱閉合」是正確的。

我會稍微修正的部分

我不會說：

“all new mathematical ideas are verified.”

我會改成：

the local new ideas appear verified:

Φ

Φ -forcing, gate numerics, and primitivity contradiction look solid; the global application still depends on the Liu–Sawhney interface.

這個差別很重要。因為「新想法」裏也包含 protected pruning / Archimedean mass preservation / target

$1/D$

$1/D$ matching；這些還沒有完全寫成正式 lemma。

關於 “post the note”

我同意可以 post / send, 但要看 post 到哪裏、怎樣說。

不建議現在用標題：

Solution to Erdős #319

也不建議說：

proof complete.

因為 Erdős Problems #319 仍列為 open, 頁面也說 comments 中沒有 claimed partial or complete solutions; 它只記錄 Adenwalla/Croot lower bound 和 problem statement formalised in Lean。

但可以寫成：

A fixed-

Q

Q partial obstruction related to Erdős Problem #319

或：

A fixed-negative-side reduction for primitive signed reciprocal circuits

並且誠實註明：

Liu–Sawhney interface checked modulo explicit parameter verification.

這樣是合理的。

你的理解非常準確

你說：

“this is a known step but not like we can pull that step out of thin air.”

完全對。

更準確講：

它是 known kind of specialist step, 不是 known completed step。

就像 Claude 的橋樑/建築 code 比喻：

橋的設計圖看起來合理; building code 是公開的; 但是否每條 code 都滿足, 仍然要逐條核對。

現在的 status 應該寫成這樣

最客觀版本：

We have a serious fixed-

Q

Q , arbitrary-

P

P proof architecture. The elementary and new local components appear sound. The remaining gap is a nontrivial but standard Liu–Sawhney parameter audit, especially the full Proposition 5.2 inequalities plus protected local-mass and Archimedean-mass preservation after pruning.

中文：

這不是「已經證完」；但也不是「還差一個神秘新 idea」。

現在差的是：把已知大型機器 Liu–Sawhney 的每個條件精準對上。

最終判斷

我同意 Claude 的 cautious version。

但我會把最後一句從：

“Post it.”

改成：

Prepare the note and circulate it honestly — first to one or two specialists or as a clearly labelled draft, not as a solved #319 claim.

這是現在最穩妥、最有價值的下一步。

Many thanks! Now i am thinking to put a marker here "Insert - contact specialists (e.g. thomas bloom)"; then proceed to trying P and Q arbitrary and growing. Can you explain in secondary school level 1. What have we achieved 2. what is missing Thx!

已思考 21s

可以。你這個 marker 很合理，但我會把它寫得更精準一點：

INSERT — Contact specialists / ask for Liu–Sawhney interface audit before claiming theorem. Possible people: Thomas Bloom; also Liu–Sawhney authors if appropriate. Current status: fixed-

Q

Q , arbitrary-

P

P proof architecture; parameter verification pending.

Thomas Bloom 是合理人選之一，因為他的研究領域包括 additive combinatorics、analytic number theory、combinatorics、harmonic analysis，而且他有 unit fractions / density conjecture 相關工作。但更直接的 interface 專家其實也包括 Yang Liu / Mehtaab Sawhney，因為我們現在卡的正是他們 Proposition 5.2 的參數對接。

用中學生版本講：我們做到了甚麼？

你可以想像 #319 是一個天秤問題。

左邊是

P

P :

$$\sum_{p \in P} 1/p$$

$$p \in P$$

$$\sum$$

$$p$$

$$1$$

右邊是

Q

Q :

$$\sum_{q \in Q} 1/q$$

$$q \in Q$$

$$\sum$$

$$q$$

$$1$$

題目問:最多可以放多少個數字, 令左右剛好平衡, 而且這個平衡不是由一個更小的平衡偷偷組成?

官方 Erdős #319 現在仍然是 OPEN; 目前公開頁面記錄的是 Adenwalla/Croot 型下界, 大約可以做到

$$(1-1/e+o(1))N$$

$(1-1/e+o(1))N$, 並沒有 claimed partial 或 complete solution。

我們目前做到的, 是找到一個「固定右邊

Q

」的攻擊方法

例如先固定:

$$Q=\{3,5,7,9,11,13,15,17\}.$$

$$Q=\{3,5,7,9,11,13,15,17\}.$$

意思是:右邊砝碼先固定好。然後我們問:

不管左邊

P

P 怎樣亂放, 只要

P

P 太大, 它是否一定會包含一小撮

B

B , 剛好等於右邊某個單一砝碼, 例如

$$1/3$$

$$1/3?$$

如果能證明:

$$\sum_{b \in B} b = 13,$$

$$b \in B$$

$$\sum$$

$$b$$

$$1$$

$$=$$

$$3$$

$$1$$

$$,$$

那麼

B

B 和右邊的

3

3 就形成一個小平衡：

$$\sum_{b \in B} b - 13 = 0.$$

$$b \in B$$

$$\sum$$

$$b$$

$$1$$

$$-$$

$$3$$

$$1$$

$$=0.$$

這會破壞 primitive condition。也就是說，原本的大平衡不是「不可拆」的。

這就是目前的主 idea。

我們真正 achieved 的東西

1. 題目被翻譯成

P, Q

P, Q 兩邊天秤

這是重要的 conceptual simplification:

$$R(P)=R(Q).$$

$$R(P)=R(Q).$$

以前題目看似是一堆正負號；現在可以看成左邊

P

P 、右邊

Q

Q 的 reciprocal balance。

2. 我們找到 fixed-

Q

Q route

目前最成熟的是：

固定

Q

Q , 證明任意很大的

P

P 都不能保持 primitive。

這比 full #319 窄很多，但是真實、有方向。

你的上傳研究筆記也正是這樣分層:fixed-

Q

Q under gate condition 是近期最高可行性;minimal-over-one 也高;arbitrary fixed-

Q

Q 、slowly growing

Q

Q 、full #319 依次更難。

3. 我們有一個

Φ

Φ -based local forcing idea

簡單講:

如果左邊

P

P 太大, 又要和固定右邊

Q

Q 平衡, 那麼

P

P 不能完全避開某些 prime-power layer。

中學生版本:

你想把天秤放平, 但你左邊砝碼太多。如果你刻意不放某一類必要形狀的砝碼, 總重量就會超出或不夠, 天秤不可能平衡。

這是目前最像「新 idea」的部分。

4. 我們知道要接哪台大型機器

那台機器是 Liu-Sawhney 的 unit-fraction reachability theorem / Proposition 5.2。它的 paper 證明, 如果一個集合的 reciprocal mass 和 local

p

p -adic 條件足夠好, 就可以找到 subset 令 reciprocal sum 等於指定分數; 作者也說他們 refine Croot 和 Bloom 的方法。

所以我們不是憑空發明全部機器; 我們是在試圖把自己的

Φ

Φ -forcing 接到他們的 technical machine。

5. 我們知道現在最該找專家查甚麼

不是叫專家「幫我看我是否解了 #319」。

而是問:

這個 fixed-

Q

Q partial theorem 的 Liu–Sawhney Proposition 5.2 interface 是否 faithful?
參數

S, K, M, η, Γ

S, K, M, η, Γ 和 pruning preservation 是否能同時滿足?

這是一個很具體、可審核的問題。

那還缺甚麼?

缺口 A: Stage 1 的 technical audit 還未完成

這是最短期缺口。

我們需要確認 Liu–Sawhney Proposition 5.2 的所有條件真的可同時滿足, 包括:

$$S \leq \min(M^2 CN, \eta MK^2 N^2 (\log N)^3),$$

$$S \leq \min($$

$$CN$$

M

2

,

N

2

$(\log N)$

3

ηMK

2

),

$K \leq M \exp(-(\log N)^{1-\delta}),$

$K \leq M \exp(-(\log N)$

$1-\delta$

),

以及其他 local mass / smoothness / Archimedean window 條件。

中學生版：

我們已經找到「可以用哪台機器」，但還要核對插頭、電壓、尺寸、說明書每一條是否吻合。這不是新概念，但不能跳過。

缺口 B: 固定

Q

Q 不等於任意

Q

Q

現在我們主要處理的是「右邊

Q

Q 固定」。

但 full #319 允許

Q

Q 是任何樣子。

也就是說，我們現在像是證明：

這一種右邊砵碼擺法不行。

之後要證：

所有固定右邊砵碼擺法都不行。

這就是 arbitrary fixed

Q

Q 。

缺口 C : 任意固定

Q

Q 仍不等於 growing

Q

Q

就算你證明所有固定

Q

Q 都不行，還有一種可能：

$$Q=QN$$

$$Q=Q$$

N

隨著

N

N 變大, 右邊也越來越複雜。

這是 growing

Q

Q obstruction。

中學生版:

你可能證明「任何固定形狀的鎖都開不了門」。但敵人可以每次

N

N 變大時換一把更複雜的新鎖。

缺口 D:full #319 需要兩邊一起處理

最後 full #319 不只是:

P

P 很大、

Q

Q 固定。

而是:

$$P=PN, Q=QN$$

$$P=P$$

N

$$,Q=Q$$

N

兩邊都可以變、都可以很大、都可以很複雜。

真正要證的是兩邊的 subset reciprocal sums 一定會撞到：

$$\Sigma(P) \cap \Sigma(Q) \neq \{0, SQ\}.$$

$$\Sigma(P) \cap \Sigma(Q)$$

$$= \{0, S$$

$$Q$$

$$\}.$$

中學生版：

左邊可以拿出很多小組合，右邊也可以拿出很多小組合。full proof 要證明：如果兩邊都太大，它們一定會有某個小組合剛好相等，於是大結構不是 primitive。

現在最好怎樣放 marker？

我建議你在筆記中放這段：

INSERT — Specialist audit checkpoint.

Current status:

We have a fixed-Q, arbitrary-P proof architecture for a partial obstruction related to Erdős #319. The local Φ -forcing, gate numerics, and primitivity contradiction appear sound. The remaining Stage 1 gap is the Liu-Sawhney Proposition 5.2 interface: full parameter table, pruning preservation, local mass preservation, and Archimedean window verification.

Action:

Contact / circulate to specialists in unit fractions / additive combinatorics / analytic number theory, e.g. Thomas Bloom or the Liu-Sawhney authors, asking specifically whether the Proposition 5.2 interface is valid.

Do not claim full Erdős #319. Label as fixed- Q partial theorem candidate.

最短總結

我們 achieved:

找到一條可能真的能證明 fixed-

Q

Q , arbitrary-

P

P partial theorem 的路線。

我們 missing:

還未完成 Liu–Sawhney parameter audit; 還未處理 arbitrary fixed

Q

Q ; 還未處理 growing

Q

Q ; 更未完成 full

P, Q

P, Q two-sided spectrum intersection。

所以你可以繼續研究 arbitrary

P, Q

P, Q / growing

Q

Q , 但 marker 要寫清楚:

Stage 1 is promising but not fully certified. Specialist audit pending.

Many thanks! "Welcome back, Leonard! Great handover note — very clean state. Let me engage seriously with both remaining fronts.

Case 1: Arbitrary Fixed Q with Large S_Q

****The obstacle:**** When $S_Q \geq \Phi_D(1-1/e)$, the singleton target $1/D$ is no longer locally forced, so the existing gate argument breaks.

****The key observation:**** You don't need to force $1/D$. You need to force *any* element of the set

$$T_Q := \{ S_{Q'} : \emptyset \neq Q' \subsetneq Q \}$$

which is a finite set of rationals in $(0, S_Q)$. The question is whether Liu-Sawhney can reach at least one element of T_Q .

Proposed approach: Smooth-denominator subset selection

****Step 1 — Identify "bad primes".**** A prime p is *bad* if P might lack sufficient p -adic local mass for target r . From your Lemma 2+3: P *cannot* avoid multiples of p when $S_Q < \Phi_p(1-1/e+\epsilon)$, i.e. for all p satisfying $S_Q < \Phi_p$. Since $\Phi_p(1-1/e) \rightarrow 1$ as $p \rightarrow \infty$ and $S_Q > 1$ always (for meaningful Q), the balance constraint already forces P to have p -adic local mass at every large prime. The genuinely bad primes are finitely many small ones where Φ_p is large.

****Step 2 — Davenport reduction.**** Let $p_1, \dots, p_\square$ be the bad primes. Define the "rough part" map

$$\rho: q \mapsto \prod_i p_i^{v_{p_i}(q)} \in G := \prod_i \mathbb{Z}/p_i^{a_i}\mathbb{Z}$$

Consider the sequence $(\rho(1/q))_{q \in Q}$ in G . If $|Q| \geq D(G)$ (the Davenport constant), then Q contains a proper non-empty subsequence $Q' \subsetneq Q$ with $\sum_{q \in Q'} \rho(1/q) = 0$ in G — meaning $S_{Q'}$ has denominator free of bad primes. Liu-Sawhney can then reach $S_{Q'}$ (since P 's local mass at all bad primes is guaranteed).

****Feasibility assessment:**** This works cleanly for the case $|Q| \geq D(G)$. The open sub-question is what happens when $|Q| < D(G)$. Note that $D(G) = O(\sum_i p_i^{a_i})$ which is a fixed constant for fixed Q . So for *fixed* Q , this is a *finite check* — you verify directly whether T_Q contains a smooth-denominator element. If yes, done. If not, $|Q|$ is bounded by $D(G)$, putting you in a manageable finite regime.

****The real challenge:**** When S_Q is large because Q is itself large (growing $|Q|$). This bleeds into Case 2.

Case 2: Growing Q (Both Sides Grow with N)

This is the genuinely hard case. Let me give you an honest decomposition.

Sub-case 2a: Both $|P|, |Q| > (1-1/e+\epsilon)N$ individually

This is actually *easy* — apply Liu-Sawhney Theorem 1.3 to P alone: P contains $P' \subseteq P$ with $R(P') = 1$. Then $R(P \setminus P') = S_Q - 1$. Apply Liu-Sawhney to Q : Q contains $Q' \subseteq Q$ with $R(Q') = 1$. Then (P', Q') with $R(P') = R(Q') = 1$ gives a proper signed zero-subcircuit. Done. ****This sub-case falls out for free.****

Sub-case 2b: $|Q|$ is small ($O(\log N)$ or fixed)

Reduces to Case 1 above. Already handled (or in progress).

Sub-case 2c: Both $|P|, |Q| \in [\omega(N), (1-1/e-\epsilon)N]$ — the true hard case

Neither side individually triggers Liu-Sawhney. This requires a genuinely new input.

****Proposed framework: Subset sum interval density****

The key lemma you need (currently unproved in the literature as far as I know):

> ****Conjecture/Target Lemma:**** If $X \subseteq [1, N]$ has $|X| \geq \delta N$ for fixed $\delta > 0$, then the subset sums of X are (C/N) -dense in $[0, R(X)]$ — i.e. for every $t \in [0, R(X)]$, there exists $B \subseteq X$ with $|R(B) - t| \leq C/(\delta N)$.

The greedy argument almost proves this: add elements of X largest-first until you first exceed t . The "overshoot" is at most $1/(\min \text{ element of } X) \leq 1$ in the worst case — too coarse. But if X has density δN in $[1, N]$, then X contains many "small" elements (near N in denominator), and the overshoot becomes $O(1/N)$. Making this rigorous is a non-trivial step requiring a density-of-small-elements argument.

If this lemma holds with error $1/N$, then for any fixed rational $S_{\{Q\}} \in (0, S_Q)$, P hits $S_{\{Q\}}$ within $1/N$, and for exact equality you still need Liu-Sawhney. But the interval lemma would reduce the problem to: "does T_Q intersect the reachable interval for P ?" and "does T_P intersect the reachable interval for Q ?" — a much cleaner formulation.

****The Fourier approach (harder but potentially sharper):****

Define for $X \subseteq [1, N]$:

$$F_X(x) = \prod_{x \in X} \left(1 + e^{2\pi i x / x}\right)$$

The number of pairs $(B \subseteq P, C \subseteq Q)$ with $R(B) = R(C)$ is the inner product $\langle F_P, F_Q \rangle$ (appropriately discretised). The trivial pair $(B=P, C=Q)$ contributes 1. If you can show this inner product exceeds 1, there must be a nontrivial matching subset — this is a signed Parseval/circle method strategy.

The issue is that controlling F_P away from $\xi=0$ requires smooth number inputs (major/minor arc separation), which is exactly Liu-Sawhney's machinery. Extending it to *two-sided* density control is the open technical frontier.

My Honest Assessment

Case	Status	Main Obstacle	Feasibility
Fixed Q, small S_Q	Proved (Theorem 4)	Parameter verification only	✓
Fixed Q, large S_Q	Partially attacked	Davenport + smooth target selection	Months of work, plausible
Growing Q, both large	Trivial	None	✓
Growing Q, one large	Reduces to fixed-Q	As above	Same as row 2
Growing Q, both medium	Open	Subset sum interval density	Hard; likely requires new lemma beyond Liu-Sawhney

****My recommendation for next step:**** Before attacking the growing-Q case, nail the "arbitrary fixed Q large S_Q " case via the Davenport approach — it's the natural next theorem and it's within reach with existing tools. The growing-Q case (both medium) is a genuine research frontier and probably the core difficulty of #319 itself.

One concrete question that would clarify Case 1: for Q with large S_Q that you've tried, does T_Q always contain a smooth-denominator element? Even computational evidence for small Q would be informative. Want to think through an example?" Does this help? Thx

這段很有幫助，因為它把下一階段拆成「arbitrary fixed

Q

Q」和「growing

Q

Q」兩個不同難度層。我要核對幾個外部基準：#319 官方狀態、Liu-Sawhney 的

Proposition 5.2、以及 Davenport/zero-sum 理論的適用性。

已思考 53s

Yes — it helps a lot as a research map, but I would not accept it as proof-level yet. It correctly identifies the next frontier, but it also contains two important overstatements that should be fixed before you rely on it.

Officially, #319 is still open; the page states there are no claimed partial or complete solutions in the comments, and records only the Adenwalla/Croot lower bound

$$(1 - 1/e + o(1))N$$

$(1 - 1/e + o(1))N$. Liu–Sawhney Proposition 5.2 is genuinely relevant because the authors explicitly say it separates “Archimedean” and

p

p -adic considerations and can reach “all possible” fractions under the right hypotheses.

Short answer

Yes, this helps — but mainly as a roadmap, not as a solved argument.

The strongest useful parts are:

1. For arbitrary fixed
2. Q
3. Q : target-selection via
4. $TQ = \{R(Q') : \emptyset \neq Q' \subsetneq Q\}$
5. T
6. Q
- 7.
8. $= \{R(Q$
9. $'$
10. $); \emptyset$
11. $= Q$
12. $'$
13. $\subsetneq Q\}$
14. is exactly the right next idea.

15. Bad-prime quotient / Davenport direction: this is a real mathematical bridge.
Davenport constant theory does say that sufficiently long sequences in a finite abelian group contain non-empty zero-sum subsequences.
16. Growing
17. Q
18. Q : the note correctly says the true hard case is when both
19. P
20. P and
21. Q
22. Q are medium-sized and neither side alone triggers a known unit-fraction theorem.

But there are two serious corrections.

Correction 1: “Bad primes are finitely many small ones” may have the inequality backwards

The text says roughly:

P

P cannot avoid multiples of

p

p when

$SQ < \Phi p$

S

Q

$< \Phi$

p

. Since

$\Phi p \rightarrow 1$

Φ

p

$\rightarrow 1$ and

$SQ > 1$

S

Q

>1 , this forces mass at every large prime.

This is suspicious as written.

If

$\Phi_p \rightarrow 1$

Φ

p

$\rightarrow 1$ and

$SQ > 1$

S

Q

>1 , then for large

p

$p,$

$\Phi_p < SQ,$

Φ

p

$< S$

Q

,

so the inequality

$$SQ < \Phi p$$

S

Q

$< \Phi$

p

will usually fail, not hold.

So either the inequality direction has been misstated, or the “large primes are automatically good” conclusion is wrong.

This matters because the bad-prime quotient method depends on knowing which prime powers are “bad” and which are “forceable.” Before using Davenport, we must first re-check the exact direction of the

Φ

Φ -forcing lemma.

Verdict: target-selection idea good; the “large primes” explanation needs correction.

Correction 2: “Both

$$|P|, |Q| > (1 - 1/e + \varepsilon)N$$

$|P|, |Q| > (1 - 1/e + \varepsilon)N$ is easy by Liu–Sawhney” is probably false

The note says:

If both

$$|P|$$

$$|P| \text{ and}$$

$$|Q|$$

$|Q|$ are individually larger than

$$(1 - 1/e + \varepsilon)N$$

$(1 - 1/e + \varepsilon)N$, apply Liu–Sawhney Theorem 1.3 to

P

P and

Q

Q , get subset sums equal to 1, done.

This is not safe.

Liu–Sawhney’s headline theorem requires large reciprocal mass, roughly

$$\sum_{n \in A} \frac{1}{n} \geq (\log N)^{4/5 + o(1)},$$

$$n \in A$$

$$\sum$$

$$n$$

$$1$$

$$\geq (\log N)$$

$$4/5 + o(1)$$

$$,$$

not merely

$$|A| \geq \delta N$$

$$|A| \geq \delta N.$$

A set can have positive density in

$$[1, N]$$

$[1, N]$ but only constant reciprocal mass. For example, if

A

A sits mostly near

N

N , then each term is about

$$1/N$$

$1/N$, so

$$\delta N$$

δN terms only contribute about

$$\delta$$

δ , not

$$(\log N)^{4/5}$$

$$(\log N)$$

$$4/5$$

.

So this “both large is trivial” claim is too optimistic.

Verdict: this subcase is not automatically solved by Liu–Sawhney Theorem 1.3.

What remains genuinely helpful?

1. Arbitrary fixed

Q

Q : yes, this is the right next route

The note’s best idea is:

Instead of forcing

$$1/D$$

$1/D$, try to force some proper subtarget

$$R(Q')$$

$$R(Q)$$

,

).

That is exactly right.

For fixed

Q

Q , the set

$$TQ=\{R(Q'):\emptyset\neq Q'\subsetneq Q\}$$

T

Q

$$=\{R(Q$$

,

$$):\emptyset$$

$$=Q$$

,

$$\subsetneq Q\}$$

is finite. If Liu–Sawhney can reach any one of these targets from

P

P , then we get a proper subcircuit:

$$R(B)=R(Q')$$

$$R(B)=R(Q$$

,

$$)$$

with

$$B\subseteq P$$

$$B \subseteq P,$$

$$Q' \subsetneq Q$$

$$Q$$

,

$\subsetneq Q$, so the original circuit is not primitive.

This is probably the correct Stage 2 after fixed-

$$Q$$

Q singleton-gate.

2. Davenport / zero-sum approach is plausible but must be formalized carefully

The idea is:

Encode the “bad prime-power denominator obstruction” into a finite abelian group, then use zero-sum theory to find

$$Q'$$

$$Q$$

,

whose target denominator avoids those bad primes.

This is a real and promising route.

But the map cannot just be informally written as

$$\rho(q) = \prod_i p_i v_{p_i}(q).$$

$$\rho(q) =$$

$$i$$

$$\prod$$

$$p$$

$$i$$

v

p

i

(q)

You need an exact additive model for the rational numbers

$1/q$

$1/q$ modulo the bad prime powers. In practice, you likely need to clear denominators outside the bad-prime part and work in a product of local rings or cyclic groups.

Otherwise “zero-sum in

G

G ” may not actually imply the reduced denominator of

$R(Q')$

$R(Q$

,

) avoids the bad primes.

Also, Davenport gives a non-empty zero-sum subsequence, but #319 needs a proper subset

$Q' \subsetneq Q$

Q

,

$\subsetneq Q$. If

$|Q| > D(G)$

$|Q| > D(G)$, properness is usually recoverable; if

$|Q| = D(G)$

$|Q|=D(G)$, the only zero-sum might be the whole sequence, so there is a boundary case to handle.

Verdict: good idea, but not yet theorem-ready.

3. Growing

Q

Q : the note correctly identifies the real hard zone

The useful insight is:

The hardest full-#319 case is not fixed

Q

Q , but

$Q=QN$

$Q=Q$

N

growing with

N

N , especially when neither side alone has enough structure to trigger Liu–Sawhney.

This agrees with our earlier hierarchy.

The proposed subset-sum interval lemma is a reasonable dream lemma, but approximation

$$|R(B)-t|\leq C/N$$

$$|R(B)-t|\leq C/N$$

is not enough by itself, because #319 requires exact rational equality, not approximate equality.

So the interval-density idea may help locate targets, but exact equality still needs

p

p -adic / denominator-lattice control.

My updated status table

Front	Does this help?	My honest status
Fixed Q Q , small SQ S Q / gate condition	Yes	Most mature; needs Liu–Sawhney parameter audit
Arbitrary fixed Q Q , large SQ S Q	Yes	Davenport / bad-prime quotient is promising
Growing Q Q , one side tiny/fixed	Partly	May reduce to fixed- Q -type methods
Growing Q Q , both sides huge	Not automatically solved	The “apply Liu–Sawhney to both” step is too fast
Growing Q Q , both medium	Yes as diagnosis	This is probably core hard case
Fourier two-sided method	Conceptually useful	Very far from proof

Bottom line

Yes, it helps. It gives the correct next research direction:

fixed $Q \rightarrow$ arbitrary fixed $Q \rightarrow$ growing $Q \rightarrow$ two-sided spectrum intersection.

fixed $Q \rightarrow$ arbitrary fixed $Q \rightarrow$ growing $Q \rightarrow$ two-sided spectrum intersection.

But I would revise the note before relying on it:

1. Recheck the inequality direction in the “bad primes” discussion.
2. Remove or weaken the claim that “both large” is trivial by Liu–Sawhney.
3. Formalize the bad-prime quotient map exactly.
4. Treat the growing-
5. Q
6. Q Fourier/interval ideas as speculative, not near-term proof.

The best immediate next experiment is exactly the one suggested at the end:

For small fixed

Q

Q , compute

TQ

T

Q

, compute reduced denominators of all

$R(Q')$

$R(Q$

,

), and test whether some proper

Q'

Q

,

avoids the bad prime powers.

That would tell us whether the Davenport/target-selection route is likely real or only elegant.

Many thanks! Is “Yes — both corrections are fixable, and importantly neither kills the core Theorem 4 argument. Let me address each precisely.

Correction 1: Inequality direction — Fixable cleanly

GPT is right. Let me restate the corrected logic:

Lemma 2 says: if P avoids multiples of p , then $R(P) \geq \Phi_p(1-1/e)$.

Since $\Phi_p(1-1/e) \rightarrow 1$ as $p \rightarrow \infty$, and $S_Q > 1$ always:

- **Small primes** ($p=2,3,5,\dots$): $\Phi_p(1-1/e)$ is large ($\Phi_2 = \infty$, $\Phi_3 \approx 1.973\dots$), so $\Phi_p > S_Q$ for most Q . P **cannot** avoid their multiples $\rightarrow$ small primes are **good** (P is forced to have their local mass).
- **Large primes** (p beyond a threshold): $\Phi_p < S_Q$, so avoidance of p -multiples is **consistent** with the balance constraint $\rightarrow$ large primes are potentially **bad**.

The fix for the Davenport step: For fixed Q , let $p_1 < p_2 < \dots$ be the prime factors of elements of Q . The bad ones are the large primes p_i where $\Phi_{p_i} < S_Q$. Here's the clean fix: for any such bad large prime p dividing some $q \in Q$, consider $Q' = Q \setminus \{q\}$. Then $S_{Q'} = S_Q - 1/q$, and since $1/q = 1/(p \cdot \text{something})$, removing q **eliminates** the large prime p from the dominant denominator structure of $S_{Q'}$. Iterating over the finitely many bad large primes in Q 's elements, you can always find a proper subset Q' whose target $S_{Q'}$ involves only good (small-prime) denominators — which P is forced to reach by Liu-Sawhney. The Davenport constant framing was suggestive but this direct subtraction argument is more elementary and works for fixed Q .

Correction 2: "Both large" case — Fixable, but reveals something important

GPT is completely right: Liu-Sawhney Theorem 1.3 requires $R(A) \geq (\log N)^{4/5}$, not just $|A| \geq \delta N$. A set concentrated near N can have $|P| \geq (1-1/e+\epsilon)N$ while $R(P)$ stays bounded ($\approx S_Q = \text{constant}$ for fixed Q).

The key observation is: **this correction doesn't affect Theorem 4 at all**, because the existing proof already uses Prop 5.2 (the general target engine), not Theorem 1.3. The "both large $\rightarrow$ Theorem 1.3" shortcut was something I introduced when discussing the growing- Q sub-case, not part of your established proof.

The fix distinguishes two growing- Q sub-cases:

Sub-case	S_Q behaviour	$R(P) = S_Q$	Theorem 1.3 applicable?	Correct tool
Both $ P , Q $ large, S_Q bounded ($\approx \text{const}$)	Constant	Constant	✗	Prop 5.2 + local mass
Both $ P , Q $ large, $S_Q \sim \log N$	Growing	$\sim \log N$	✓	Theorem 1.3 on P or Q

When $S_Q \sim \log N$ (which requires $|Q|$ to grow in a spread-out way), $R(P) = S_Q \sim \log N$ means P must contain many small elements, and Theorem 1.3 genuinely applies. This case is actually *easier* than expected.

When S_Q is bounded, neither side has large reciprocal mass, and you're back to Prop 5.2 as the right engine — which is exactly what the Theorem 4 pipeline already uses.

Net assessment

Both corrections are real but *non-fatal*:

- Correction 1 fixes a direction error in the heuristic framing but the corrected version (large primes are bad, remove them from Q to get a smooth-denominator target) actually gives a *cleaner proof strategy* for arbitrary fixed Q .
- Correction 2 removes an invalid shortcut in the growing- Q "both large" case, but the correct replacement (case-split on whether S_Q grows) is tractable and the bounded- S_Q sub-case was already handled by the Prop 5.2 machinery.

The recommended immediate experiment (from GPT, and I agree) is exactly right: for small fixed Q containing a large prime, compute T_Q , check whether Q minus the large-prime elements gives a proper subset Q' with smooth-denominator $S_{Q'}$, and verify computationally that Liu-Sawhney's local conditions for that target are met. Want to try a concrete example together?" ok

這段確實比前一版更接近正確，但我不會把它當成「已修好 arbitrary fixed

Q

Q / growing

Q

」。最值得保留的是 target-selection 思路；最需要小心的是「移除含大質數的

q

q 就清掉 bad prime」這一步。

已思考 26s

Yes — it helps, but I would mark it as:

useful revised roadmap, not yet a proof.

It fixes the two earlier errors in spirit, but the new version still has two places where it jumps too quickly.

Officially, Erdős #319 is still open, with only the Adenwalla/Croot lower bound recorded; so any wording still needs to remain “partial theorem candidate / roadmap,” not “proof of #319.” Liu–Sawhney Proposition 5.2 is a real relevant machine, but its use still requires exact hypotheses, target denominator, local mass, and parameter-window verification. Davenport-style zero-sum reasoning is also real mathematics: the Davenport constant is exactly about guaranteeing non-empty zero-sum subsequences in a finite abelian group.

My verdict on this new answer

1. Correction 1 is directionally fixed, but the “remove large-prime

q

q ” step is too fast

The corrected logic:

small primes are good, large primes may be bad

is plausible if your

Φ_p

Φ

p

-forcing lemma has that inequality direction.

But this sentence is not automatically true:

remove

q

q , and the large prime

p

p disappears from the denominator of

SQ'

S

Q

,

.

Why not? Because:

- there may be several elements of
- Q
- Q divisible by the same large prime
- p
- p ;
- after summing reciprocals, denominators can interact by cancellation;
- removing all bad-prime elements might make
- Q'
- Q
- ,
- empty, too small, or not the target you need;
- even if the reduced denominator avoids bad primes, you still need Liu–Sawhney’s target
- x/QA
- x/Q
- A
-
- and Archimedean window to match.

So I would replace the direct subtraction claim with a safer computational/theoretical statement:

Enumerate all proper non-empty

$Q' \subsetneq Q$

Q

,

$\subsetneq Q$, compute the reduced denominator of

$R(Q')$

$R(Q$

,

), and test whether it only uses forceable prime powers.

That is the correct experiment. The Davenport/bad-prime quotient idea is still promising, but it has to be formalized through the actual reduced denominators of

$R(Q')$

$R(Q$

,

), not just by deleting one large-prime element.

2. Correction 2 is mostly fine, but the growing-

Q

Q case is still not “tractable”

It is correct that the invalid shortcut does not damage the fixed-

Q

Q Theorem 4 route. That route uses Proposition 5.2, not just Liu–Sawhney’s headline Theorem 1.3.

But for growing

Q

Q , the new table is still too optimistic.

If

$SQ \sim \log N$,

S

Q

$\sim \log N$,

then yes, the reciprocal mass is certainly above the Liu–Sawhney threshold

$(\log N)^{4/5+o(1)}$

$(\log N)$

$$4/5+o(1)$$

for large

N

N , so a Theorem 1.3-style input may become available.

But if

SQ

S

Q

grows slowly, for example

$$(\log N)^{0.3}$$

$$(\log N)$$

$$0.3$$

, or stays bounded while

$$|Q| \rightarrow \infty$$

$|Q| \rightarrow \infty$, then Theorem 1.3 does not automatically apply. That is still part of the hard growing-

Q

Q zone.

So the corrected statement should be:

growing

Q

Q splits into high reciprocal-mass, medium reciprocal-mass, and bounded

reciprocal-mass regimes. Only the high-mass regime is likely easy via Liu–Sawhney Theorem 1.3.

What I would keep from this answer

The best part is this:

$$TQ = \{R(Q') : \emptyset \neq Q' \subsetneq Q\}.$$

T

$\mathcal{Q}$

$=\{R(\mathcal{Q}$

,

$):\emptyset$

$=\mathcal{Q}$

,

$\subseteq \mathcal{Q}\}$.

That is exactly the right target-selection object.

For arbitrary fixed

$\mathcal{Q}$

$\mathcal{Q}$, the next goal should be:

find some

$\mathcal{Q}'\subsetneq \mathcal{Q}$

$\mathcal{Q}$

,

$\subsetneq \mathcal{Q}$ such that

$R(\mathcal{Q}')$

$R(\mathcal{Q}$

,

) has a denominator compatible with the prime powers that

$\mathcal{P}$

$\mathcal{P}$ is forced to contain.

If such

$\mathcal{Q}'$

$\mathcal{Q}$

,

exists, then Liu–Sawhney may produce

$$B \subseteq P$$

$$B \subseteq P \text{ with}$$

$$R(B)=R(Q'),$$

$$R(B)=R(Q$$

,

),

and then

$$B,Q'$$

$$B,Q$$

,

give a proper subcircuit, killing primitivity.

Clean revised status

Claim	My status
Core fixed- Q Q Theorem 4 is not killed	Agree
Small-prime / large-prime correction is fixable	Agree
Directly removing large-prime q q always works	Too fast
Davenport / bad-prime quotient is promising	Agree
Both-large growing case is solved by Theorem 1.3	Only if reciprocal mass is high enough
Immediate experiment on TQ T 2	Strongly agree

Best next step

Yes, try a concrete example — but do it this way:

1. Choose a fixed
2. Q
3. Q with
4. $SQ > 1$
5. S
6. Q
- 7.
8. > 1 and some large-prime denominators.
9. Enumerate all non-empty proper
10. $Q' \subsetneq Q$
11. Q
12. '
13. $\subsetneq Q$.
14. Compute
15. $R(Q')$
16. $R(Q$
17. '
18.) exactly as a reduced fraction.
19. List the prime powers in its reduced denominator.
20. Check whether those prime powers are “good” under your
21. Φ
22. Φ -forcing condition.
23. If yes, that
24. Q'
25. Q
26. '
27. is a candidate target for Liu–Sawhney.

So final verdict:

The response is useful and mostly directionally right, but the arbitrary fixed-

Q

Q proof is not fixed yet. The correct next move is computational target-selection over

TQ

T

Q

, not assuming large-prime deletion automatically works.

貼上的 Markdown (1)(6).md

檔案

Dear GPT sorry to bother! Now with Davenport/ bad prime approach (or any approaches we discussed yesterday for the bad prime methods or the 16 methods discussed ytd) do we have potential in making real progress/ even solve arbitrary or growing Q with these above? Please feel free to use as much time and invariance not afraid to make mistake free from boundary and constrains mode thx. for "" Thx!

我會用研究進度的角度誠實分級: 哪些方向可能帶來真進展、哪些只是可攻的部分定理、哪些離完整 #319 仍屬 speculative。

新的重點是不平均押注十六法: 先看 bad-prime target selection 與 Liu–Sawhney reachability, 再把 p-adic constraints 作輔助; entropy/Fourier/containers 屬長線。關鍵不變量是約化後

$R(Q')$

$R(Q$

,

) 分母保留下來的 prime-power 複雜度。

已思考 8m 9s

According to the document from 2026-06-04 you uploaded, the best “next” idea beyond fixed-

Q

Q is indeed the bad-prime quotient target-selection route: it is concrete, finite, directly attacks arbitrary fixed

Q

Q , and may scale conceptually toward slowly growing

Q

Q .

My honest answer is:

Yes, there is real potential for progress on arbitrary fixed Q .

Yes, there is real potential for progress on arbitrary fixed Q .

But:

`\boxed{ \text{For growing } Q \text{ and full #319, the same ideas are promising but not yet enough.} }`

Erdős #319 itself is still officially open; the page states there are no partial or complete solutions claimed in comments and records the Adenwalla/Croot lower bound

$$|A| \geq (1 - 1/e + o(1))N.$$

$$|A| \geq (1 - 1/e + o(1))N.$$

The page also says the problem statement has been formalised in Lean, but that is not a Lean proof of a solution.

1. Where the strongest real progress potential is

The strongest route is:

Φ -forcing+bad-prime quotient+Liu–Sawhney reachability.

Φ -forcing+bad-prime quotient+Liu–Sawhney reachability.

In simple terms:

- Φ
- Φ -forcing tells us which prime-power “ingredients”
- P
- P is forced to contain.
- Bad-prime quotient tries to choose a target
- $R(Q')$
- $R(Q)$
- $'$
- $)$ whose denominator avoids the ingredients
- P
- P might not contain.

- Liu–Sawhney then tries to make
- P
- P actually hit that target.

If this works, then we get:

$$R(B)=R(Q')$$

$$R(B)=R(Q$$

,

)

for some

$$B \subseteq P$$

$B \subseteq P$ and non-empty proper

$$Q' \subsetneq Q$$

$$Q$$

,

$\subsetneq Q$. Then

$$B \cup Q'$$

$$B \cup Q$$

,

is a smaller zero-subcircuit, so the original

$$P, Q$$

P, Q circuit was not primitive.

That is exactly the kind of mechanism #319 needs.

2. What this could realistically prove first

The first plausible new theorem is not full #319. It is:

Arbitrary fixed-

$$Q$$

Q theorem.

For every fixed finite negative side

$$Q$$

Q , any sufficiently large positive side

P

P satisfying

$$R(P)=R(Q)$$

$$R(P)=R(Q)$$

cannot remain primitive.

This would be serious progress.

It would mean:

no fixed right-hand pattern

Q

Q can beat the Croot/Adenwalla-style barrier.

That is much stronger than the current singleton-gate theorem, because it would not depend on choosing

$1/D$

$1/D$ as the target.

Your uploaded roadmap says exactly this: Stage 3 should be “arbitrary fixed

Q

Q ” via smoothable target selection, with likely tools being bad-prime quotient, zero-sum/Davenport, and pair-cancellation.

3. Why Davenport/bad-prime is a real idea

The key object is:

$$TQ=\{R(Q'):\emptyset\neq Q'\subsetneq Q\}.$$

T

Q

$$=\{R(Q$$

,

$$):\emptyset$$

$$=Q$$

,

$$\subseteq Q\}.$$

This is the list of all possible proper targets coming from the

Q

Q -side.

The problem becomes:

Can we find one

$$Q' \subsetneq Q$$

Q

,

$\subseteq Q$ whose denominator only uses prime powers that

P

P is forced to support?

The bad-prime quotient idea turns this into a finite abelian group problem. Roughly:

$$q \mapsto L/q$$

$$q \mapsto L/q$$

modulo the bad prime-power components, where

$$L = \text{lcm}(Q)$$

$L = \text{lcm}(Q)$. If a subset sums to zero in that finite group, then the corresponding

$$R(Q')$$

$$R(Q)$$

,

) may have denominator avoiding the bad prime powers.

This is exactly the ecosystem of Davenport constants and zero-sum theory. The Davenport constant

$D(G)$

$D(G)$ is the threshold length forcing a non-empty zero-sum subsequence in a finite abelian group.

So this is not just metaphor. It is a real mathematical translation:

target selection $\rightarrow$ zero-sum subsequence problem.

target selection $\rightarrow$ zero-sum subsequence problem.

4. The main warning: Davenport alone is not enough

Davenport gives a non-empty zero-sum subsequence. But #319 needs more:

1. The subset must be proper, not the whole
2. Q
3. Q .
4. The zero-sum condition must really imply the reduced denominator of
5. $R(Q')$
6. $R(Q$
7. '
8.) avoids the bad prime powers.
9. The target
10. $R(Q')$
11. $R(Q$
12. '
13.) must lie in the Archimedean range where Liu–Sawhney can reach it.
14. The
15. P
16. P -side must still satisfy the required smoothness, bounded-
17. Ω
18. Ω , and local-mass hypotheses after pruning.

So Davenport is not a magic wand. It is a target-selection engine. It must be coupled with exact rational denominator checks and Liu–Sawhney interface verification.

Liu–Sawhney’s paper is the right machine because it proves strong unit-fraction reachability theorems and explicitly refines Croot and Bloom’s methods. Bloom’s

earlier density result is formally verified in Lean, but that does not automatically formalise Liu–Sawhney Proposition 5.2 or your #319 application. Bloom’s paper itself says the main results were completely Lean-verified with Bhavik Mehta, and also shows how much formal work even “obvious” asymptotic checks can require.

5. For arbitrary fixed

Q

Q: my probability-style assessment

I would grade the approaches like this.

Potential for arbitrary fixed		
Method	Q	Why
Fixed- Q Q Liu–Sawhney interface	Very high	Nearest to a real partial theorem
Minimal-over-one Q Q	High	Natural primitive core; mostly technical estimates
Bad-prime quotient target selection	High	Concrete finite reduction; directly attacks arbitrary fixed
Davenport / zero-sum theory	Medium-high	Gives threshold-style existence; needs properness and denominator control
Exact computation over TQ T 2	Very high	Can immediately test whether the target-selection conjecture is true in examples

p-adic layer peeling		Uses primitivity directly, but hard to Mediumturn congruences into rational equalities
Reachability interval extraction	Medium-high	Could unify many targets if Prop 5.2 gives a clean interval corollary
Entropy/Fourier	Medium-low short term	Conceptually important, but exact equality is hard
Hypergraph containers	Low short term	Needs supersaturation theorem first
Chevalley–Warning / polynomial method	Low-medium	Good local mod- p tool, but exact rational lift is hard
Graver/toric/convex-dual	Low short term	Elegant language, but does not yet see $1-1/e$ $1-1/e$ sharply

So for arbitrary fixed

Q

Q , I do see real progress potential.

The most realistic pipeline is:

compute $TQ \rightarrow$ identify bad prime powers $\rightarrow$ find smoothable $Q' \rightarrow$ apply
Liu–Sawhney $\rightarrow$ break primitivity.

compute T

Q

$\rightarrow$ identify bad prime powers $\rightarrow$ find smoothable Q

,

→ apply Liu–Sawhney → break primitivity.

6. For slowly growing

Q

Q : some potential, but only in controlled regimes

Growing

$Q = Q_N$

$Q = Q$

N

is much harder because fixed-

Q

Q constants may disappear.

For fixed

Q

Q , you can say:

$$c_Q > 0.$$

c

Q

$$> 0.$$

For growing

Q_N

Q

N

, that may become:

$$c_{Q_N} \rightarrow 0.$$

c

Q

N

$$\rightarrow 0.$$

Then protected pruning may delete exactly the prime-power layers you wanted to keep.

So the first growing-

Q

Q theorem should not try full generality. It should try controlled growth, for example:

$$|Q_N| = O(1),$$

$|Q$

N

$$| = O(1),$$

then:

$$|Q_N| = o(\log \log N),$$

$|Q$

N

$$| = o(\log \log N),$$

or:

$$\max_N Q_N = N^{o(1)}.$$

$\max_Q$

N

$=N$

$o(1)$

Your uploaded roadmap also puts this as Stage 4: slowly growing

Q

Q , where constants degrade slowly enough to survive pruning.

This is plausible, but not immediate.

A possible theorem might look like:

If

QN

Q

N

grows slowly and its bad-prime quotient complexity is

$o(|QN|)$

$o(|Q$

N

$|)$, then target selection still finds a proper

Q'

Q

,

, so primitivity fails.

That would be meaningful. It would not solve #319, but it would push beyond fixed

Q

Q .

7. For full growing

Q

Q : the real hard wall

Full #319 allows both sides to vary:

$$P=PN,Q=QN.$$

$$P=P$$

N

$$,Q=Q$$

N

Then the problem is no longer “can

P

P hit one target from

Q

$Q?$ ”

It becomes:

$$\Sigma(P)\cap\Sigma(Q)\neq\{0,S\},$$

$$\Sigma(P)\cap\Sigma(Q)$$

$$=\{0,S\},$$

where

$$\Sigma(P)=\{R(P'):P'\subseteq P\}.$$

$$\Sigma(P)=\{R(P$$

,

$);P$

,

$$\subseteq P\}.$$

Your uploaded document correctly identifies full #319 as a two-sided subset-sum spectrum intersection problem.

That means a full solution likely needs a theorem like:

If

$$|P| + |Q| > (1 - 1/e + \varepsilon)N$$

$$|P| + |Q| > (1 - 1/e + \varepsilon)N \text{ and}$$

$$R(P) = R(Q)$$

$R(P) = R(Q)$, then the subset-sum spectra of

P

P and

Q

Q must intersect somewhere other than

0

0 and the full sum.

That is the real endgame.

The bad-prime quotient idea may contribute to this, but it probably cannot solve the full two-sided case by itself.

8. Which of the 16 methods are most worth pushing?

I would divide them into four tiers.

Tier 1: Push now

These can produce real output soon.

A. Liu–Sawhney interface table

This is still the first checkpoint. Without it, even fixed-

Q

Q remains a sketch.

B. Bad-prime quotient target-selection search

This is the strongest new direction. It is finite, programmable, and theorem-shaped.

C. Minimal-over-one

Q

$\mathcal{Q}$

This is a natural class and likely easier than arbitrary fixed

Q

$\mathcal{Q}$. Your earlier documents correctly emphasize that it may be the true primitive “hard core.”

D. Exact

TQ

T

$\mathcal{Q}$

experiments

For each

Q

$\mathcal{Q}$, compute all proper

Q'

$\mathcal{Q}$

,

, reduce

$R(Q')$

$R(\mathcal{Q}$

,

), and check denominator prime powers.

These four should be the main focus.

Tier 2: Develop as proof infrastructure

These may become important after the first computations.

E. Reachability intervals

Instead of using Liu–Sawhney to hit one target, extract a corollary saying

P

P can hit many targets in an interval/lattice.

F. p -adic layer peeling

This uses the equality

$$R(P)=R(Q)$$

$$R(P)=R(Q)$$

directly. It may be essential for growing

Q

Q , because fixed-target methods may not see primitivity strongly enough. Your document correctly notes that top

p

p -adic layers must cancel modulo

p

p , but turning local congruence into a rational subcircuit is hard.

G. Two-sided reachable spectra

Build:

$$R(P) \subseteq \Sigma(P), R(Q) \subseteq \Sigma(Q),$$

$$R(P) \subseteq \Sigma(P), R(Q) \subseteq \Sigma(Q),$$

then prove the reachable sets intersect.

This is probably the real full-proof language.

Tier 3: Use for experiments and intuition

H. NTT / modular subset spectra

Good for testing whether dense subset spectra become full modulo small moduli.

I. SAT / MILP small-

N

N search

Good for finding counterexamples to naive conjectures.

J. Davenport constants / stronger zero-sum inverse theory

Useful once the correct finite group is defined.

Tier 4: Long-term speculative

K. Entropy/Fourier

L. Hypergraph containers

M. Absorption

N. Chevalley–Warning

O. Graver/toric ideals

P. Convex/tropical dual certificate

These are not useless, but I would not lead with them. They are language and possible future machinery, not near-term proof engines.

9. Do we have potential to solve arbitrary fixed

Q

$Q?$

Yes. I think this is the strongest claim we can responsibly make:

Arbitrary fixed Q is now a plausible research target.

Arbitrary fixed Q is now a plausible research target.

The reason is that for fixed

Q

Q , everything is finite:

- finite target set
- TQ
- T
- Q
-
- ;
- finite bad-prime set;
- finite denominator quotient group;
- fixed local-mass constants;
- fixed pruning loss margin.

That is exactly the kind of environment where a bad-prime quotient / Davenport / computation-to-theorem route can work.

But we should not yet say “almost solved.” The missing theorem is:

For every fixed

Q

Q , there exists a proper target

$Q' \subsetneq Q$

Q

,

$\subsetneq Q$ whose denominator avoids the bad prime powers forced absent/present by

P

P .

That is not proven yet.

10. Do we have potential to solve growing

Q

Q ?

Some potential, but only after controlling the complexity of the bad-prime quotient.

For growing

Q

Q , the key quantity is not just

$|Q|$

$|Q|$. It is:

bad-prime quotient complexity.

bad-prime quotient complexity.

If

Q

Q grows but its bad-prime complexity stays small, Davenport may still force a good

Q'

Q

,

.

If

Q

Q grows and the bad-prime quotient grows faster, then Davenport may become too weak.

So a serious slowly-growing theorem would need a tradeoff:

$|Q| > D(GB) \Rightarrow \text{good target exists.}$

$|Q| > D(G$

B

) $\Rightarrow$ good target exists.

Then you need to prove that a primitive circuit with too many total elements cannot have

$D(GB)$

$D(G$

B

)

too large.

That is a real possible bridge from fixed

Q

Q to growing

Q

Q .

11. Could this solve full #319?

My honest answer:

Not yet, but it points toward the correct kind of full proof.

Not yet, but it points toward the correct kind of full proof.

Full #319 probably needs a two-sided theorem:

$R(P) \cap R(Q) \neq \{0, S\}$.

$R(P) \cap R(Q)$

$= \{0, S\}$.

The current best vision is exactly what your document says:

Φ -forcing+bad-prime quotient zero-sum+Liu–Sawhney reachability intervals.

Φ -forcing+bad-prime quotient zero-sum+Liu–Sawhney reachability intervals.

That combination may get from fixed

Q

Q to arbitrary fixed

Q

Q , and possibly to slowly growing

Q

Q . For full #319, one likely also needs p-adic layer peeling or a two-sided spectrum theorem.

12. My recommended next concrete move

Do not start with entropy, containers, or full growing

Q

Q .

Start with this experiment:

For every fixed

$Q \subseteq [2, M]$

$Q \subseteq [2, M]$ with

$S_Q > 1$

S

Q

> 1 :

1. Compute all non-empty proper
2. $Q' \subsetneq Q$
3. Q
4. '
5. $\subsetneq Q$.
6. Compute
7. $R(Q')$
8. $R(Q$
9. '
10.) as a reduced fraction.
11. Compute denominator prime powers of
12. $R(Q')$
13. $R(Q$
14. '
15.).
16. Compute which prime powers are “good” under
17. Φ
18. Φ -forcing.
19. Check whether some
20. Q'
21. Q
22. '
23. uses only good prime powers.
24. If no such
25. Q'
26. Q
27. '
28. exists, record
29. Q
30. Q as a rigid obstruction candidate.
31. Build the bad-prime quotient group
32. GB
33. G
34. B
35. .
36. .
37. Test whether zero-sum theory predicts the observed result.

This experiment is not just data. It directly tests the central conjecture for arbitrary fixed

Q

Q .

Final verdict

Yes, there is real potential.

But I would rank the promise like this:

Goal	&	Potential with current methods
fixed- Q under gate condition	&	high, near-term
minimal-over-one fixed Q	&	high
arbitrary fixed Q	&	medium-high, plausible
slowly growing Q	&	medium, needs new complexity bound
full growing P, Q	&	low-medium, needs two-sided spectrum theorem
full #319	&	not near, but roadmap improved

The most important sentence is:

Bad-prime quotient target selection is not a proof yet, but it is the first genuinely concrete mechanism we have seen for escaping the fixed-

Q

Q limitation.

That is worth pushing.

Many thanks! sorry i forgot have i posted but "Yes — both corrections are fixable, and importantly neither kills the core Theorem 4 argument. Let me address each precisely.

Correction 1: Inequality direction — Fixable cleanly

GPT is right. Let me restate the corrected logic:

Lemma 2 says: if P avoids multiples of p , then $R(P) \geq \Phi_p(1-1/e)$.

Since $\Phi_p(1-1/e) \rightarrow 1$ as $p \rightarrow \infty$, and $S_Q > 1$ always:

- **Small primes** ($p=2,3,5,\dots$): $\Phi_p(1-1/e)$ is large ($\Phi_2 = \infty$, $\Phi_3 \approx 1.973\dots$), so $\Phi_p > S_Q$ for most Q . P **cannot** avoid their multiples $\rightarrow$ small primes are **good** (P is forced to have their local mass).
- **Large primes** (p beyond a threshold): $\Phi_p < S_Q$, so avoidance of p -multiples is **consistent** with the balance constraint $\rightarrow$ large primes are potentially **bad**.

The fix for the Davenport step: For fixed Q , let $p_1 < p_2 < \dots$ be the prime factors of elements of Q . The bad ones are the large primes p_i where $\Phi_{p_i} < S_Q$. Here's the clean fix: for any such bad large prime p dividing some $q \in Q$, consider $Q' = Q \setminus \{q\}$. Then $S_{Q'} = S_Q - 1/q$, and since $1/q = 1/(p \cdot \text{something})$, removing q **eliminates** the large prime p from the dominant denominator structure of $S_{Q'}$. Iterating over the finitely many bad large primes in Q 's elements, you can always find a proper subset Q' whose target $S_{Q'}$ involves only good (small-prime) denominators — which P is forced to reach by Liu-Sawhney. The Davenport constant framing was suggestive but this direct subtraction argument is more elementary and works for fixed Q .

Correction 2: "Both large" case — Fixable, but reveals something important

GPT is completely right: Liu-Sawhney Theorem 1.3 requires $R(A) \geq (\log N)^{4/5}$, not just $|A| \geq \delta N$. A set concentrated near N can have $|P| \geq (1-1/e+\epsilon)N$ while $R(P)$ stays bounded ($\approx S_Q = \text{constant}$ for fixed Q).

The key observation is: **this correction doesn't affect Theorem 4 at all**, because the existing proof already uses Prop 5.2 (the general target engine), not Theorem 1.3. The "both large $\rightarrow$ Theorem 1.3" shortcut was something I introduced when discussing the growing- Q sub-case, not part of your established proof.

The fix distinguishes two growing- Q sub-cases:

Sub-case	S_Q behaviour	$R(P) = S_Q$	Theorem 1.3 applicable?	Correct tool
Both $ P , Q $ large, S_Q bounded ($\approx \text{const}$)	Constant	Constant	✗	Prop 5.2 + local mass
Both $ P , Q $ large, $S_Q \sim \log N$	Growing	$\sim \log N$	✓	Theorem 1.3 on P or Q

When $S_Q \sim \log N$ (which requires $|Q|$ to grow in a spread-out way), $R(P) = S_Q \sim \log N$ means P must contain many small elements, and Theorem 1.3 genuinely applies. This case is actually **easier** than expected.

When S_Q is bounded, neither side has large reciprocal mass, and you're back to Prop 5.2 as the right engine — which is exactly what the Theorem 4 pipeline already uses.

Net assessment

Both corrections are real but **non-fatal**:

- Correction 1 fixes a direction error in the heuristic framing but the corrected version (large primes are bad, remove them from Q to get a smooth-denominator target) actually gives a **cleaner proof strategy** for arbitrary fixed Q .
- Correction 2 removes an invalid shortcut in the growing- Q "both large" case, but the correct replacement (case-split on whether S_Q grows) is tractable and the bounded- S_Q sub-case was already handled by the Prop 5.2 machinery.

The recommended immediate experiment (from GPT, and I agree) is exactly right: for small fixed Q containing a large prime, compute T_Q , check whether Q minus the large-prime elements gives a proper subset Q' with smooth-denominator $S_{\{Q'\}}$, and verify computationally that Liu-Sawhney's local conditions for that target are met. Want to try a concrete example together?"

已思考 15s

Yes — 你這段可以作為 handover / working note, 但我會建議你在 post 或交給 specialist 前做兩個保守修正。整體方向是有幫助的; 尤其和你之前的 deep map 一致: 從 fixed-

Q

Q interface 推到 target selection, 再推 bad-prime quotient / zero-sum route。

官方狀態仍要保留: Erdős #319 目前仍標示 open, 頁面只記錄 Adenwalla/Croot 的

$(1-1/e+o(1))N$

$(1-1/e+o(1))N$ lower bound, 並說沒有 comments claimed partial or complete solutions。Liu-Sawhney paper 則是真正相關的外部工具, 因為它證明了 unit-fraction reachability 型結果, 並說其方法 refine Croot and Bloom。

我的 verdict

這段可以保留，但不要寫成：

corrections fixed, arbitrary fixed

Q

Q essentially solved.

應改成：

corrections fixed at the roadmap level; arbitrary fixed

Q

Q now has a plausible target-selection strategy, but still needs exact

TQ

T

Q

/ reduced-denominator verification and Liu–Sawhney interface audit.

對你貼的兩個 correction 的評價

Correction 1: 方向修正是對的

這部分基本正確：

$$\Phi p(1-1/e) \rightarrow 1.$$

Φ

p

$$(1-1/e) \rightarrow 1.$$

所以如果

$SQ > 1$

S

Q

> 1 , 那麼 large

p

p 時通常會有：

$$\Phi_p < SQ.$$

Φ

p

$< S$

Q

因此 large primes 可能是 bad primes; small primes 反而較容易被 balance forced。

這是有用修正。

但我仍會保留一個 caveat:

「移除含 bad large prime 的

q

q 就一定清掉該 bad prime」這句太快。

因為:

1. 同一個 large prime 可能出現在多個
2. $q \in Q$
3. $q \in Q$;
4. $R(Q')$
5. $R(Q$
6. '
7.) 約分後的 denominator 可能有 cancellation, 也可能保留意外 prime powers;
8. 移除所有 bad-prime elements 後,
9. Q'
10. Q
11. '
12. 可能太小、空、或 target 不在 Liu-Sawhney 可達範圍;
13. 我們真正需要的是 reduced denominator of
14. $R(Q')$
15. $R(Q$
16. '

17.), 不是單個

18. q

19. q 的 denominator。

所以最安全寫法是：

The direct subtraction idea is a useful heuristic. The rigorous version is to enumerate all non-empty proper

$Q' \subsetneq Q$

Q

,

$\subsetneq Q$, reduce

$R(Q')$

$R(Q$

,

), and check whether its denominator avoids the bad prime powers.

Davenport constant / zero-sum 理論仍然有價值，因為 Davenport constant 正是 finite abelian group 裏保證 non-empty zero-sum subsequence 的門檻。但要先把「bad-prime quotient group」定義得完全準確。

Correction 2: Theorem 4 不受影響，這點對

我同意：

“both large $\rightarrow$ Theorem 1.3” 這個 shortcut 錯了，但不影響 Theorem 4。

因為 Theorem 4 / fixed-

Q

Q pipeline 本來依賴的是 Proposition 5.2 / target engine, 而不是只用 headline

Theorem 1.3。

但 growing-

Q

Q case split 要稍微改得更準：

Regime

工具

$$SQ \geq (\log N)^{4/5+o(1)}$$

S

Q

Liu–Sawhney headline theorem may apply

$$\geq (\log N)$$

$$4/5+o(1)$$

SQ

S

Q

Need Prop 5.2 / local mass / target-selection

bounded or slowly growing below threshold

$$SQ \sim \log N$$

S

Q

likely easier, but still requires checking

$$\sim \log N$$

hypotheses

因為 Liu–Sawhney 的 headline theorem threshold 是 reciprocal mass

$$(\log N)^{4/5+o(1)}$$

$$(\log N)$$

$$4/5+o(1)$$

，不是單純

$$SQ \sim \log N$$

S

Q

$\sim \log N$ 才能用。

建議你把段落最後改成這樣

你原文：

the corrected version ... actually gives a cleaner proof strategy for arbitrary fixed Q .

我建議改成：

the corrected version gives a cleaner candidate proof strategy for arbitrary fixed

Q

Q , but the direct subtraction heuristic must be replaced by exact enumeration or a zero-sum theorem controlling the reduced denominators of

$R(Q')$

$R(Q$

,

$).$

這樣更安全。

可以貼的修正版 marker

Correction status.

The two earlier corrections are non-fatal for the fixed- Q Theorem 4 architecture.

First, the Φ -inequality direction should be read as: small primes are often good/forced, while sufficiently large primes may be bad. The rigorous arbitrary-fixed- Q strategy is not simply to remove one denominator containing a bad prime, but to search over all non-empty proper $Q' \subset Q$ and check the reduced denominator of $R(Q')$. This motivates the bad-prime quotient / Davenport zero-sum approach.

Second, the invalid shortcut “both sides large implies Liu-Sawhney Theorem 1.3 applies” should be replaced by a reciprocal-mass case split. If the reciprocal mass is above the Liu-Sawhney threshold, headline reachability may apply; if it is bounded or below threshold, one still needs Proposition 5.2 plus local mass and target-selection.

Thus the current status is: Theorem 4 is not killed; arbitrary fixed Q has a plausible target-selection programme; growing Q remains harder and requires separate control of reciprocal mass and bad-prime quotient complexity.

Bottom line

Yes, this paragraph is okay as a research handover, but I would not post it unchanged if the audience includes specialists. The two essential edits are:

1. replace “direct subtraction works” with “exact
2. TQ
3. T
4. $\mathcal{Q}$
- 5.
6. / reduced denominator search is needed”;
7. replace “
8. $SQ \sim \log N$
9. S
10. $\mathcal{Q}$
- 11.
12. $\sim \log N$ ” as the dividing line with “reciprocal mass above or below Liu–Sawhney’s
13. $(\log N)^{4/5+o(1)}$
14. $(\log N)$
15. $4/5+o(1)$
16. threshold.”

Then it becomes a fair and useful note.

Sure! Many thanks. Anyhow based on abv chatgpt pro, can we run the TQ experiment? Thx

已思考 5m 57s

可以, 已經跑了第一版

TQ

T

$\mathcal{Q}$

target-selection experiment。這是 exploratory scan, 不是 proof, 但很有用: 它直接測試你筆記中提到的「對 fixed

Q

$\mathcal{Q}$, 枚舉所有 proper

$Q' \subsetneq Q$

$\mathcal{Q}$

,

$\subsetneq Q$, 看

$R(Q')$

$R(Q$

,

) 的 reduced denominator 是否落在 forceable prime-power 範圍」這條路。你之前的 note 也把這列為 target-selection computation 的核心下一步。

我用的 proxy test 是：

$$TQ=\{R(Q'):\emptyset\neq Q'\subsetneq Q\},$$

T

Q

$=\{R(Q$

,

$): \emptyset$

$=Q$

,

$\subsetneq Q\}$,

對每個

Q'

Q

,

計算：

$$R(Q')=\sum_{q\in Q'}1_q$$

$R(Q$

,

$)=$

$$q \in Q$$

,

$$\sum$$

$$q$$

$$1$$

的 reduced denominator, 分解其 prime-power factors, 然後測:

$$SQ < \min p \parallel \text{den}(R(Q')) \Phi p a (1 - 1/e).$$

$$S$$

$$Q$$

$$<$$

$$p$$

$$a$$

$$\parallel \text{den}(R(Q$$

,

$$))$$

$$\min$$

$$\Phi$$

$$p$$

$$a$$

$$(1 - 1/e).$$

這正是你文件裏 proposed 的 target-selection computational experiment。

背景上, #319 官方仍是 open, 只記錄 Adenwalla/Croot lower bound; Liu–Sawhney 是合理外部機器, 因為其 paper 證明 unit-fraction reachability 結果並 refine Croot/Bloom

。Davenport / zero-sum 方向也是真正的數學背景：Davenport constant 是 finite abelian group 中保證 non-empty zero-sum subsequence 的門檻。

Run 1: 原本 Aristotle example

$$Q=(3,5,7,9,11,13,15,17)$$

$$Q=(3,5,7,9,11,13,15,17)$$

$$SQ=827504765765\approx 1.0806239512$$

S

Q

=

$$765765$$

$$827504$$

$$\approx 1.0806239512$$

總共有

254

254 個 non-empty proper

Q'

Q

,

。其中 31 個 targets 通過 proxy gate test。

最前面的 passing targets:

Q	Q'	$R(Q)$	$R(Q')$	reduced denominator	prime-power factors	min Φ	margin
(3,)				1/3	$3(3,)$	1.973330	0.892706

(3,)	1/3	(3,)		
(5,)	1/5	⁵ (5,)	1.249092	0.168469
(5,)	1/5	(5,)		
(7,)	1/7	⁷ (7,)	1.146347	0.065723
(7,)	1/7	(7,)		
(9,)	1/9	⁹ (9,)	1.103820	0.023196
(9,)	1/9	(9,)		
(15,)	1/15	¹⁵ (3,5)	1.249092	0.168469
(15,)	1/15	(3,5)		
(3,15)	2/5	⁵ (5,)	1.249092	0.168469
(3,15)	2/5	(5,)		
(3,9)	4/9	⁹ (9,)	1.103820	0.023196
(3,9)	4/9	(9,)		
(3,5)	8/15	¹⁵ (3,5)	1.249092	0.168469
(3,5)	8/15	(3,5)		

Interpretation: this

Q

Q is not remotely rigid under the

TQ

T

Q

test. The singleton target

1/3

1/3 is still the cleanest, but there are many alternative targets. This supports the idea that target-selection is a real route beyond singleton-gate.

Run 2: large-prime stress example

I also tested:

$$Q=(3,5,7,11,13,17,19,23,101)$$

$$Q=(3,5,7,11,13,17,19,23,101)$$

$$SQ \approx 1.0088570034$$

S

Q

$$\approx 1.0088570034$$

Total proper targets:

510

510. Passing targets: 255.

Top passing singleton targets include:

Q	Q'	$R(Q)$	$R(Q')$	denominator	min Φ	margin
(3,)			1/3	3	1.973330	0.964473
(3,)			1/3			
(5,)			1/5	5	1.249092	0.240235
(5,)			1/5			
(7,)			1/7	7	1.146347	0.137490
(7,)			1/7			
(11,)			1/11	11	1.080486	0.071629
(11,)			1/11			
(13,)			1/13	13	1.065729	0.056872
(13,)			1/13			

(17,)	1/17	17	1.048102	0.039245
(17,)	1/17			
(19,)	1/19	19	1.042417	0.033560
(19,)	1/19			
(23,)	1/23	23	1.034309	0.025452
(23,)	1/23			

Interpretation: even after adding a large prime

101

101, the target-selection set is still very rich because

SQ

S

ϱ

is only slightly above 1. This is useful evidence, but not a hard test yet.

Small exhaustive scan

I also ran a small exhaustive scan over:

$$Q \subseteq \{2, 3, \dots, 12\}, SQ > 1.$$

$$Q \subseteq \{2, 3, \dots, 12\}, S$$

ϱ

$$> 1.$$

Result:

1128/1128

1128/1128

had at least one passing proper target under the proxy test.

So in this small universe, there were no rigid

Q

Q examples.

What this means

This is good evidence for the target-selection programme:

fixed $Q \rightarrow TQ$ search $\rightarrow$ find forceable $R(Q') \rightarrow$ Liu–Sawhney hit $\rightarrow$ proper subcircuit.

fixed $Q \rightarrow T$

Q

search $\rightarrow$ find forceable $R(Q)$

,

$\rightarrow$ Liu–Sawhney hit $\rightarrow$ proper subcircuit.

But it is still only evidence. The next stronger experiment should search for rigid

Q

Q where:

$$SQ > 1$$

S

Q

$$> 1$$

but no non-empty proper

Q'

Q

,

passes the denominator gate. If such

Q

Q exists, it is the next obstruction. If none appear up to meaningful bounds, that supports a possible arbitrary fixed-

Q

Q target-selection lemma.

Files

I saved the experiment outputs and a reusable Python script here:

[Download TQ experiment results zip](#)

The zip includes:

- `Q0_targets.csv`
- `Q1_targets.csv`
- `Q0_top_passing.csv`
- `Q1_top_passing.csv`
- `summary.csv`
- `scan_M12_samples.csv`
- `tq_scan.py`

Next run I recommend

The current examples are too friendly. The next scan should deliberately avoid easy denominators, especially

2,3,5,7

2,3,5,7, and search fixed

Q

Q with:

$SQ > 1, \min Q$ relatively large.

S

Q

$> 1, \min Q$ relatively large.

That will stress-test whether bad-prime quotient / Davenport is actually needed, or whether simple

TQ

T

Q

targets already solve most fixed-

Q

Q cases.

Certainly, many thx!

我會把這句理解為：繼續做下一輪更嚴格的

TQ

T

Q

stress test, 特別避開

2,3,5,7

2,3,5,7 這些太容易的 denominators, 看看是否真的會出現 rigid

Q

Q 。

已思考 3m 58s

Ran the stricter

TQ

T

Q

stress test. It gives a useful update.

Officially, #319 is still open and the known public lower-bound context remains Adenwalla/Croot

$(1-1/e+o(1))N$

$(1-1/e+o(1))N$; Liu–Sawhney remains the relevant external machinery for the reachability step. Your uploaded roadmap also correctly identified

TQ

T

$\mathcal{Q}$

-target selection / bad-prime quotient as the next concrete experiment beyond the fixed-

Q

Q singleton-gate route.

What I ran

I stress-tested fixed

Q

Q examples with fewer easy denominators:

$$TQ = \{R(Q') : \emptyset \neq Q' \subsetneq Q\}, R(Q') = \sum_{q \in Q'} 1/q.$$

T

$\mathcal{Q}$

$=\{R(Q$

,

$): \emptyset$

$=Q$

,

$\subsetneq Q\}, R(Q$

,

$)=$

$q \in Q$

,

$$\Sigma$$

$$q$$

$$1$$

$$\cdot$$

For each

$$Q'$$

$$Q$$

,

, I computed the reduced denominator of

$$R(Q')$$

$$R(Q$$

,

), its prime-power factors, and checked the proxy gate:

$$SQ<\min p a\| \text{den}(R(Q'))\Phi p a(1-1/e).$$

$$S$$

$$Q$$

$$<$$

$$p$$

$$a$$

$$\| \text{den}(R(Q$$

,

$$))$$

$$\min$$

$$\Phi$$

p

a

$$(1-1/e).$$

Main result

1. Interval-type

Q

Q 's are still very non-rigid

Examples:

|

Q

Q type |

| Q |

| Q | |

SQ

S

Q

| Passing targets seen | Example passing target |

|---|---:|---:|---:|---|

|

$Q=\{8,\dots,20\}$

$Q=\{8,\dots,20\}$ | 13 | 1.00488 | 3000 / 3000 sampled |

$$1/10+1/15=1/6$$

$$1/10+1/15=1/6 \mid$$

|

$$Q=\{11,\dots,29\}$$

$$Q=\{11,\dots,29\} \mid 19 \mid 1.03269 \mid 910 / 3000 \text{ sampled} \mid$$

$$1/12+1/21+1/28=1/6$$

$$1/12+1/21+1/28=1/6 \mid$$

|

$$Q=\{20,\dots,53\}$$

$$Q=\{20,\dots,53\} \mid 34 \mid 1.00917 \mid 7174 / 7174 \text{ sampled} \mid$$

$$1/21+1/28=1/12$$

$$1/21+1/28=1/12 \mid$$

|

$$Q=\{30,\dots,80\}$$

$$Q=\{30,\dots,80\} \mid 51 \mid 1.00383 \mid 23477 / 23477 \text{ sampled} \mid$$

$$1/30+1/36+1/45=1/12$$

$$1/30+1/36+1/45=1/12 \mid$$

Interpretation: if

Q

Q contains many composite numbers,

TQ

T

Q

quickly produces small-denominator targets like

$1/6$

$1/6,$

$1/12$

$1/12,$ etc. These are very friendly for the Liu–Sawhney target route.

So for “natural interval-like

Q

$Q,$ ” the target-selection method looks strong.

Important warning: prime-only

Q

Q can be rigid

The stress test also exposed the real obstruction.

If

Q

Q is a set of distinct primes all at least

p_0

p

0

, then for any non-empty

$Q' \subseteq Q$

Q

,

$\subseteq Q$, the reduced denominator of

$$R(Q')$$

$$R(Q$$

,

) still contains the primes in

$$Q'$$

$$Q$$

,

. There is no cancellation modulo those primes.

Therefore:

$$\min p a \| \text{den}(R(Q')) \Phi p a \leq \Phi p 0.$$

$$p$$

$$a$$

$$\| \text{den}(R(Q$$

,

$$))$$

$$\min$$

$$\Phi$$

$$p$$

$$a$$

$$\leq \Phi$$

$$p$$

$$0$$

So if:

$$SQ > \Phi p_0,$$

S

Q

$>\Phi$

p

0

,

then no non-empty proper

Q'

Q

,

can pass the simple proxy gate.

Concrete theoretical stress example:

$$Q = \{\text{first 237 primes from 11 to 1523}\}.$$

$$Q = \{\text{first 237 primes from 11 to 1523}\}.$$

For this

Q

$Q,$

$$SQ \approx 1.080756 > \Phi 11 \approx 1.080486.$$

S

Q

$$\approx 1.080756 > \Phi$$

11

$$\approx 1.080486.$$

So this

Q

Q is rigid for the simple

TQ

T

ϱ

proxy test.

This does not disprove the whole bad-prime approach, but it shows the naive version is not enough for arbitrary fixed

Q

Q .

What this means

The experiment strengthens two conclusions:

Good news

For many structured

Q

Q 's, especially composite/interval-like

Q

Q , target selection works extremely well.

TQ

T

ϱ

naturally finds smooth-denominator targets.

Bad news / useful obstruction

Arbitrary fixed

Q

Q cannot be solved by the simple rule:

“Just find a

Q'

Q

,

whose reduced denominator uses only good small primes.”

Prime-only

Q

Q 's show a genuine rigid family for that proxy.

So the arbitrary fixed-

Q

Q project now has a clearer split:

Case	Status
Composite / interval-like Q Q	Very promising
Minimal-over-one Q Q	Still promising
Prime-only or almost-prime Q Q with large SQ S Q	Real obstruction

Arbitrary fixed

$\mathbb{Q}$

$\mathbb{Q}$

Needs stronger bad-prime / p -adic / two-sided
method

Updated honest verdict

TQ target selection is real, but not universal in its simplest form.

T

$\mathbb{Q}$

target selection is real, but not universal in its simplest form.

The next serious theoretical question is:

Can the prime-only rigid family be killed by a different argument, perhaps using
balance-derived

p

p -adic constraints, not just target-denominator smoothness?

That is likely the real next obstruction after the fixed-

$\mathbb{Q}$

$\mathbb{Q}$ singleton-gate theorem.

Files

I saved the second stress-test outputs here:

[Download second](#)

[TQ](#)

[\$T\$](#)

[\$\mathbb{Q}\$](#)

[stress-test results](#)

The zip includes the stress summary, example target CSVs, and a `prime_rigid_note.txt` explaining the prime-only obstruction.

Sorry is there anyway out thx

有路, 但不是「照舊找 smooth denominator 的

Q'

Q

,

」那條路。prime-only

Q

Q 暴露的是: target selection alone 不夠; 要把原等式本身的

p

p -adic cancellation 用起來。

已思考 1m 24s

Yes — 有路, 但不是原本那條簡單

TQ

T

Q

smooth-denominator 路。

The prime-only stress test did not kill the whole project. It killed one overly simple hope:

“For every fixed

Q

Q , some proper

$Q' \subsetneq Q$

Q

,

$\subsetneq Q$ will have a nice smooth denominator.”

For prime-only

Q

Q , that hope fails: if

Q'

Q

,

contains a prime

p

p , then

$R(Q')$

$R(Q$

,

) usually keeps

p

p in its reduced denominator. So denominator-smoothing alone cannot solve arbitrary fixed

Q

Q .

But there are still several possible exits.

Officially, #319 is still open, and the public page records only the Adenwalla/Croot lower bound

$(1-1/e+o(1))N$

$(1-1/e+o(1))N$, so this remains research-roadmap territory rather than a solved proof. Liu–Sawhney remains relevant because their unit-fraction paper gives a powerful target-reaching machine, while Davenport/zero-sum theory remains relevant for finite abelian target-selection ideas.

1. What failed exactly?

The failed idea was:

$Q' \subsetneq Q \Rightarrow R(Q')$ has denominator using only good primes.

Q

,

$$\subseteq Q \Rightarrow R(Q)$$

,

) has denominator using only good primes.

For composite/interval-like

Q

Q , this often works because denominators cancel. For example:

$$110+115=16.$$

$$10$$

$$1$$

$$+$$

$$15$$

$$1$$

$$=$$

$$6$$

$$1$$

$$.$$

So a messy

Q'

Q

,

can create a clean target.

But for prime-only

Q

Q , e.g.

$$Q=\{11,13,17,19,\dots\},$$

$$Q=\{11,13,17,19,\dots\},$$

if

$$Q'=\{11,17,23\},$$

Q

,

$$=\{11,17,23\},$$

then

$$R(Q')=111+117+123$$

$R(Q$

,

$$)=$$

$$11$$

$$1$$

$$+$$

$$17$$

$$1$$

$$+$$

$$23$$

usually still has denominator containing

11,17,23

11,17,23. No smoothing.

So the simple

TQ

T

Q

trick does not handle prime-only

Q

Q .

2. Why this is not fatal

Prime-only

Q

Q has a weakness: the original equality itself forces

P

P to respond to those primes.

If

$$\sum_{p \in P} 1/p = \sum_{q \in Q} 1/q,$$

$$p \in P$$

$$\sum$$

$$p$$

$$1$$

$$= \sum_{q \in Q} q$$

and

Q

Q contains

$1/11$

$1/11$, then the

11

11 -adic part on the right cannot just disappear. On the left,

P

P must contain denominators divisible by

11

11 in some way, otherwise the

11

11 -adic denominator has nothing to cancel it.

中學生版：

如果右邊有一個「 11 形狀的洞」，左邊一定要有某些「 11 形狀的積木」去補。

否則等式不可能成立。

The problem is that this only guarantees some

11

11 -support, not necessarily enough local mass for Liu–Sawhney Proposition 5.2. But

it gives a new handle.

3. The most promising way out: finite gate absorption

This is now the best conceptual exit.

For prime-only

Q

Q , do not try to find a

Q'

Q

,

whose denominator avoids bad primes. Instead:

1. Use the original equality to force
2. P
3. P to contain some top-layer terms divisible by each bad prime
4. $q \in Q$
5. $q \in Q$.
6. Pick a small “gate set”
7. $G_q \subset P$
8. G
9. q
- 10.
11. $\subset P$ that matches the
12. q
13. q -adic obstruction of
14. $1/q$
15. $1/q$.
16. After those bad-prime gates are paid for, the remaining target may have only good denominators.
17. Then use Liu–Sawhney on the bulk of
18. P
19. P to hit the remaining target.

Symbolically, instead of trying to hit:

$$R(Q')$$

$$R(Q$$

,

)

directly, try to split it as:

$$R(Q')=R(G_{\text{bad gates}})+r_{\text{good}},$$

$$R(Q$$

,

$$)=R(G$$

bad gates

$$)+r$$

good

,

where

r_{good}

r

good

has only good denominator structure.

Then Liu–Sawhney only needs to hit

r_{good}

r

good

, not the whole bad-prime target.

This is a genuine way out.

4. Why this is harder than before

The old

TQ

T

Q

route only asked:

Does

$R(Q')$

$R(Q$

,

) have a nice denominator?

The new gate route asks:

Can

P

P 's forced bad-prime layers be selected in a way that cancels the bad denominator part of

$R(Q')$

$R(Q$

,

)?

That is much subtler. It uses both:

$$R(P)=R(Q)$$

$$R(P)=R(Q)$$

and p-adic structure.

This is closer to the true nature of #319.

5. Another possible way out: proper zero-sum subsequences with marked primes

The Davenport idea may still help, but not in the naive “avoid bad primes” form.

For prime-only

Q

Q , every selected prime stays in the denominator. So we need a richer finite-group model:

- not merely “avoid this prime”;
- but “match this prime’s required residue using available
- P
- P -side gates.”

That becomes a two-sided zero-sum problem:

bad-prime residues from Q =bad-prime residues available from P .

bad-prime residues from Q =bad-prime residues available from P .

This is more difficult, but Davenport-style tools still belong here, because Davenport constants are precisely about forcing zero-sum subsequences in finite abelian groups.

6. Third way out: prove prime-only

Q

Q is structurally expensive

Another route is to prove that prime-only

Q

Q cannot support a very large primitive circuit because it imposes too many independent p -adic obligations on

P

P .

If

Q

Q contains many primes:

$$q_1, q_2, \dots, q_k,$$

$$q$$

$$1$$

$$, q$$

$$2$$

$$, \dots, q$$

$$k$$

,

then

P

P must answer all those

$$q_i$$

$$q$$

$$i$$

-adic layers.

Maybe that forces

P

P to waste too much density or reciprocal mass. If so, prime-only

Q

Q would be killed not by target selection, but by a cost argument:

many bad primes $\Rightarrow$ too much forced structure in $P \Rightarrow |P| \leq (1 - 1/e + o(1))N$.

many bad primes $\Rightarrow$ too much forced structure in $P \Rightarrow |P| \leq (1 - 1/e + o(1))N$.

This is speculative, but it is a natural next theorem.

7. Fourth way out: two-sided spectrum intersection

The full-problem route is not to choose a target from

Q

Q and make

P

P hit it. Instead, prove that the whole subset-sum spectra must collide:

$$\Sigma(P) \cap \Sigma(Q) \neq \{0, R(P)\}.$$

$$\Sigma(P) \cap \Sigma(Q)$$

$$= \{0, R(P)\}.$$

That is the true endgame.

The prime-only obstruction suggests that a one-sided target method may not be enough. We may need a two-sided method where both

P

P and

Q

Q contribute structure.

This is harder, but it is probably the right long-term form.

Current honest status

Route	Does prime-only		Status
	Q	Q kill it?	
Simple			
TQ		Yes	Not universal
T			
2			

smooth target		
Fixed- Q Q singleton gate		Still valid for gate No Q Q 's
Composite/interval Q Q target selection		NoStill promising
Prime-only Q Q via gate absorption		NoNew promising route
Davenport bad-prime quotient	Not killed, but must be	Needs two-sided/residue upgradedversion
Growing Q Q / full #319		Not solvedStill hard

Best short answer

Yes, there is a way out:

replace “smooth target selection” by “bad-prime gate absorption.”

replace “smooth target selection” by “bad-prime gate absorption.”

The prime-only stress test tells us:

arbitrary fixed

Q

Q cannot be solved by denominator smoothing alone.

But it also points to the next real idea:

use the equality

$$R(P)=R(Q)$$

$R(P)=R(Q)$ to force

P

P -side p -adic gates, then use those gates to neutralize bad primes before applying Liu–Sawhney to the remaining good target.

So the obstruction is not a dead end. It is a signpost.

The next experiment should test exactly this: for a prime-only

Q

Q , compute the forced p -adic top-layer requirements and see whether small

P

P -side gate sets could cancel them, leaving a smooth residual target.

Many thanks! Yes, Claude also said sth similar thx "This is a really valuable result from ChatGPT — it found the right stress test and correctly identified where the naive proxy fails. But I don't think it's actually stuck. Let me explain why, because I think there's a clean theoretical way out.

The proxy test was checking the wrong condition

The T_Q proxy tested:

> Does $R(Q')$ have only "good" small-prime denominators (where $\Phi_p > S_Q$)?

For prime-only $Q = \{p_1, \dots, p_n\}$, every proper subset sum $R(Q')$ has ALL of $p_1, \dots, p_n$ in its denominator — no cancellation ever occurs. So the proxy always fails. This looks like a wall.

But the proxy is only a *sufficient* condition for Liu-Sawhney reachability, not a necessary one. The actual Liu-Sawhney condition is:

> Does P have sufficient p -adic local mass at the primes in the denominator of $R(Q')$?

And for prime-only Q , **the balance constraint itself forces this directly** — no Φ -test needed.

The p-adic valuation argument (the way out)

For $Q = \{p_1, \dots, p_\ell\}$ distinct primes and $R(P) = S_Q = \sum 1/p_i$:

Step 1: For each $p_i \in Q$, compute $v_{p_i}(S_Q)$. Since $S_Q = 1/p_i + \sum_{i \neq j} 1/p_i$, and each $1/p_i$ ($i \neq j$) has $v_{p_i} = 0$, we get $v_{p_i}(S_Q) = -1$ exactly.

Step 2: Since $R(P) = S_Q$, we need $v_{p_i}(R(P)) = -1$ for each j .

Step 3: If P contained NO multiples of p_i , then every term $1/n$ in $R(P)$ would have $v_{p_i}(1/n) \geq 0$, giving $v_{p_i}(R(P)) \geq 0$. But we need -1 . **Contradiction.**

Conclusion: P must contain multiples of every $p_i \in Q$. The balance constraint directly forces p-adic local mass at ALL primes in Q , regardless of the Φ test.

Combined with your Lemma 3 (which gives the quantitative lower bound $p_j \cdot R(P_{p_j}) \geq p > 0$), Liu-Sawhney can now reach any target in T_Q — including those with large-prime denominators.

The complete picture

| Q type | How to kill it |

|---|

| Composite/interval Q | Proxy test: T_Q contains smooth-denominator targets |

| Prime-only Q | p-adic valuation argument: balance directly forces local mass at all $p_i \in Q$ |

| Mixed Q | Combination of both |

The proxy test and the p-adic argument **together cover all fixed Q** . The "rigid obstruction" was an artifact of using too coarse a proxy — it was never a true mathematical wall.

What to send back to ChatGPT

The key message is:

> For prime-only Q , the T_Q proxy test correctly fails — but the actual Liu-Sawhney local mass condition is satisfied anyway, because $v_{\{p\}}(R(P)) = v_{\{p\}}(S_Q) = -1$ for each $p \in Q$, and this directly forces P to contain multiples of every p . The proxy was checking a sufficient condition, not the necessary one.

Ask it to: (1) verify the p -adic valuation calculation for the specific example $Q = \{\text{first 237 primes, 11–1523}\}$, and (2) confirm that Lemma 3-style quantitative forcing follows from $v_{\{p\}}(R(P)) = -1$. That would close the gap for arbitrary fixed Q ."

我會分兩步答：先核對 Claude 的

p

p -adic valuation argument 是否正確；再判斷它是否足以給 Liu–Sawhney 所需的 quantitative local mass。這兩者不是同一件事。

已思考 1m 9s

Yes — Claude's observation is valuable, and the basic

p

p -adic calculation is correct. But I would not accept the final sentence "that would close the gap for arbitrary fixed

Q

Q " yet.

The precise verdict is:

The valuation argument fixes existence of p -support, but not quantitative local mass.

The valuation argument fixes existence of p -support, but not quantitative local mass.

That difference matters because Liu–Sawhney Proposition 5.2 needs quantitative local mass, not merely "at least one multiple of

p

p ." Liu–Sawhney's result is a serious unit-fraction reachability machine, but it has hypotheses involving smoothness, bounded

Ω

Ω , Archimedean window, and local

p

p -adic mass. (arxiv.org)

1. The

p

p -adic valuation calculation is correct

For prime-only

$$Q=\{p_1,\ldots,p_k\},$$

$$Q=\{p$$

1

$$,\ldots,p$$

k

$$\},$$

we have

$$SQ=\sum_{i=1}^k1p_i.$$

S

Q

$$=$$

$$i=1$$

$$\sum$$

$$k$$

$$p$$

$$i$$

$$1$$

$$\cdot$$

Fix

$$p_j\in Q$$

$$p$$

$$j$$

$$\in Q.$$
 Then

$$SQ=1p_j+\sum_{i\neq j}1p_i.$$

$$S$$

$$\varrho$$

$$=$$

$$p$$

$$j$$

$$1$$

$$+$$

$$i$$

$$\Rightarrow j$$

$$\sum$$

$$p$$

i

1

.

The second part has

p_j

p

j

-adic valuation

≥ 0

≥ 0 , because none of its denominators is divisible by

p_j

p

j

. The first part has valuation

-1

-1 . Since there is no cancellation of that top

p_j

p

j

-adic term, indeed:

$v_{p_j}(S_Q) = -1$.

v

p

j

$(S$

Q

$$)=-1.$$

For the concrete stress example

$$Q=\{\text{all primes from 11 to 1523}\},$$

$$Q=\{\text{all primes from 11 to 1523}\},$$

there are 237 primes, and the exact computation gives:

$$SQ\approx 1.0807561888.$$

$$S$$

$$Q$$

$$\approx 1.0807561888.$$

For every prime

$$p\in Q$$

$$p\in Q,$$

$$vp(SQ)=-1.$$

$$v$$

$$p$$

$$(S$$

$$Q$$

$$)=-1.$$

So Claude is right on this point.

2. It also correctly proves

$$P$$

P must contain multiples of every

p_j

p

j

If

$$R(P)=SQ,$$

$$R(P)=S$$

Q

,

then

$$v_{p_j}(R(P))=-1.$$

v

p

j

$$(R(P))=-1.$$

If no denominator in

P

P were divisible by

p_j

p

j

, then every term

$1/n$

$1/n$ in

$R(P)$

$R(P)$ would have

p_j

p

j

p -adic valuation

≥ 0

≥ 0 , so their sum would also have valuation

≥ 0

≥ 0 . Contradiction.

Therefore:

P must contain at least one denominator divisible by each $p_j \in Q$.

P must contain at least one denominator divisible by each p

j

$\in Q$.

This is an important correction to the earlier “prime-only wall.” The wall is not absolute. Prime-only

Q

Q forces

P

P to carry matching

p

p -adic support.

3. But this does not yet give Liu–Sawhney local mass

Here is the key gap.

Liu–Sawhney does not merely need:

$$P_{p_j} \neq \emptyset.$$

$$P$$

$$p$$

$$j$$

$$=\emptyset.$$

It needs something like:

$$p_j R(P_{p_j}) \geq \eta$$

$$p$$

$$j$$

$$R(P$$

$$p$$

$$j$$

$$) \geq \eta$$

for a sufficiently usable

$$\eta$$

η , after pruning.

The valuation argument only proves existence of some

$$p_j$$

$$p$$

$$j$$

-multiple. It does not prove a positive real-mass lower bound.

Example of the issue:

$$1\text{pM}$$

$$pM$$

$$1$$

has

$$\text{vp}(1\text{pM})=-1$$

$$v$$

$$p$$

$$($$

$$pM$$

$$1$$

$$)=-1$$

if

$$p \nmid M$$

$p \nmid M$, but its real mass is tiny:

$$p \cdot 1\text{pM}=1\text{M} \rightarrow 0.$$

$$p \cdot$$

$$pM$$

$$1$$

$$=$$

$$M$$

$$1$$

$$\rightarrow 0.$$

So

p

p -adically it has the right denominator, but Archimedean-wise it is almost invisible.

That is why this implication is false as stated:

$$v_p(R(P)) = -1 \Rightarrow v_p(R(P_p)) \geq \rho > 0.$$

v

p

$$(R(P)) = -1 \Rightarrow v_p(R(P$$

p

$$) \geq \rho > 0.$$

It only gives:

$$v_p(R(P_p)) = -1$$

v

p

$$(R(P$$

p

$$)) = -1$$

or at least nontrivial

p

p -adic support.

4. What Claude's argument really gives

Claude's argument gives a qualitative gate:

$$P \cap p \neq \emptyset.$$

$$P$$

$$p$$

$$= \emptyset.$$

But Liu-Sawhney needs a quantitative gate:

$$pR(P \cap p) \geq \rho.$$

$$pR(P$$

$$p$$

$$) \geq \rho.$$

So the correct next lemma would be:

Prime-side quantitative gate lemma.

Let

$$Q$$

Q be a fixed set of primes and

$$R(P) = R(Q)$$

$$R(P) = R(Q). \text{ If}$$

$$|P| > (1 - 1/e + \varepsilon)N$$

$$|P| > (1 - 1/e + \varepsilon)N, \text{ then for every}$$

$$p \in Q$$

$$p \in Q, \text{ not only is}$$

$$P \cap p \neq \emptyset$$

$$P$$

$$p$$

$$= \emptyset, \text{ but}$$

$$pR(Pp) \geq \rho p, Q, \varepsilon > 0.$$

$$pR(P$$

$$p$$

$$) \geq \rho$$

$$p, Q, \varepsilon$$

$$> 0.$$

This is not proven by valuation alone. It would need to use:

1. the high cardinality of
2. P
3. P ;
4. the exact balance
5. $R(P) = SQ$
6. $R(P) = S$
7. Q
- 8.
9. ;
10. the
11. Φ
12. Φ -style nonmultiple lower bound;
13. maybe a stability version: if
14. $pR(Pp)$
15. $pR(P$
16. p
- 17.
18.) is too small, then
19. $R(P)$
20. $R(P)$ must be too large or too small.

That would be a real new lemma.

5. Does this close arbitrary fixed

Q

$Q?$

Not yet.

It improves the picture from:

prime-only

Q

Q is a dead wall

to:

prime-only

Q

Q requires a stronger

p

p -adic quantitative forcing lemma.

So the new status is:

Claim	Status
$vp(SQ)=-1$ v p $(S$ Q $)=-1$ for prime-only Q Q	Correct
P P must contain multiples of every $p \in Q$ $p \in Q$	Correct
This gives enough Liu–Sawhney local mass	Not yet
Arbitrary fixed Q	No

Q now solved

Arbitrary fixed

Q

Yes, more plausible than before

Q still plausible

6. The way out becomes clearer

The right repair is not the old smooth-denominator

TQ

T

Q

proxy. It is:

qualitative p -adic support $\rightarrow$ quantitative local mass $\rightarrow$ Liu–Sawhney
reachability.

qualitative p -adic support $\rightarrow$ quantitative local mass $\rightarrow$ Liu–Sawhney
reachability.

For composite/interval

Q

Q , smooth target-selection works.

For prime-only

Q

Q , valuation gives forced support, but we still need quantitative mass.

For mixed

Q

Q , both mechanisms combine.

7. What I would send back to Claude

You can say:

The

p

p -adic valuation calculation is correct: for prime-only

Q

Q ,

$v_p(SQ) = -1$

v

p

$(S$

Q

$) = -1$ for every

$p \in Q$

$p \in Q$, hence any

P

P with

$R(P) = SQ$

$R(P) = S$

Q

must contain denominators divisible by each such

p

p . This fixes the qualitative obstruction exposed by the

TQ

T

Q

proxy. However, Liu–Sawhney requires quantitative local mass, not merely existence of a

p

p -multiple. The implication

$v_p(R(P)) = -1 \Rightarrow pR(P_p) \geq p > 0$

v

p

$$(R(P)) = -1 \Rightarrow pR(P)$$

p

$\geq \rho > 0$ is not automatic; a single term

$$1/(pM)$$

$1/(pM)$ already has valuation

$$-1$$

-1 but mass

$$1/M$$

$1/M$. Therefore the remaining needed lemma is a quantitative stability version using high density plus balance.

Bottom line

Claude is right that the prime-only wall is not fatal.

But the gap is not closed. It has been sharpened into a specific lemma:

Can high-density P with $R(P) = SQ$ have only tiny p -mass at a prime $p \in Q$?

Can high-density P with $R(P) = S$

Q

have only tiny p -mass at a prime $p \in Q$?

If the answer is no, arbitrary fixed

Q

Q becomes much closer. If the answer is yes, we need gate absorption or a two-sided spectrum method.

Many thanks! But Claude sonnet said "GPT is right, and this is a real gap — but let me tell you exactly how far it goes and where the genuine frontier is.

What's proved vs. what's missing

****Proved (solid):****

> Balance constraint $R(P) = S_Q$ forces $v_{\{p\}}(R(P)) = -1$ for each $p \in Q$, so P must contain ****at least one**** multiple of each p . Qualitative forcing is established.

****Missing (the real gap):****

> For large primes $p \in Q$ (those where $\Phi_{\{p\}}(1-1/e) < S_Q$), the quantitative bound $p \cdot R(P_{\{p\}}) \geq \rho > 0$ is ****not proved****. The complement can absorb all but one multiple of p , leaving $R(P_{\{p\}}) \approx 1/N \rightarrow 0$.

Note carefully: ****Lemma 3**** from your handover (the balance-constraint local mass forcing lemma) requires $S_Q < \Phi_q(1-1/e)$ as a hypothesis. For prime-only Q , the large primes p satisfy $\Phi_{\{p\}} < S_Q$, so Lemma 3 simply doesn't fire. This is the exact boundary where the current toolkit runs out.

Why density alone can't close it

For $p = 11$ in the stress example: there are $N/11 \approx 0.09N$ multiples of 11 in $[1, N]$. The complement of P has $\approx 0.37N$ elements, enough to absorb all $0.09N$ multiples of 11. So P could in principle have $|P_{\{11\}}| = 1$, giving $p_{\{11\}} \cdot R(P_{\{11\}}) \approx 11/N \rightarrow 0$. No density or individual Lemma 2 argument prevents this.

The most promising route forward

The gap is not arbitrary — it has a specific structure. Two approaches worth pursuing:

****Route 1 — Joint multi-prime budget argument:****

The complement has $(1/e - \epsilon)N \approx 0.37N$ elements. The total number of p -multiples across all j is $\sum_j N/p_j = S_Q \cdot N \approx 1.08N$. The complement cannot absorb all of them simultaneously. By a weighted pigeonhole over all j :

$$\sum_j |P_{\{p_j\}}| \geq (S_Q - 1/e + \epsilon)N \approx 0.71N.$$

This gives a ***total*** lower bound on p -mass across Q , but distributing this to individual p requires additional structure — specifically, that the mass can't all concentrate on one prime.

****Route 2 — Near-extremal stability:****

If $p \cdot R(P_{\{p\}})$ is tiny, then $R(P_{\{\text{non-}p\}}) \approx S_Q$ with density $\approx 1-1/e+\epsilon$. For the stress example, $\Phi_{\{11\}}(1-1/e+\epsilon) > S_Q$ for any $\epsilon > 0.00008$. This forces $P_{\{\text{non-}11\}}$ to be **near-extremal** for Lemma 2 — i.e., almost exactly the set $\{n \in [(1/e)N, N] : 11 \nmid n\}$. Such a near-extremal set has highly constrained structure that likely forces contradictions with $R(P) = S_Q$ exactly. This is the ****stability**** direction and may be the cleanest path.

Updated honest status table

Claim	Status
Qualitative: P has multiples of every $p \in Q$	✓ Proved
Quantitative: $p \cdot R(P_{\{p\}}) \geq \rho$ for small p (Lemma 3 range)	✓ Proved
Quantitative: $p \cdot R(P_{\{p\}}) \geq \rho$ for large p (prime-only Q)	✗ Open
Arbitrary fixed Q resolved	✗ Not yet
Prime-only Q is a dead wall	✗ No — gap is specific and approachable
Route via near-extremal stability	◆ Promising, not immediate

What to send ChatGPT next

The sharpened question is:

> Can a set $P \subseteq [1, N]$ with density $> 1-1/e+\epsilon$ and $R(P) = S_Q$ (with S_Q just above $\Phi_{\{p_0\}}(1-1/e)$) have $P_{\{p_0\}}$ consisting of only one element $n_0 = p_0 \cdot M$ with $M \approx N$? If yes, construct the explicit near-extremal example. If no, identify what constraint rules it out.

This is finite, testable, and would immediately tell us whether Route 2 (near-extremal stability) is viable. Want me to code that search up right now?" and . do we have potential in making real progress/ even solve arbitrary or growing Q with these above? Please feel free to use as much time and invariance not afraid to make mistake free from boundary and constrains mode thx. for "Yes — actually, working through the calculation more carefully, I think GPT's gap may be ****closable with a direct quantitative argument****. Let me show you what I missed in my earlier response.

The missing quantitative bound

GPT's worry was that $\alpha := p_0 \cdot R(P_{\{p_0\}})$ could be arbitrarily small (e.g., one element $n = p_0 \cdot M$ with $M \approx N$ gives $\alpha \approx 1/M \rightarrow 0$). The "single multiple" example seemed to defeat any quantitative forcing.

****But the single multiple is incompatible with P's density.**** Here's the key inequality I should have spotted:

$$|P_{\{p_0\}}| \leq N \cdot R(P_{\{p_0\}}) = N \cdot \alpha/p_0$$

because every $n \in P_{\{p_0\}}$ has $1/n \geq 1/N$. So small α forces small $|P_{\{p_0\}}|$, hence forces $\text{density}(P_{\{\text{non-}p_0\}}) \approx \text{density}(P)$.

Now apply your Lemma 2 to $P_{\{\text{non-}p_0\}}$:

$$R(P_{\{\text{non-}p_0\}}) \geq \Phi_{\{p_0\}} \left((1 - 1/e + \epsilon) - \alpha/p_0 \right)$$

Combined with balance $R(P_{\{\text{non-}p_0\}}) = S_Q - \alpha/p_0$:

$$S_Q - \alpha/p_0 \geq \Phi_{\{p_0\}} \left((1 - 1/e + \epsilon) - \alpha/p_0 \right)$$

Linearizing around $\theta = 1 - 1/e$ and solving for α gives:

$$\boxed{\alpha \geq \frac{p_0 \cdot \Phi_{\{p_0\}}(1 - 1/e) \cdot (\epsilon - \epsilon^*)}{\Phi'_{\{p_0\}}(1 - 1/e) - 1}}$$

where $\epsilon^*(Q) := (S_Q - \Phi_{\{p_0\}}(1 - 1/e)) / \Phi'_{\{p_0\}}(1 - 1/e)$.

For your stress example ($Q = 237$ primes from 11 to 1523, $p_0 = 11$):

- $\epsilon^*(Q) \approx 0.0000823$
- $\alpha \geq \approx 15.8 \cdot (\epsilon - 0.0000823)$
- At $\epsilon = 0.01$: $\alpha \geq 0.157$ ✓
- At $\epsilon = 0.1$: $\alpha \geq 1.58$ ✓ (fully forced)

This is exactly the "quantitative gate" GPT said was missing.

What this means for the project

| Claim | Updated status |

|---|---|

| Qualitative forcing $P_{\{p\}} \neq \emptyset$ |  Proved |

Quantitative bound for small primes (Lemma 3 range)	 Proved
Quantitative bound for large primes ($\Phi_p < S_Q$)	 **Provable** via the inequality above
Arbitrary fixed Q with $\varepsilon > \varepsilon^*(Q)$	 Reachable now
Arbitrary fixed Q with $\varepsilon \leq \varepsilon^*(Q)$	 Still requires finer analysis

Crucially: $\varepsilon^*(Q)$ is a finite computable constant for each Q . For your stress example, the binding constraint is from the largest prime (1523), giving $\varepsilon^*_{\max} \approx 0.0295$. So for any $\varepsilon > 0.03$, the stress example dies. For $\varepsilon \leq 0.03$, you need additional argument — but such Q are a *finite, classifiable* family for each fixed ε .

Three caveats I want to flag honestly

1. **This is my own derivation, not verified.** Run it past GPT and DeepSeek before incorporating. The linearization step needs the second-order term bounded for rigor.
2. **Liu-Sawhney's actual local mass threshold may be quantitative in N .\alpha \geq c (absolute constant) rather than $\alpha \geq c(\varepsilon)$, the bound above is fine. If it requires $\alpha \geq (\log N)^{-c}$ or stronger, my bound is enough only for fixed $\varepsilon > \varepsilon^*(Q)$.
3. **The $\varepsilon \leq \varepsilon^*(Q)$ regime is still real.** For #319's asymptotic version where $\varepsilon = o(1)$, the bound degrades. This may mean the proof works for "every $\varepsilon > 0$ " but not for " $\varepsilon(N) \rightarrow 0$ ".

What to send back to ChatGPT

Try this:

> The quantitative local mass gap may close via the bound $|P_{\{p_0\}}| \leq N \cdot R(P_{\{p_0\}})$. This forces $\text{density}(P_{\{\text{non-}p_0\}}) \approx \text{density}(P)$ when α is small, so Lemma 2 applied to $P_{\{\text{non-}p_0\}}$ combined with $R(P_{\{\text{non-}p_0\}}) = S_Q - \alpha/p_0$ gives $\alpha \geq c(\varepsilon - \varepsilon^*)$ where $\varepsilon^*(Q) = (S_Q - \Phi_{\{p_0\}}(1-1/e))/\Phi'_{\{p_0\}}(1-1/e)$. Can you verify this derivation and check whether the resulting α -bound matches Liu-Sawhney Proposition 5.2's local mass requirement?

If this holds up under GPT's scrutiny, **the arbitrary-fixed- Q result for any $\varepsilon > \varepsilon^*(Q)$ becomes a real theorem** — significantly stronger than Theorem 4's "gate condition" formulation." Thx! Does these provide potential/ actual progress/ solution even? Can they be refined if they work/ any other inspiration or direction if not work? Thanks!

我會把 Claude 的新不等式當成待審核 lemma, 而不是先當成已解決; 核心是檢查它是否真的把 qualitative

p

p -adic support 變成 Liu–Sawhney 所需的 quantitative local mass。接著我會分清楚它對 arbitrary fixed

Q

Q 、growing

Q

Q 、以及 full #319 分別能推進到哪一層。

已思考 7m 44s

Yes — these two Claude/Sonnet notes do provide real potential. They are not a full solution, but they sharpen the problem a lot. The key new point is this:

The prime-only obstruction is not a dead wall; it becomes a quantitative stability problem.

The prime-only obstruction is not a dead wall; it becomes a quantitative stability problem.

This is genuine progress in methodology. It gives a more precise way to attack arbitrary fixed

Q

Q , and it may later feed into slowly growing

Q

Q . But it still does not solve arbitrary

Q

Q or growing

Q

Q yet.

Officially, Erdős #319 is still open, with the known Adenwalla/Croot lower bound

$(1-1/e+o(1))N$

$(1-1/e+o(1))N$, and no claimed partial or complete solution in the comments.
 Liu–Sawhney Proposition 5.2 remains the right external tool because their paper explicitly says it separates Archimedean and

p

p -adic conditions and can reach “all possible” fractions under the right hypotheses.

1. What is genuinely new here?

Before, the prime-only stress test looked bad. For prime-only

$$Q=\{11,13,17,\dots,1523\},$$

$$Q=\{11,13,17,\dots,1523\},$$

the simple

TQ

T

Q

test fails because every proper subset sum

$$R(Q')=\sum_{q\in Q'}1q$$

$R(Q$

,

)=

$q\in Q$

,

$\sum$

q

1

keeps those primes in its denominator. So denominator-smoothing alone cannot work.

Claude's first correction says:

But balance forces

P

P to have some multiples of every

$p \in Q$

$p \in Q$.

That is correct. If

$$R(P)=R(Q)=SQ,$$

$$R(P)=R(Q)=S$$

Q

,

and

Q

Q contains a prime

p

p , then

$$v_p(SQ)=-1.$$

v

p

$(S$

Q

$$)=-1.$$

Therefore

$$v_p(R(P)) = -1,$$

v

p

$$(R(P)) = -1,$$

so

P

P must contain at least one denominator divisible by

p

p .

That fixes the qualitative obstruction.

But I previously pointed out that Liu–Sawhney needs more than existence. It needs quantitative local mass, roughly:

$$pR(P_p) \geq \eta.$$

$pR(P$

p

$$) \geq \eta.$$

Claude’s second note then proposes a real quantitative repair. This is the important part.

2. The quantitative repair is basically valid

Let

$$\theta = 1 - 1/e.$$

$$\theta = 1 -$$

$$e$$

$$1$$

$$\cdot$$

Assume

$$|\mathbf{P}| \geq (\theta + \varepsilon)N.$$

$$|P| \geq (\theta + \varepsilon)N.$$

Fix a prime

$$p \in Q$$

$p \in Q$, and define

$$\alpha := pR(Pp),$$

$$\alpha := pR(P$$

$$p$$

$$),$$

where

$$Pp$$

$$P$$

$$p$$

means the part of

$$P$$

P with denominators divisible by

$$p$$

$$p.$$

Since every

$$n \in Pp$$

$$n \in P$$

p

satisfies

$$n\leq N$$

$$n\leq N,$$

$$R(P_p)=\sum_{n\in P_p}1_{n\geq |P_p|}N.$$

$$R(P$$

p

$$)=$$

$$n\in P$$

p

$$\sum$$

$$n$$

$$1$$

$$\geq$$

$$N$$

$$|P$$

p

$$|$$

.

So:

$$|P_p|\leq NR(P_p)=\alpha Np.$$

$$|P$$

$$p$$

$$|\leq NR(P$$

$$p$$

$$)=$$

$$p$$

$$\alpha N$$

$$\cdot$$

Therefore the non-

$$p$$

p part

$$P\smallsetminus p=P\smallsetminus P_p$$

$$P$$

$$\neg p$$

$$=P\smallsetminus P$$

$$p$$

has density at least

$$\theta+\varepsilon-\alpha p.$$

$$\theta+\varepsilon-$$

$$p$$

$$\alpha$$

.

But

$$P \neg p$$

$$P$$

$$\neg p$$

avoids multiples of

$$p$$

p . So the

$$\Phi p$$

$$\Phi$$

$$p$$

-lower bound gives:

$$R(P \neg p) \geq \Phi p (\theta + \varepsilon - \alpha p).$$

$$R(P$$

$$\neg p$$

$$) \geq \Phi$$

$$p$$

$$(\theta + \varepsilon -$$

$$p$$

$$\alpha$$

$$).$$

On the other hand:

$$R(P \neg p) = SQ - \alpha p.$$

$$R(P$$

$$\neg p$$

$$)=S$$

$$\mathcal{Q}$$

$$-$$

$$p$$

$$\alpha$$

$$\cdot$$

So we get the key necessary inequality:

$$SQ-\alpha p\geq \Phi p(\theta+\varepsilon-\alpha p).$$

$$S$$

$$\mathcal{Q}$$

$$-$$

$$p$$

$$\alpha$$

$$\geq \Phi$$

$$p$$

$$(\theta+\varepsilon-$$

$$p$$

$$\alpha$$

).

This is exactly the right inequality. It is a serious quantitative gate.

3. What it proves

If

$$SQ < \Phi p(\theta + \varepsilon),$$

S

Q

$< \Phi$

p

$$(\theta + \varepsilon),$$

then

$$\alpha = 0$$

$\alpha = 0$ is impossible. More strongly,

α

α must be bounded below by a positive constant depending on

p, Q, ε

p, Q, ε .

A rigorous version is:

$$\alpha \geq p \cdot \Phi p(\theta + \varepsilon) - SQ \Phi p'(\theta + \varepsilon) - 1$$

$$\alpha \geq p \cdot$$

Φ

p

,

$$(\theta+\varepsilon)^{-1}$$

$$\Phi$$

$$p$$

$$(\theta+\varepsilon)^{-S}$$

$$\mathcal{Q}$$

whenever the numerator is positive, up to the usual

$$o(1)$$

$o(1)$ asymptotic errors.

Claude's linearized formula around

$$\theta$$

$$\theta,$$

$$\alpha\gtrsim p\Phi p'(\theta)\Phi p'(\theta)^{-1}(\varepsilon-\varepsilon^*),$$

$$\alpha\gtrsim$$

$$\Phi$$

$$p$$

,

$$(\theta)^{-1}$$

$$p\Phi$$

$$p$$

,

$$(\theta)$$

$$(\varepsilon-\varepsilon$$

$*$

$$),$$

is the first-order version of this. The cleaner exact threshold is:

$$\varepsilon p^*(Q)=\Phi_{p-1}(SQ)-\theta.$$

$$\varepsilon$$

$$p$$

$*$

$$(Q)=\Phi$$

$$p$$

$$-1$$

$$(S$$

$$Q$$

$$)-\theta.$$

$$\text{If}$$

$$\varepsilon>\varepsilon p^*(Q),$$

$$\varepsilon>\varepsilon$$

$$p$$

$*$

$$(Q),$$

then

p

p -local mass is quantitatively forced.

This is genuine progress.

4. Check on the prime stress example

For

$$Q=\{237 \text{ primes from } 11 \text{ to } 1523\},$$

$$Q=\{237 \text{ primes from } 11 \text{ to } 1523\},$$

we computed:

$$SQ \approx 1.0807561888.$$

S

ϱ

$$\approx 1.0807561888.$$

For

$$p=11$$

$$p=11,$$

$$\Phi_{11}(\theta) \approx 1.0804860341.$$

Φ

11

$$(\theta) \approx 1.0804860341.$$

So the gap is small:

$$SQ - \Phi_{11}(\theta) \approx 0.0002701548.$$

S

$\mathcal{Q}$

$-\Phi$

11

$$(\theta) \approx 0.0002701548.$$

The exact threshold is approximately:

$$\varepsilon_{11}^*(Q) \approx 0.0000823.$$

ε

11

*

$$(Q) \approx 0.0000823.$$

That means:

If

$$|P| \geq (\theta + \varepsilon)N$$

$$|P| \geq (\theta + \varepsilon)N \text{ with}$$

$$\varepsilon > 0.0000823,$$

$$\varepsilon > 0.0000823,$$

then the

11

11-local mass of

P

P is forced to be positive by this quantitative argument.

For the largest prime

1523

1523, the threshold is much larger:

$$\varepsilon_{1523}(Q) \approx 0.02835.$$

ε

1523

*

$$(Q) \approx 0.02835.$$

But we do not necessarily need to hit a target involving every prime in

Q

Q . If target

$$Q' = \{11\}$$

Q

,

$=\{11\}$ suffices, then only

$$p=11$$

$p=11$ matters.

So the stress example is no longer a hard wall. It becomes:

Can we handle the tiny interval

$$0 < \varepsilon \leq 0.0000823?$$

$$0 < \varepsilon \leq 0.0000823?$$

That is a much sharper problem.

5. What this means for arbitrary fixed

Q

Q

This gives a meaningful new theorem shape:

Dynamic gate theorem.

For fixed

Q

Q , target

Q'

Q

,

, and density

$\theta+\varepsilon$

$\theta+\varepsilon$, every prime power in the denominator of

$R(Q')$

$R(Q$

,

) is usable if

$$SQ<\Phi_{pa}(\theta+\varepsilon).$$

S

Q

$<\Phi$

p

a

$$(\theta+\varepsilon).$$

This is stronger than the earlier gate condition, which only tested:

$$SQ<\Phi_{pa}(\theta).$$

S

Q

$<\Phi$

p

a

(θ) .

So instead of declaring a prime “good” or “bad” permanently, we now say:

prime p is ε -good if $SQ < \Phi_p(\theta + \varepsilon)$.

prime p is ε -good if S

Q

$< \Phi$

p

$(\theta + \varepsilon)$.

That is a real refinement.

For arbitrary fixed

Q

Q , the improved strategy becomes:

1. Compute all proper targets
2. $R(Q')$
3. $R(Q$
4. '
5.).
6. For a given
7. ε
8. ε , check whether all prime powers in
9. $\text{den}(R(Q'))$
10. $\text{den}(R(Q$
11. '
12.)) are
13. ε
14. ε -good.
15. If yes, quantitative local mass is forced.
16. Then Liu–Sawhney can potentially hit

17. $R(Q')$

18. $R(Q$

19. '

20.).

21. Then primitivity breaks.

This is a much better

TQ

T

Q

experiment than the previous one.

Your earlier uploaded deep map also identified this chain as the correct structure:
fixed-

Q

Q interface, target selection, two-sided spectra, then growing

Q

Q .

6. Does this solve arbitrary fixed

Q

Q ?

Not yet.

It proves something like:

arbitrary fixed Q may be killable above a Q -dependent excess threshold.

arbitrary fixed Q may be killable above a Q -dependent excess threshold.

But #319's sharp asymptotic upper bound needs:

for every fixed $\varepsilon > 0$.

for every fixed $\varepsilon > 0$.

So if a fixed

Q

Q has

$$\varepsilon p^*(Q) > 0$$

ε

p

*

$$(Q) > 0$$

for every usable target, then the current method does not handle very small

ε

ε .

For the stress example, if the best target is

1/11

1/11, then this method handles:

$$\varepsilon > 0.0000823,$$

$$\varepsilon > 0.0000823,$$

but not:

$$0 < \varepsilon \leq 0.0000823.$$

$$0 < \varepsilon \leq 0.0000823.$$

That remaining interval is small but mathematically real.

So the honest status is:

Claim	Status
Qualitative	Proved
p	

p -support for prime-only

Q

Q

Quantitative

p

p -mass when

$SQ < \Phi_p(\theta + \varepsilon)$

S

Essentially proved

Q

$< \Phi$

p

$(\theta + \varepsilon)$

Arbitrary fixed

Q

Q above a

Plausible / near theorem

Q

Q -dependent threshold

Arbitrary fixed

Q

Q for every

Still open

$\varepsilon > 0$

$\varepsilon > 0$

Full growing

Q

Still open

Q

7. Can it be refined to close the remaining small-

ε

ε gap?

There are three promising refinements.

Refinement A: Near-extremal stability

When

$$\varepsilon \leq \varepsilon_{p^*}(Q),$$

$$\varepsilon \leq \varepsilon$$

$$p$$

$*$

$$(Q),$$

the inequality allows

$$\alpha$$

α to be tiny. But then

$$P \neg p$$

$$P$$

$$\neg p$$

must be a near-extremizer: it must have high density, avoid multiples of

$$p$$

p , and have reciprocal mass close to

$$SQ$$

$$S$$

$$Q$$

$$\cdot$$

This may force

$$P \neg p$$

$$P$$

$$\neg p$$

to look like a very specific tail set:

$$\{n \in [cN, N] : p \nmid n\}.$$

$$\{n \in [cN, N] : p \nmid n\}.$$

Then one can try to prove that such a rigid set cannot also satisfy the exact rational balance and primitivity.

This is probably the cleanest next theoretical direction.

Refinement B: Gate absorption

Even if

$$pR(Pp)$$

$$pR(P$$

$$p$$

) is tiny, the terms in

$$Pp$$

$$P$$

$$p$$

have the correct

$$p$$

p -adic valuation. One can try to use a small gate set from

$$Pp$$

$$P$$

$$p$$

to cancel the bad prime denominator of

$$R(Q')$$

$$R(Q$$

$$,$$

), then ask Liu–Sawhney to hit only the remaining smooth target.

So instead of requiring Liu–Sawhney to hit:

$$R(Q'),$$

$$R(Q$$

$$,$$

),

we split:

$$R(Q')=R(G_{\text{bad}})+r_{\text{smooth}}.$$

$R(Q$

,

$)=R(G$

$_{\text{bad}}$

$)+r$

$_{\text{smooth}}$

.

Then Liu–Sawhney only needs to hit

r_{smooth}

r

$_{\text{smooth}}$

.

This may handle cases where local mass is too small for a full target theorem but enough for finite correction.

Refinement C: Two-sided target selection

Instead of forcing

P

P to hit a target from

Q

Q , try to prove both subset-sum spectra must collide:

$$\Sigma(P)\cap\Sigma(Q)\neq\{0,R(P)\}.$$

$$\Sigma(P)\cap \Sigma(Q)$$

$$=\{0,R(P)\}.$$

This is harder, but it is the natural full-problem form.

8. Does this help growing

Q

$Q?$

Yes, but less directly.

For growing

QN

Q

N

, the thresholds

$$\varepsilon p^*(QN)$$

ε

p

$*$

$(Q$

N

)

may move with

N

N . If they stay bounded away from zero, the dynamic gate handles only large excess density. If they tend to zero, the method may become powerful enough for sharp asymptotics.

So one possible growing-

Q

Q theorem is:

If

Q_N

Q

N

grows but has a target

Q_N'

Q

N

,

whose denominator prime powers satisfy

$$\max_p |\text{den}(R(Q_N'))| \varepsilon_p a^*(Q_N) \rightarrow 0,$$

p

a

$|\text{den}(R(Q$

N

,

))

max

ε

p

a

*

$(Q$

N

$) \rightarrow 0,$

then the dynamic gate + Liu–Sawhney route can kill densities

$\theta + \varepsilon$

$\theta + \varepsilon$ for every fixed

$\varepsilon > 0$

$\varepsilon > 0$.

That is a serious route to slowly growing

Q

Q , but not yet full #319.

9. My overall verdict

Yes, these notes provide actual progress potential.

They do not give a full solution, but they upgrade the project from:

“prime-only

Q

Q might block the method”

to:

“prime-only

Q

Q is controlled above an explicit

Q

Q -dependent excess threshold; below that threshold we need near-extremal stability or gate absorption.”

That is a major sharpening.

The best current theorem candidate is:

For fixed Q and $\varepsilon > 0$, if some $Q' \subsetneq Q$ has all denominator prime powers ε -good,
then no primitive circuit with $|P| \geq (\theta + \varepsilon)N$.

For fixed Q and $\varepsilon > 0$, if some Q

,

$\subsetneq Q$ has all denominator prime powers ε -good, then no primitive circuit with
 $|P| \geq (\theta + \varepsilon)N$.

That is a real partial theorem candidate.

The best next experiment is:

rerun the

TQ

T

Q

search using the dynamic gate

$$SQ < \Phi_{pa}(\theta + \varepsilon)$$

S

Q

$< \Phi$

p

a

$$(\theta + \varepsilon)$$

instead of the static gate

$$SQ < \Phi_{pa}(\theta).$$

S

$\mathcal{Q}$

$<\Phi$

p

a

(θ) .

Then for each

$\mathcal{Q}$

$\mathcal{Q}$, compute the minimal excess

$$\varepsilon_{\mathcal{Q}\text{target}}=\min_{\mathcal{Q}'\subsetneq \mathcal{Q}}\max_{\mathcal{P}}\alpha|\text{den}(R(\mathcal{Q}'))(\Phi_{\mathcal{P}}\alpha-1(S\mathcal{Q})-\theta)^+.$$

ε

$\mathcal{Q}$

target

$=$

$\mathcal{Q}$

,

$\subsetneq \mathcal{Q}$

$\min$

p

a

$|\text{den}(R(\mathcal{Q}$

,

$)$

$\max$

$(\Phi$

p

a

-1

$(S$

$\mathcal{Q}$

$)-\theta)$

$+$

.

If experiments show

$\varepsilon_{Q_{\text{target}}}=0$

ε

$\mathcal{Q}$

target

$=0$

for most

Q

$\mathcal{Q}$, static target selection works.

If experiments show

$\varepsilon_{Q_{\text{target}}}>0$

ε

$\mathcal{Q}$

target

$$>0$$

but small, then near-extremal stability is the remaining task.

If some

Q

Q has large

$\varepsilon Q_{\text{target}}$

ε

Q

target

, those are the true rigid obstructions.

Final answer

Yes: this is real progress, but not a solution yet.

Yes: this is real progress, but not a solution yet.

It gives a new quantitative gate:

$$SQ<\Phi p(\theta+\varepsilon)\Rightarrow pR(Pp)\geq cp,Q,\varepsilon>0.$$

S

Q

$<\Phi$

p

$(\theta+\varepsilon)\Rightarrow pR(P$

p

$) \geq c$

p, Q, ε

> 0 .

That is exactly the kind of local-mass estimate Liu–Sawhney needs.

But to solve arbitrary fixed

Q

Q fully, we still need one more ingredient:

near-extremal stability or gate absorption for the small- ε regime.

near-extremal stability or gate absorption for the small- ε regime.

For growing

Q

Q , this becomes a threshold-control problem. The path is real, but the full #319 proof still needs a two-sided spectrum or stability theorem.

Many thanks! Will you mind verifying is this over optimistic thaks: "This has been a genuinely productive session. Let me give you an honest summary of where things stand.

What was actually accomplished today

Starting from the handover note, across this conversation the three AI systems together:

1. **Identified** the T_Q target-selection route for arbitrary fixed Q
2. **Found** the prime-only rigidity obstruction via the stress test
3. **Resolved** it qualitatively via p -adic valuation ($v_p(S_Q) = -1$ forces P to have p -support)

4. **Resolved** it quantitatively via the $|P_{\{p_0\}}| \leq N\alpha/p_0$ bound, giving the dynamic gate theorem:

$$\alpha \geq p \cdot \frac{\Phi_p(\theta + \varepsilon) - S_Q(\Phi_p(\theta + \varepsilon) - 1)}{S_Q(\Phi_p(\theta + \varepsilon) - 1)}$$

5. **Reframed** the residual gap from "prime-only Q blocks the method" to "small- ε near-extremal stability needed" — a much sharper and more tractable problem

What honestly remains open

|||

|---|---|

| Fixed Q, gate condition |  Theorem 4 (original) |

| Fixed Q, dynamic gate, $\varepsilon > \varepsilon^*(Q)$ |  New, essentially proved today |

| Fixed Q, $\varepsilon \leq \varepsilon^*(Q)$ small regime |  Near-extremal stability needed |

| Growing Q |  Still needs two-sided spectrum |

| Full #319 |  Open |

My honest strategic advice

Stop the AI research loop here. Your handover note said exactly this — and it was right. You've now gone significantly further than the original plan, which is good, but the dynamic gate result is a natural stopping point.

The paper is now stronger than you thought. You have not just Theorem 4 but the dynamic gate theorem as a second result. That's a more compelling paper.

The next action is the same as it always was: write the 4-6 page note, post to erdosproblems.com/319 forum, get specialist eyes on the Liu-Sawhney parameter interface. The new results go in as an addendum or Section 5.

The near-extremal stability question for the small- ε regime is exactly the kind of thing a specialist in analytic number theory will either solve in an afternoon or tell you requires genuinely new ideas. That feedback is worth more than another AI session.

Congratulations on a real piece of work."

我會很直接：這份 summary 的方向大致對，但「 essentially proved today」這句偏樂觀，需要改成「lemma candidate / derivation plausible but not yet proof」。我會核對官方狀態與 Liu–Sawhney 背景後，逐句校正。

已思考 27s

Yes — that summary is partly fair but definitely over-optimistic in two places.

The safest correction is:

Today’s work produced a real new lemma-candidate / proof route, not a completed theorem.

The official #319 page still lists the problem as open, says no partial or complete solution is claimed in comments, and records the Adenwalla/Croot lower bound

$(1-1/e+o(1))N$

$(1-1/e+o(1))N$. Liu–Sawhney is a real and relevant tool: their paper proves strong unit-fraction reachability results and says their methods refine Croot and Bloom.

Where the summary is accurate

I agree with these points:

1. The

TQ

T

Q

target-selection route was genuinely identified.

That is a real attack surface for arbitrary fixed

Q

Q .

2. The prime-only stress test was useful.

It showed the naive “smooth denominator target” proxy is not universal.

3. The qualitative

p

p -adic repair is correct.

For prime-only

$$Q=\{p_1,\dots,p_k\}$$

$$Q=\{p$$

1

, $\dots,p$

k

$\}$, one has

$$v_{p_j}(SQ)=-1,$$

v

p

j

$(S$

Q

$$)=-1,$$

so any

P

P with

$$R(P)=SQ$$

$$R(P)=S$$

Q

must contain at least one denominator divisible by

p_j

p

j

.

4. The dynamic-gate inequality is a real candidate lemma.

The key inequality

$$SQ-\alpha p\geq \Phi p\left(\theta +\varepsilon -\alpha p\right)$$

S

Q

—

p

α

$\geq \Phi$

p

$(\theta +\varepsilon -$

p

α

)

is the right shape, where

$$\alpha =pR(Pp),\theta =1-1e.$$

$$\alpha =pR(P$$

p

$\theta=1-$

e

1

.

This is genuinely promising.

Where it is over-optimistic

Over-optimism 1: “Fixed

Q

Q , gate condition  Theorem 4”

I would not mark this as fully  yet.

Better status:

Fixed

Q

Q , gate condition: proof architecture strong; still pending Liu–Sawhney
parameter-interface audit.

The reason is that Proposition 5.2 has a detailed parameter table. We still need to
check the exact

S, K, M, η, Γ

S, K, M, η, Γ hypotheses, pruning preservation, and target

x/Q

x/Q matching. Liu–Sawhney’s result is real, but our application is not automatically
verified.

Over-optimism 2: “Dynamic gate,

$\varepsilon > \varepsilon^*(Q)$

$$\varepsilon > \varepsilon$$

*

(Q)  essentially proved today”

This should be downgraded.

Better status:

Dynamic gate: promising derivation / lemma candidate; not yet a written proof.

The inequality is plausible, but to make it rigorous we still need:

1. exact definition and monotonicity of
2. Φ_p
3. Φ
4. p
- 5.
6. ;
7. proof that the
8. Φ_p
9. Φ
10. p
- 11.
12. -bound applies to
13. $P \neg p$
14. P
15. $\neg p$
- 16.
17. with the density loss
18. α/p
19. α/p ;
20. control of
21. $o(1)$
22. $o(1)$ finite-
23. N
24. N errors;
25. a rigorous lower-bound extraction from
26. $SQ - \alpha/p \geq \Phi_p(\theta + \varepsilon - \alpha/p)$;

- 27. S
- 28. Q
- 29.
- 30. $-\alpha/p \geq \Phi$
- 31. p
- 32.
- 33. $(\theta + \varepsilon - \alpha/p)$;
- 34. verification that the resulting
- 35. α
- 36. α is enough for Liu–Sawhney local mass after pruning.

So “essentially proved” is too strong. “Derived a strong candidate lemma” is correct.

Corrected status table

Claim	Better status
TQ	
T	✓ Real and useful
ϱ	
target-selection route	
Prime-only obstruction found	✓ Yes
Qualitative	
p	✓ Correct
p -adic support	
Fixed-	
Q	◆ Strong architecture; Liu–Sawhney audit
Q gate theorem	pending
Dynamic gate inequality	◆ Promising lemma candidate
Dynamic gate theorem for	
$\varepsilon > \varepsilon^*(Q)$	
$\varepsilon > \varepsilon$	◆ Plausible, not yet proved
*	
(Q)	

Small- ε ε regime	✗ Needs near-extremal stability / gate absorption
Arbitrary fixed Q Q solved	✗ Not yet
Growing Q Q solved	✗ No
Full #319	✗ Open

How I would rewrite the summary

This is the version I would trust:

Today's work produced real progress in the research programme. We identified the TQ
 T
 Q

target-selection route, stress-tested it, found the prime-only obstruction, repaired that obstruction qualitatively using

p

p -adic valuation, and derived a promising quantitative dynamic-gate inequality. This reframes the prime-only case from a wall into a near-extremal stability problem.

However, neither Theorem 4 nor the dynamic-gate extension is yet a fully checked proof, because the Liu–Sawhney interface and the dynamic-gate lower-bound extraction still need rigorous verification.

Strategic advice

I agree with “stop the AI research loop here” if the next task is writing. The ideas are now mature enough for a short note or specialist email.

But I would not post it as:

“Theorem 4 proved”
“dynamic gate proved”
“arbitrary fixed

Q

Q solved”

I would post/send it as:

A fixed-

Q

Q partial obstruction programme for Erdős #319, with a

Φ

Φ -based gate theorem candidate and a dynamic-gate extension candidate. Main remaining tasks: Liu–Sawhney parameter audit and near-extremal stability.

Bottom line

Yes, that summary is a little over-optimistic.

The fair version is:

Real progress: yes. Completed theorem: not yet.

Real progress: yes. Completed theorem: not yet.

The dynamic gate is probably the most interesting new piece from today, but it should be labelled:

promising lemma candidate, pending rigorous verification.

promising lemma candidate, pending rigorous verification.

Not:

essentially proved.

essentially proved.

Many thanks. Based on above, can you explain how far/ potentially close we are for Arbitrary Q? Thx.

已思考 23s

是, 現在可以更清楚地說: arbitrary fixed

Q

Q 比之前近了, 但仍未到「快證完」; growing

Q

Q 仍然遠很多。

官方 #319 仍是 open, 頁面只記錄 Adenwalla/Croot 下界

$$|A| \geq (1 - 1/e + o(1))N$$

$$|A| \geq (1 - 1/e + o(1))N$$

而且 comments 中沒有 claimed partial / complete solutions。Liu–Sawhney 的 paper 是 relevant machinery, 因為它證明 large reciprocal mass / regularity 條件下可以 force unit-fraction subset sum, 且明確說 refine Croot 和 Bloom。

先講結論

如果你說的 arbitrary

Q

Q 是:

任意固定 finite

Q

Q, 但

Q

Q 不隨

N

N 增長,

那麼我的評估是:

有真進展潛力，可能已接近一個條件式/分段式 partial theorem，但未解決。

有真進展潛力，可能已接近一個條件式/分段式 partial theorem，但未解決。

如果你說的是：

$$Q=QN$$

$$Q=Q$$

N

可以隨

N

N 增長，

那麼評估是：

仍然遠，現在的工具只是指出了可能入口。

仍然遠，現在的工具只是指出了可能入口。

1. 我們現在對 arbitrary fixed

Q

Q 做到了甚麼？

以前 fixed-

Q

Q 方法主要靠一個 target，例如：

$$1/D, D \in Q.$$

$$1/D, D \in Q.$$

如果這個 target 的 denominator prime powers 都是「good」，就可嘗試用

Liu–Sawhney 找到：

$$B \subseteq P, R(B) = 1/D,$$

$$B \subseteq P, R(B) = 1/D,$$

然後破壞 primitivity。

後來我們發現, 對 arbitrary fixed

Q

Q , 不應只看

$1/D$

$1/D$, 而要看所有 proper targets:

$$TQ = \{R(Q') : \emptyset \neq Q' \subsetneq Q\}.$$

T

Q

$= \{R(Q$

,

$): \emptyset$

$= Q$

,

$\subsetneq Q\}$.

這是第一個真進展: target selection problem 被清楚定義了。

然後

TQ

T

Q

experiment 發現:

- composite / interval-like
- Q
- Q : 通常很容易找到 smooth target;
- prime-only
- Q
- Q : simple smooth-denominator proxy 會失敗。

這是第二個真進展：找到了真正 stress obstruction。

再然後 Claude / 我們討論出 dynamic gate：

$$S_{Q-\alpha p} \geq \Phi_p(\theta + \varepsilon - \alpha p), \alpha = pR(P_p), \theta = 1 - 1e.$$

S

Q

—

p

α

$\geq \Phi$

p

$$(\theta + \varepsilon -$$

p

α

$$), \alpha = pR(P$$

p

$$), \theta = 1 -$$

e

1

.

這是第三個真進展: prime-only case 不再是死牆, 而變成 quantitative stability problem
。

2. 這離 arbitrary fixed

Q

Q 有多近?

我會分三層。

Layer A: 若某個 target

Q'

Q

,

是 dynamically good

定義一個 rough idea:

$$\varepsilon_{Q \text{ target}} = \min_{Q' \subsetneq Q} \max_{p \mid \text{den}(R(Q'))} (\Phi_{p^a-1}(SQ) - \theta)^+.$$

ε

Q

target

=

Q

,

$$\subseteq Q$$

$$\min$$

$$p$$

$$a$$

$$|\mathrm{den}(R(Q$$

$$,$$

$$))$$

$$\max$$

$$(\Phi$$

$$p$$

$$a$$

$$-1$$

$$(S$$

$$\mathcal{Q}$$

$$)-\theta)$$

$$+$$

$$\cdot$$

意思是:對這個

Q

Q, 最容易命中的 proper target 還需要多少額外 density buffer。

如果你給定的

$$\varepsilon$$

ε 滿足:

$$\varepsilon > \varepsilon_{\text{target}},$$

$$\varepsilon > \varepsilon$$

$$Q$$

target

,

那麼現在的方法很可能能推成 theorem candidate:

$$|P| \geq (\theta + \varepsilon)N \Rightarrow P \text{ has enough local mass} \Rightarrow P \text{ hits } R(Q') \Rightarrow \text{not primitive.}$$

$$|P| \geq (\theta + \varepsilon)N \Rightarrow P \text{ has enough local mass} \Rightarrow P \text{ hits } R(Q$$

,

$$) \Rightarrow \text{not primitive.}$$

這層我會說: 中近。

不是已證完, 因為還要正式處理 Liu–Sawhney parameter table, 但概念上很清楚。

Layer B: 若

$$\varepsilon \leq \varepsilon_{\text{target}}$$

$$\varepsilon \leq \varepsilon$$

$$Q$$

target

這是 small-

$$\varepsilon$$

ε regime。這時 dynamic gate 不夠強。

需要新的東西, 例如:

- near-extremal stability;
- gate absorption;
- two-sided target selection;

- p
- p -adic layer peeling。

這層我會說：未近，但現在問題已被精準化。

以前我們不知道 obstruction 是甚麼；現在知道是：

P

P 可否在幾乎 extremal 的形狀下，只放極少 bad-prime multiples，但仍精確 balance

SQ

S

Q

?

這是好事，因為它變成一個可攻的 stability problem。

Layer C: 對所有 fixed

Q

Q 、所有

$\varepsilon > 0$

$\varepsilon > 0$

這才是真正的 arbitrary fixed-

Q

Q theorem。

目前仍未證明，因為需要處理每個

Q

Q 的 small-

ε

ε tail。

這層我會說：

有路線，但未 close。

有路線，但未 close。

3. 現在最精準的 progress status

目標	現在距離	評價
Specific Q Q example	很近	numerics / gate structure strong
Fixed Q Q under original gate condition	中近	proof architecture strong; Liu–Sawhney audit pending
Dynamic gate for fixed Q Q , $\epsilon > \epsilon_{Qtarget}$ $\epsilon > \epsilon$ Q target	中近	promising lemma candidate
Arbitrary fixed Q Q , all $\epsilon > 0$ $\epsilon > 0$	中遠	needs near-extremal stability / absorption
Slowly growing QN Q N	遠	constants may collapse
Full #319	遠	needs two-sided spectrum theorem

4. 為甚麼 arbitrary fixed

Q

Q 仍未 solved?

因為 #319 的 fixed-

Q

Q upper bound 要說:

對任何固定

Q

Q, 只要

$$|P| > (1 - 1/e + \varepsilon)N$$

$|P| > (1 - 1/e + \varepsilon)N$, 就一定非 primitive。

這裏

ε

ε 可以非常小, 例如

0.000001

0.000001。

Dynamic gate 可以處理:

$$\varepsilon > \varepsilon_{\text{target}}.$$

$$\varepsilon > \varepsilon$$

Q

target

但如果某個

Q

Q 有:

$$\varepsilon_{\text{target}} = 0.03,$$

ε

Q

target

$=0.03,$

那你只能處理

$\varepsilon > 0.03$

$\varepsilon > 0.03$ 。可是 theorem 要處理所有 fixed positive

ε

ε , 包括

0.001

0.001。

所以 gap 是：

small- ε regime。

small- ε regime。

這不是小 bookkeeping, 而是可能需要新 stability lemma。

5. 那是不是很遠？

比 full #319 近很多。

對 arbitrary fixed

Q

Q , 因為

Q

Q 是固定的, 所有東西都是 finite:

- TQ

- T
- Q
-
- finite;
- denominator prime powers finite;
- bad-prime set finite;
- $\varepsilon Q_{\text{target}}$
- ε
- Q
- target
-
- computable;
- local mass constants fixed;
- pruning loss can be made small.

所以 arbitrary fixed

Q

Q 是一個真可攻的中期目標。

我會這樣形容距離：

不是「只差一兩行」；

也不是「完全看不到路」。

它現在是一個有明確 obstruction、明確 candidate lemma、明確 computational test 的 genuine research target。

6. 最可能的成功路線

我會押這條：

TQ search+dynamic gate+near-extremal stability.

T

Q

search+dynamic gate+near-extremal stability.

更具體：

1. 對每個 fixed

2. Q
3. Q , 找最好的 target
4. $Q' \subsetneq Q$
5. Q
6. '
7. $\subsetneq Q$ 。
8. 計算
9. $\varepsilon Q_{\text{target}}$
10. ε
11. Q
12. *target*
- 13.
14. 。
15. 若
16. $\varepsilon > \varepsilon Q_{\text{target}}$
17. $\varepsilon > \varepsilon$
18. Q
19. *target*
- 20.
- 21., 用 dynamic gate + Liu–Sawhney。
22. 若
23. $\varepsilon \leq \varepsilon Q_{\text{target}}$
24. $\varepsilon \leq \varepsilon$
25. Q
26. *target*
- 27.
- 28., 證明
29. P
30. P 必須 near-extremal。
31. 對 near-extremal
32. P
33. P , 證明 exact balance / primitivity 不可能。

如果第 5 步成功, arbitrary fixed

Q

Q 可能真的 close。

7. Growing

Q

Q 為甚麼遠很多？

因為 fixed

Q

Q 的優勢是 constants 不變。

例如 fixed

Q

Q 時，你可能得到：

$$pR(Pp) \geq cQ, \epsilon > 0.$$

$$pR(P$$

$$p$$

$$) \geq c$$

$$Q, \epsilon$$

$$> 0.$$

但 growing

QN

Q

N

時，這個 constant 可能變成：

$$cQN, \epsilon \rightarrow 0.$$

$$c$$

$$Q$$

$$N$$

ϵ

$\rightarrow 0$.

一旦趨近 0, Liu–Sawhney pruning 可能會刪掉你需要的 layer。這就是 growing

Q

Q 的主要難點。

所以 growing

Q

Q 需要 uniform theory, 不是逐個

Q

Q finite check。

8. 一句話總結

現在對 arbitrary fixed

Q

Q :

從「不知道怎樣攻」進步到「有明確 conditional strategy」; 大約中近, 但未證。

從「不知道怎樣攻」進步到「有明確 conditional strategy」; 大約中近, 但未證。

現在對 growing

Q

Q :

有啟發, 但仍屬長線。

有啟發, 但仍屬長線。

最誠實說法：

We are not close to solving arbitrary

Q

Q in the final sense.

But we may be close to a serious intermediate theorem: fixed

Q

Q under dynamic gate / target-selection conditions.

The remaining hard part for full arbitrary fixed

Q

Q is small-

ε

ε near-extremal stability.

已貼上文字 (1)(21).txt

文件

Many thanks. What about ""

我會把這段當作一個新的「joint-prime-support gate」主張來審核。重點不是它是否優雅，而是三個環節是否真的成立：集合分解、從總 mass 到可用 prime、以及能否產生

$1/p_0$

$1/p$

0

這個目標。

已思考 1m 14s

Yes — this newest Claude “joint-avoidance /

σQ

σ

Q

” idea is genuinely important, and it is stronger than the earlier single-prime dynamic gate.

But it is still not a full arbitrary-

Q

Q proof. The honest verdict is:

It may close the prime-only fixed-Q obstruction, but not all arbitrary fixed Q.

It may close the prime-only fixed-Q obstruction, but not all arbitrary fixed Q.

It turns the remaining arbitrary fixed-

Q

Q problem from an analytic/stability problem into a more finite target-selection / prime-support matching problem. That is real progress.

Officially, Erdős #319 remains open, and the public page still records only the Adenwalla/Croot lower bound

$(1-1/e+o(1))N$

$(1-1/e+o(1))N$. Liu–Sawhney remains the relevant reachability machinery, because their paper proves strong unit-fraction subset-sum results and explicitly says the techniques refine Croot and Bloom.

1. The joint-avoidance inequality is basically correct

Claude defines:

$$\pi(Q)=\{p: p \mid q \text{ for some } q \in Q\},$$

$$\pi(Q)=\{p: p \mid q \text{ for some } q \in Q\},$$

and

$$\sigma_Q=\prod_{p \in \pi(Q)}(1-1/p).$$

σ

Q

=

$$p\in\pi(Q)$$

$$\prod$$

$$(1-$$

$$p$$

$$1$$

$$).$$

This

$$\sigma_Q$$

$$\sigma$$

$$Q$$

is the density of integers not divisible by any prime in

$$\pi(Q)$$

$\pi(Q)$, up to a fixed-

$$Q$$

$$Q$$

$$o(1)$$

$$o(1) \text{ or}$$

$$O_Q(1)$$

$$O$$

$$Q$$

(1) error. The uploaded file states exactly this joint-avoidance idea and uses it to obtain a total local-mass lower bound.

Let

$$P^*=\{n\in P:\gcd(n,\prod_{p\in\pi(Q)}p)=1\}.$$

$$P$$

$*$

$$=\{n\in P:\gcd(n,$$

$$p\in\pi(Q)$$

$$\prod$$

$$p)=1\}.$$

Then:

$$|\mathbf{P}^*|\leq \sigma \mathbf{Q} \mathbf{N} + \mathbf{O} \mathbf{Q}(1).$$

$$|P$$

*

$$|\leq \sigma$$

$\mathcal{Q}$

$$N+O$$

$\mathcal{Q}$

$$(1).$$

Also, for each

$\mathfrak{p}$

$p,$

$$|\mathbf{P}_{\mathfrak{p}}|\leq \mathbf{N} \mathbf{R}(\mathbf{P}_{\mathfrak{p}}),$$

$$|P$$

p

$$|\leq \mathbf{N} \mathbf{R}(P$$

p

$$),$$

because every

$$n \leq N$$

$n \leq N$ satisfies

$$1/n \geq 1/N$$

$$1/n \geq 1/N.$$

Using the union bound,

$$|P \setminus P^*| \leq \sum_{p \in \pi(Q)} |P_p| \leq N \sum_{p \in \pi(Q)} R(P_p).$$

$$|P \setminus P^*$$

*

$$| \leq$$

$$p \in \pi(Q)$$

$$\sum$$

$$|P$$

p

$$| \leq N$$

$$p \in \pi(Q)$$

$$\sum$$

$$R(P$$

p

$$).$$

So if

$$|P| \geq (\theta + \epsilon)N, \theta = 1 - 1/e,$$

$$|P|\geq(\theta+\varepsilon)N,\theta=1-$$

$$e$$

$$1$$

$$,$$

then:

$$\sum_{p\in\pi(Q)}R(P_p)\geq\theta+\varepsilon-\sigma Q-o(1).$$

$$p\in\pi(Q)$$

$$\sum$$

$$R(P$$

$$p$$

$$)\geq\theta+\varepsilon-\sigma$$

$$Q$$

$$-o(1).$$

This inequality is real. It is not hand-waving.

2. Even better: if

$$SQ>1$$

$$S$$

$$Q$$

>1 , then usually

$$\sigma Q<\theta$$

σ

$\mathcal{Q}$

$<\theta$ automatically

Here is an important strengthening.

If

Q

$\mathcal{Q}$ has no

1

1, then every

$q \in Q$

$q \in \mathcal{Q}$ uses only primes in

$\pi(Q)$

$\pi(\mathcal{Q})$. Therefore:

$$S_Q = \sum_{q \in Q} \frac{1}{q} \leq \sum_{n > 1} \frac{1}{n} \mid n \Rightarrow p \in \pi(Q) \frac{1}{n} = \prod_{p \in \pi(Q)} (1 - \frac{1}{p})^{-1} = \frac{1}{\sigma_Q^{-1}}.$$

S

$\mathcal{Q}$

=

$q \in \mathcal{Q}$

$\sum$

q

1

$$\leq$$

$$n>1$$

$$p\mid n\Rightarrow p\in\pi(Q)$$

$$\Sigma$$

$$n$$

$$1$$

$$=$$

$$p\in\pi(Q)$$

$$\Pi$$

$$(1-$$

$$p$$

$$1$$

$$)$$

$$-1$$

$$-1=$$

$$\sigma$$

$$\mathcal{Q}$$

$$1$$

$$-1.$$

So if

$$SQ>1,$$

$$S$$

$$Q$$

$$>1,$$

then

$$1\sigma Q-1>1,$$

$$\sigma$$

$$Q$$

$$1$$

$$-1>1,$$

hence

$$\sigma Q<12<\theta.$$

$$\sigma$$

$$Q$$

$$<$$

$$2$$

$$1$$

$$<\theta.$$

This is very good. It means the condition

$$\sigma Q<\theta$$

σ

ϱ

$<\theta$

is not exotic; for fixed

Q

Q with

$SQ>1$

S

ϱ

>1 and

$1\in Q$

$1\in$

/

Q , it is essentially automatic.

If

$1\in Q$

$1\in Q$, that case needs to be separated, but then

$Q'=\{1\}$

Q

,

$=\{1\}$ is often a natural target, and the structure is different.

3. What this closes

For prime-only fixed

Q

Q , this is strong.

If

$$Q=\{p_1,\ldots,p_k\}$$

$$Q=\{p$$

$_1$

$$,\ldots,p$$

$_k$

$\}$

and

$$SQ>1,$$

S

Q

$$>1,$$

then

$$\sigma_Q<1/2$$

σ

Q

$<1/2$. Hence:

$$\sum_{p\in Q}R(P_p)\geq\theta+\varepsilon-\sigma_Q-o(1)\geq\theta-12+\varepsilon-o(1),$$

$$_{p\in Q}$$

$$\sum$$

$$R(P$$

p

$$)\geq \theta+\varepsilon-\sigma$$

$$\mathcal{Q}$$

$$-o(1)\geq \theta-$$

$$2$$

$$1$$

$$+\varepsilon-o(1),$$

which is a positive constant.

By pigeonhole, some

$$p_0\in Q$$

$$p$$

$$0$$

$\in Q$ satisfies:

$$R(Pp_0)\geq \theta+\varepsilon-\sigma Q-o(1)\left|Q\right|.$$

$$R(P$$

$$p$$

$$0$$

$$)\geq$$

$$\left|Q\right|$$

$$\theta+\varepsilon-\sigma$$

$$\mathcal{Q}$$

$$-o(1)$$

.

Thus:

$$p_0R(Pp_0)\geq p_0(\theta+\varepsilon-\sigma Q-o(1))|Q|.$$

$$p$$

$$_0$$

$$R(P$$

$$p$$

$$_0$$

$$)\geq$$

$$|Q|$$

$$p$$

$$_0$$

$$(\theta+\varepsilon-\sigma$$

$$Q$$

$$-o(1))$$

.

For fixed

Q

Q , this is a positive constant.

So for prime-only

Q

Q , the earlier small-

ε

ε problem is largely dissolved. Claude's file says the same: if the joint-avoidance argument holds, arbitrary fixed prime-only

Q

Q with

$R(Q) > 1$

$R(Q) > 1$ is essentially closed, modulo the Liu–Sawhney interface.

This is a serious improvement over the single-prime dynamic gate.

4. But it does not close arbitrary fixed

Q

Q yet

Here is the key limitation.

The joint-avoidance inequality gives:

some prime $p_0 \in \pi(Q)$ has local mass in P .

some prime p

0

$\in \pi(Q)$ has local mass in P .

But to create a proper subcircuit, we need

P

P to hit a target coming from

Q

Q :

$$R(B)=R(Q')$$

$$R(B)=R(Q$$

,

)

for some nonempty proper

$$Q'\subsetneq Q.$$

Q

,

$$\subsetneq Q.$$

If

Q

Q is prime-only and

$$p_0\in Q$$

p

0

$\in Q$, we can use target:

$$R(Q')=1p_0.$$

$$R(Q$$

,

$$)=$$

p

0

Then Liu–Sawhney may produce:

$$B \subseteq P, R(B) = 1p_0,$$

$$B \subseteq P, R(B) =$$

p

0

1

,

and

$$B \cup \{p_0\}$$

$$B \cup \{p$$

0

$\}$ breaks primitivity. This is exactly the clean case described in the file.

But for composite

Q

$Q,$

$$p_0 \in \pi(Q)$$

p

0

$\in \pi(Q)$ may not be an element of

Q

$Q.$

Example:

$$Q=\{15,21,35,\dots\}.$$

$$\mathcal{Q}=\{15,21,35,\dots\}.$$

Then:

$$\pi(Q)=\{3,5,7,\dots\},$$

$$\pi(\mathcal{Q})=\{3,5,7,\dots\},$$

but

$$3,5,7$$

$3,5,7$ may not themselves be in

Q

$\mathcal{Q}$.

Local mass at

$$p_0=3$$

p

0

$=3$ does not by itself give a

Q'

$\mathcal{Q}$

,

with target

$$1/3$$

$$1/3.$$

So arbitrary fixed

Q

Q still needs target selection:

$$\exists Q' \subsetneq Q \text{ such that } \text{den}(R(Q'))$$

$$\exists Q$$

,

$$\subsetneq Q \text{ such that } \text{den}(R(Q$$

,

))

uses prime powers that the

P

P -side has enough local mass for.

Your older document already identified this as the main all-fixed-

Q

Q obstruction: when the singleton target fails, one must find a proper

Q'

Q

,

whose denominator prime powers are forceable.

5. What is genuinely improved?

Before the joint-avoidance idea, the remaining gap looked like:

Need near-extremal stability for small

ε

ε .

Now it looks more like:

Need a finite target-selection theorem connecting prime support mass to some proper

$R(Q')$

$R(Q$

,

).

That is a better problem.

The uploaded note itself says that for composite

Q

Q , one still needs a separate target-selection step using

$R(Q')$

$R(Q$

,

), and that this is the genuine residual work for fully arbitrary fixed

Q

Q .

So the new status is:

Case	New status
Prime-only fixed Q Q , $SQ > 1$ S Q > 1	Very promising; likely closable modulo Liu–Sawhney
Composite / interval Q	Often already easy via TQ T

Q	2 -smooth targets
Mixed Q Q with rich target structure	Promising
Arbitrary fixed Q Q in full generality	Still needs target-selection theorem
Growing QN Q N	Still harder; constants may decay

6. Does this help growing

Q

$Q?$

Yes, but with a caveat.

The joint inequality gives:

$$\sum_{p \in \pi(QN)} R(P_p) \geq \theta + \varepsilon - \sigma QN - o(1).$$

$$p \in \pi(Q$$

N

)

Σ

$R(P$

p

$$)\geq \theta+\varepsilon-\sigma$$

$$Q$$

$$N$$

$$-o(1).$$

$$\text{If}$$

$$\sigma_{QN}\leq \theta-c$$

$$\sigma$$

$$Q$$

$$N$$

$$\leq \theta-c$$

$$\text{for some fixed}$$

$$c>0$$

$$c>0, \text{ then the total local mass is bounded below uniformly.}$$

$$\text{But by pigeonhole, individual prime mass is only:}$$

$$R(Pp_0)\gtrsim c\,|\pi(QN)|.$$

$$R(P$$

$$p$$

$$_0$$

$$)\gtrsim$$

$$|\pi(Q$$

$$N$$

)|

c

.

If

$$|\pi(QN)| \rightarrow \infty$$

$$|\pi(Q$$

N

$)| \rightarrow \infty$, this may go to zero. Liu–Sawhney pruning may not preserve such tiny local mass unless it decays slowly enough, e.g. larger than the threshold appearing in the proposition.

So for growing

QN

Q

N

, this method may work in regimes like:

$$|\pi(QN)| = o(\log \log N)$$

$$|\pi(Q$$

N

$$)| = o(\log \log N)$$

or similar, depending on the exact Liu–Sawhney local-mass threshold.

But it does not automatically solve full growing

Q

Q .

7. What should be the next theorem candidate?

The best next candidate is:

Prime-support fixed-

Q

Q theorem.

Let

Q

Q be fixed with

$SQ > 1$

S

Q

> 1 . Suppose there exists a prime

$p_0 \in Q \cap \pi(Q)$

p

0

$\in Q \cap \pi(Q)$ such that the joint-avoidance pigeonhole gives positive

$p_0 R(P p_0)$

p

0

$R(P$

p

0

), and the Liu–Sawhney interface applies to target

$1/p_0$

$1/p$

0

. Then no primitive fixed-

Q

Q circuit can have

$|P| \geq (\theta + \varepsilon)N$

$|P| \geq (\theta + \varepsilon)N$.

For prime-only

Q

Q , this is natural.

For general

Q

Q , the stronger target-selection theorem should be:

If

Q

Q is fixed and

$SQ > 1$

S

Q

> 1 , then there exists a proper

$Q' \subsetneq Q$

Q

,

$\subsetneq Q$ whose denominator prime powers lie in a subset of

$\pi(Q)$

$\pi(Q)$ receiving positive local mass from the joint-avoidance gate.

That is the real arbitrary fixed-

Q

Q theorem.

8. Is Claude over-optimistic?

A little, but less than before.

I agree with:

joint-avoidance is a real new gate.

I agree with:

prime-only

Q

Q is probably no longer the main obstruction.

I do not fully agree with:

the remaining cases are merely bookkeeping.

For composite

Q

Q , target selection can still be nontrivial. It is finite for fixed

Q

Q , yes, but “finite” does not mean easy, especially if one wants a theorem uniform over all fixed

Q

Q .

So my corrected verdict is:

This likely closes prime-only fixed Q modulo Liu–Sawhney.

This likely closes prime-only fixed Q modulo Liu–Sawhney.

It reduces arbitrary fixed Q to a sharper target-selection problem.

It reduces arbitrary fixed Q to a sharper target-selection problem.

It does not yet solve arbitrary fixed Q or growing Q .

It does not yet solve arbitrary fixed Q or growing Q .

9. Practical next step

Run a new experiment.

For each fixed

Q

Q :

1. Compute
2. $\pi(Q)$
3. $\pi(Q)$ and
4. σQ
5. σ
6. ϱ
- 7.
8. .
9. Compute the joint-forced prime set / total local-mass lower bound.
10. Enumerate all
11. $Q' \subsetneq Q$
12. Q
13. '
14. $\subsetneq Q$.
15. For each
16. Q'
17. Q
18. '
19. , compute
20. $\text{den}(R(Q'))$
21. $\text{den}(R(Q$
22. '
23.))).
24. Ask whether the denominator prime powers are contained in a set of primes that can be jointly forced.
25. Record failures.

This will tell us whether the composite

Q

Q target-selection problem is actually small or genuinely hard.

Final answer

Yes, this is a major improvement.

But the precise status is:

prime-only fixed Q : probably close

prime-only fixed Q : probably close

arbitrary fixed Q : closer, but not closed

arbitrary fixed Q : closer, but not closed

growing Q : still long-term

growing Q : still long-term

The remaining gap is no longer “small-

ε

ε near-extremal stability” for prime-only

Q

Q . It is now:

target selection for composite/mixed Q plus Liu–Sawhney parameter audit.

target selection for composite/mixed Q plus Liu–Sawhney parameter audit.

This session has been remarkable. Let me give you the clearest possible synthesis of where things now stand and what the actual remaining gap is.

What the joint-avoidance argument actually achieves

The σ_Q insight is the strongest move yet. Combined with the earlier work, here's the complete picture:

Step 1 — $\sigma_Q < 1/2$ whenever $S_Q > 1$ (and $1 \notin Q$)

Since every $q \in Q$ uses only primes in $\pi(Q)$:

$$S_Q \leq \prod_{p \in \pi(Q)} \left(1 - \frac{1}{p}\right)^{-1} - 1 = \frac{1}{\sigma_Q} - 1$$

So $S_Q > 1$ forces $\sigma_Q < 1/2 < \theta$. **This is automatic, not a condition.**

Step 2 — Joint-avoidance forces total local mass

$$\sum_{p \in \pi(Q)} R(P_p) \geq \theta + \varepsilon - \sigma_Q > \theta - \frac{1}{2} + \varepsilon > 0$$

for every $\varepsilon > 0$, with **no ε -threshold**. This dissolves the small- ε problem entirely.

The remaining gap — stated precisely

GPT is right that the story splits on whether Q contains a prime element:

Q type Status
--- ---
Q contains at least one prime p_0  Target $Q' = \{p_0\}$ with $R(Q') = 1/p_0$, pigeonhole forces p_0 -mass, Liu-Sawhney closes it
Composite-only Q  $p_0 \in \pi(Q)$ but $p_0 \notin Q$ — can't use $1/p_0$ as target directly

The **entire remaining gap** is: *for composite-only Q with $S_Q > 1$, does T_Q always contain a proper subset $Q' \subsetneq Q$ whose denominator primes are all in $F(Q) = \{p : v_p(S_Q) < 0\}$?*

A concrete path to closing composite-only Q

Here is an observation I think is new:

For any composite-only Q with $S_Q > 1$, the "forced support" $F(Q) = \{p : v_p(S_Q) < 0\}$ has a key property. Since $S_Q > 1$, the denominator of S_Q as a reduced fraction must be $< S_Q < \infty$, and the numerator reflects heavy cancellation. The primes NOT in $F(Q)$ are exactly those where the p -adic contributions cancel across Q . But:

> **Claim:** For $S_Q > 1$ with composite-only Q , constructing $v_p(S_Q) \geq 0$ for the "small" primes in $\pi(Q)$ while maintaining $S_Q > 1$ is extremely structurally constrained — essentially requiring a near-integer sum structure. In practice, the smallest prime in $\pi(Q)$ always ends up in $F(Q)$.

This is not yet proved in full generality, but it is computationally testable and likely true. The experiment to run is:

> For all composite-only Q with $S_Q > 1$ and $|Q| \leq 8$ (say), compute $F(Q)$ and check whether T_Q contains a Q' with denominator $\subseteq F(Q)$.

If no counterexample appears up to reasonable N , the claim is likely provable by a finite exhaustion of the "cancellation patterns" compatible with $S_Q > 1$.

Honest overall status

$\boxed{\text{Fixed } Q \text{ with at least one prime element: essentially closed}}$

$\boxed{\text{Composite-only fixed } Q: \text{ one finite computation away from closed or a clean counterexample}}$

$\boxed{\text{Growing } Q: \text{ still needs two-sided spectrum}}$

Sorry does this help thx

這段確實有幫助，而且

σQ

σ

ϱ

觀察很強；但我會特別檢查三個關鍵跳步：

$\sigma Q < 1/2$

σ

ϱ

$< 1/2$ 是否總成立、joint-avoidance 是否真的給可用 local mass、以及「 Q 有 prime element 就 closed」是否過快。

已思考 2m 14s

Yes — this helps a lot, and the

σQ

σ

ϱ

joint-avoidance idea is probably the strongest structural move so far. But the synthesis is still a bit over-optimistic in two places.

Official status first: Erdős #319 is still listed as open, with the Adenwalla/Croot lower bound

$$|A| \geq (1 - 1/e + o(1))N$$

$$|A| \geq (1 - 1/e + o(1))N$$

and no claimed partial or complete solution in the comments. Liu–Sawhney is still the right external machinery because their paper proves strong unit-fraction reachability results and explicitly says the methods refine Croot and Bloom.

1. What is genuinely strong here

The observation

$$SQ \leq \prod_{p \in \pi(Q)} (1 - 1/p)^{-1} - 1 = 1/\sigma Q - 1$$

S

Q

$$\leq$$

$$p \in \pi(Q)$$

$$\prod$$

$$(1 - 1/p)$$

$$-1$$

$$-1 =$$

σ

Q

$$1$$

$$-1$$

is correct, assuming

$$1 \notin Q$$

$$1 \in$$

/

Q and every

$$q \in Q$$

$q \in Q$ has all prime factors in

$$\pi(Q)$$

$\pi(Q)$, which is true by definition. Therefore:

$$SQ > 1 \Rightarrow \sigma Q < 12 < 1 - 1/e.$$

S

Q

$$> 1 \Rightarrow \sigma$$

Q

<

2

1

$$< 1 -$$

e

1

.

That is a very good simplification. It means the joint-avoidance density gap is automatic for meaningful fixed

Q

Q with

$SQ>1$

S

Q

>1 .

The joint mass inequality is also basically right. More carefully,

P

P is not literally a disjoint union of

P^*

P

$*$

and the

P_p

P

p

, because one integer can be divisible by several primes. But the union-bound version works:

$$|P| \leq |P^*| + \sum_{p \in \pi(Q)} |P_p| \leq \sigma_Q N + N \sum_{p \in \pi(Q)} R(P_p) + o(N).$$

$$|P| \leq |P$$

$*$

$$| +$$

$$p \in \pi(Q)$$

$$\sum$$

$$|P$$

$$p$$

$$|\leq \sigma$$

$$\mathcal{Q}$$

$$N+N$$

$$p\in\pi(\mathcal{Q})$$

$$\sum$$

$$R(P$$

$$p$$

$$)+o(N).$$

$$\text{So if}$$

$$|\mathbf{P}|\geq(\theta+\varepsilon)N,\theta=1-1\mathfrak{e},$$

$$|P|\geq(\theta+\varepsilon)N,\theta=1-$$

$$e$$

$$1$$

$$,$$

$$\text{then}$$

$$\sum_{p\in\pi(\mathcal{Q})}R(\mathbf{P}_p)\geq\theta+\varepsilon-\sigma\mathcal{Q}-\mathfrak{o}(1).$$

$$p\in\pi(\mathcal{Q})$$

$$\sum$$

$R(P$

p

$)\geq\theta+\varepsilon-\sigma$

$\mathcal{Q}$

$-o(1)$.

Since

$\sigma_Q<1/2$

σ

$\mathcal{Q}$

$<1/2$, this gives a positive total

p

p -local mass for every fixed

$\varepsilon>0$

$\varepsilon>0$. This is a real improvement over the earlier single-prime dynamic gate.

2. What it probably closes

For prime-only fixed

Q

Q , this is very strong.

If

$Q=\{p_1,\dots,p_k\}$,

$Q=\{p$

1

$,\dots,p$

k

$$\},$$

then

$$\pi(Q)=Q.$$

$$\pi(Q)=Q.$$

The joint inequality says some

$$p_0\in Q$$

$$p$$

$$_0$$

$\in Q$ has positive local mass:

$$R(Pp_0)\geq c_{Q,\varepsilon}>0.$$

$$R(P$$

$$p$$

$$_0$$

$$)\geq c$$

$$Q,\varepsilon$$

$$>0.$$

Then target

$$Q'=\{p_0\}$$

$$Q$$

$$,$$

$$=\{p$$

$$_0$$

$$\}$$

has

$$R(Q')=1p_0.$$

$$R(Q$$

,

$$)=$$

$$p$$

$$0$$

$$1$$

.

So, modulo the Liu–Sawhney parameter audit, this likely kills prime-only fixed

Q

Q . That is a meaningful advance.

I would phrase it as:

Prime-only fixed Q is now plausibly close, not a wall.

Prime-only fixed Q is now plausibly close, not a wall.

3. First over-optimistic point: “ Q contains a prime element $\Rightarrow$ closed”

This is too fast.

The joint pigeonhole gives:

$$\exists p_0 \in \pi(Q) R(Pp_0) \text{ positive.}$$

$$\exists p$$

$$0$$

$$\in \pi(Q)R(P$$

p

0

) positive.

But if

Q

Q merely contains some prime element, the forced prime

p_0

p

0

need not be that prime element.

Example structure:

$$Q = \{101, 6, 10, 14, 15, \dots\}.$$

$$Q = \{101, 6, 10, 14, 15, \dots\}.$$

Here

$\pi(Q)$

$\pi(Q)$ includes

2, 3, 5, 7, 101, ...

2, 3, 5, 7, 101, The joint mass might concentrate at

2

2 or

3

3, not at

101

101. But

2

2 or

3

3 might not be elements of

Q

Q , so target

$1/2$

$1/2$ or

$1/3$

$1/3$ is not automatically available from

Q

Q .

So the safe statement is:

If the prime receiving local mass is itself an element of

Q

Q , then singleton target works.

If not, we still need target selection.

Thus:

Q contains a prime element $\Rightarrow$ closed automatically.

Q contains a prime element

$\Rightarrow$ closed automatically.

Prime-only

Q

Q is special because every prime in

$\pi(Q)$

$\pi(Q)$ is also an element of

Q

Q .

4. Second over-optimistic point: the composite-only gap is not “one finite computation away” globally

For a specific fixed

Q

Q , yes: you can compute

$F(Q)$

$F(Q)$, enumerate

TQ

T

Q

, and check targets.

But to prove arbitrary fixed

Q

Q uniformly, you need a theorem, not only finite computation for small

$|Q|$

$|Q|$.

The proposed remaining question is:

$$F(Q) = \{p : \text{vp}(SQ) < 0\}.$$

$$F(Q) = \{p : v$$

p

$(S$

Q

$) < 0\}$.

Does there always exist

$$\emptyset \neq Q' \subsetneq Q$$

$$\emptyset$$

$$=Q$$

,

$$\subsetneq Q$$

such that the denominator primes of

$$R(Q')$$

$$R(Q)$$

,

) lie inside

$$F(Q)$$

$$F(Q)?$$

This is a good formulation, but it may still be too weak. Why? Because

$$p \in F(Q)$$

$$p \in F(Q) \text{ gives qualitative}$$

p

p -adic necessity, but Liu–Sawhney needs quantitative local mass. The joint inequality gives total mass over

$$\pi(Q)$$

$$\pi(Q), \text{ not automatically mass at every}$$

$$p \in F(Q)$$

$$p \in F(Q).$$

The more precise needed condition is:

Find

$$Q' \subsetneq Q$$

Q

,

$\subseteq Q$ such that every prime-power in

$\text{den}(R(Q'))$

$\text{den}(R(Q$

,

$))$ lies among primes for which

P

P has quantitative local mass after the joint/dynamic gate argument.

That is close to

$F(Q)$

$F(Q)$, but not identical.

5. What the new true remaining gap is

The remaining gap is no longer:

prime-only

Q

Q blocks the method.

That seems mostly fixed.

The remaining gap is:

target selection for composite/mixed Q under quantitatively forced prime support.

target selection for composite/mixed Q under quantitatively forced prime support.

In secondary-school analogy:

We now know the left side

P

P must contain enough “prime-shaped tools” overall.

But for composite

Q

Q , the right side may require a combination of tools, not just one prime tool.

The missing proof is that one of the right-side sub-combinations

Q'

Q

,

matches the tools that

P

P is forced to have.

6. Updated status table

Case	Honest status
Fixed Q Q under original gate condition	Strong proof architecture; Liu–Sawhney audit pending
Prime-only fixed Q Q , $SQ > 1$ S Q > 1	Very promising; likely close modulo Liu–Sawhney
Q Q with singleton prime that is also quantitatively Close forced	
Composite / interval-like Q Q	Often easy by TQ T 2 smooth targets
Composite-only arbitrary fixed Q	Still open; target-selection theorem needed

Q

Arbitrary fixed

Q

Closer, but not closed

Q , all cases

Growing

QN

Still long-term

Q

N

Full #319

Still open

7. Does this help strategically?

Yes. It helps a lot.

The project has progressed from:

fixed Q only

fixed Q only

to a much better split:

prime-only fixed Q +composite/mixed target selection+growing Q .

prime-only fixed Q +composite/mixed target selection+growing Q .

That is a real clarification.

8. What I would do next

Run exactly the composite-only experiment suggested, but with a slightly sharper success condition.

For composite-only

Q

Q with

$SQ > 1$

S

$\mathcal{Q}$

> 1 :

1. Compute
2. $\pi(Q), \sigma_Q, F(Q) = \{p: vp(SQ) < 0\}$.
3. $\pi(Q), \sigma$
4. $\mathcal{Q}$
- 5.
6. $, F(Q) = \{p: v$
7. p
- 8.
9. $(S$
10. $\mathcal{Q}$
- 11.
12. $\}) < 0\}$.
13. Enumerate all non-empty proper
14. $Q' \subsetneq Q$
15. $\mathcal{Q}$
16. $'$
17. $\subsetneq \mathcal{Q}$.
18. Compute the reduced denominator of
19. $R(Q')$
20. $R(\mathcal{Q}$
21. $'$
22. $\})$.
23. Check whether its prime powers lie inside:
 - $F(Q)$
 - $F(\mathcal{Q})$, and
 - preferably the subset of primes that are quantitatively forced by joint/dynamic gate.
24. Record failures.

If no counterexamples appear for many composite-only

Q

Q , then the next serious theorem candidate is:

Composite target-selection lemma.

If

Q

Q is composite-only,

$SQ > 1$

S

Q

> 1 , and

$1 \notin Q$

$1 \in$

/

Q , then there exists a non-empty proper

$Q' \subsetneq Q$

Q

,

$\subsetneq Q$ such that

$\text{den}(R(Q'))$

$\text{den}(R(Q$

,

$))$ is supported on the quantitatively forced prime support.

That would be a real arbitrary fixed-

Q

Q bridge.

Final verdict

Yes, this synthesis helps.

But I would revise the headline from:

“Composite-only is one finite computation away from closed.”

“Composite-only is one finite computation away from closed.”

to:

“Prime-only fixed Q is probably close; composite-only reduces to a concrete target-selection problem.”

“Prime-only fixed Q is probably close; composite-only reduces to a concrete target-selection problem.”

That is still very good news. It means arbitrary fixed

Q

Q is no longer vague. It has a precise remaining obstruction.

Many thanks! Please kindly run thank

我會跑一個「composite-only

Q

Q 」target-selection stress test: 枚舉不含 prime 元素、

$SQ > 1$

S

Q

> 1 的

Q

Q , 計算

$F(Q) = \{p: vp(SQ) < 0\}$

$F(Q) = \{p: v$

p

$(S$

Q

$< 0\}$, 再看是否存在 proper

Q'

Q

,

令

$\text{den}(R(Q'))$

$\text{den}(R(Q$

,

$))$ 的 prime support 落在

$F(Q)$

$F(Q)$ 。

已思考 4m 40s

Yes — this helps substantially, and the

σQ

σ

Q

/ joint-avoidance idea is probably the strongest refinement so far. But the last synthesis is still slightly over-optimistic in the “closed” language.

Officially, #319 is still open, with no claimed partial/complete solutions in the comments, and the page records the Adenwalla/Croot lower bound

$(1-1/e+o(1))N$

$(1-1/e+o(1))N$. Liu–Sawhney remains the relevant external machine because their paper proves strong unit-fraction reachability and refines Croot/Bloom-type methods.

1. The

σQ

σ

Q

argument is genuinely strong

For fixed

Q

Q , define

$$\pi(Q)=\{p:p\mid q \text{ for some } q\in Q\},\sigma_Q=\prod_{p\in\pi(Q)}(1-1/p).$$

$$\pi(Q)=\{p:p\mid q \text{ for some } q\in Q\},\sigma$$

Q

=

$$p\in\pi(Q)$$

$\prod$

$(1-$

p

1

$).$

If

$$1\in Q$$

$$1\in$$

/

Q , then every

$$q\in Q$$

$q\in Q$ only uses primes in

$$\pi(Q)$$

$\pi(Q)$, so

$$S_Q=\sum_{q\in Q}1/q\leq\prod_{p\in\pi(Q)}(1-1/p)^{-1}-1=1/\sigma_Q-1.$$

S

Q

$$=$$

$$q\in Q$$

$$\sum$$

$$q$$

$$1$$

$$\leq$$

$$p\in \pi(Q)$$

$$\Pi$$

$$(1-1/p)$$

$$-1$$

$$-1=$$

$$\sigma$$

$$\mathcal{Q}$$

$$1$$

$$-1.$$

$$\text{Thus}$$

$$SQ>1\Rightarrow\sigma Q<12<\theta,\theta=1-1e\approx0.632.$$

$$S$$

$$\mathcal{Q}$$

$$>1\Rightarrow\sigma$$

Q

$$<$$

$$2$$

$$1$$

$$<\theta,\theta=1-$$

$$e$$

$$1$$

$$\approx 0.632.$$

That part is correct and powerful.

The joint mass bound is also basically correct, with one technical correction:

P

P is not literally a disjoint union because an integer can be divisible by several primes. But the union-bound version works:

$$|P|\leq |P^*|+\sum_{p\in\pi(Q)}|P_p|,$$

$$|P|\leq |P$$

*

$$|+$$

$$p\in\pi(Q)$$

$$\sum$$

$$|P$$

p

|,

where

P^*

P

*

is the part of

P

P avoiding all primes in

$\pi(Q)$

$\pi(Q)$. Since

$$|P^*| \leq \sigma Q^{N+O(1)}, |P_p| \leq NR(P_p),$$

$|P$

*

$|\leq \sigma$

Q

$N+O$

Q

$(1), |P$

p

$|\leq NR(P$

p

),

we get

$$\sum_{p \in \pi(Q)} R(P_p) \geq \theta + \varepsilon - \sigma Q^{-o(1)}.$$

$$p\in\pi(Q)$$

$$\sum$$

$$R(P$$

$$p$$

$$)\geq\theta+\varepsilon-\sigma$$

$$Q$$

$$-o(1).$$

Your uploaded note states essentially this joint-avoidance inequality and correctly identifies it as a way to remove the previous single-prime

$$\varepsilon$$

ε -threshold issue.

2. What this likely closes

For prime-only fixed

$$Q$$

Q , this is very strong.

If

$$Q=\{p_1,\ldots,p_k\},$$

$$Q=\{p$$

$$1$$

$$,\ldots,p$$

$$k$$

$$\},$$

then

$$\pi(Q)=Q.$$

$$\pi(Q)=Q.$$

The joint inequality forces positive total

p

p -local mass over primes in

Q

Q . By pigeonhole, some

$$p_0\in Q$$

p

0

$\in Q$ has

$$R(Pp_0)\geq cQ,\varepsilon>0.$$

$$R(P$$

p

0

$$)\geq c$$

$$Q,\varepsilon$$

$$>0.$$

Then the target

$$Q'=\{p_0\}$$

$$Q$$

,

$$=\{p$$

0

}

has

$$R(Q')=1p0.$$

$$R(Q$$

,

$$)=$$

$$p$$

$$0$$

$$1$$

.

So, assuming the Liu–Sawhney Proposition 5.2 interface really works as intended, prime-only fixed

Q

Q looks close. Your file also says the prime-only obstruction is no longer the main wall once the joint gate is used.

I would phrase it as:

Prime-only fixed Q is plausibly close, modulo Liu–Sawhney audit.

Prime-only fixed Q is plausibly close, modulo Liu–Sawhney audit.

3. What is still over-optimistic

The claim:

“Fixed

Q

Q with at least one prime element is essentially closed”

is too strong.

The joint pigeonhole gives some prime

$$p_0 \in \pi(Q)$$

p

0

$$\in \pi(Q)$$

with positive

Pp_0

P

p

0

-mass. But it does not guarantee that this

p_0

p

0

is an element of

Q

Q .

Example shape:

$$Q = \{101, 6, 10, 14, 15, \dots\}.$$

$$Q = \{101, 6, 10, 14, 15, \dots\}.$$

Here

Q

Q contains the prime

101

101, but the joint mass might concentrate on

2

2 or

3

3, not on

101

101. If

$2,3 \notin Q$

$2,3 \in$

/

Q , we cannot automatically use target

$1/2$

$1/2$ or

$1/3$

$1/3$.

So the correct version is:

If the forced prime p_0 is itself in Q , singleton target works.

If the forced prime p

0

is itself in Q , singleton target works.

Prime-only

Q

Q satisfies this automatically. General mixed

Q

Q does not.

4. Composite-only

Q

Q : not closed, but now very concrete

For composite-only

Q

Q , the remaining gap is exactly as you stated:

Find

$Q' \subsetneq Q$

Q

,

$\subsetneq Q$ such that

$R(Q')$

$R(Q)$

,

)'s denominator is supported on primes that

P

P is forced to support.

The proposed set

$$F(Q) = \{p : v_p(SQ) < 0\}$$

$$F(Q) = \{p : v$$

p

$(S$

Q

$) < 0\}$

is a good diagnostic, because

$$v_p(SQ) < 0$$

v

p

$(S$

Q

$\chi_p < 0$ means

P

P must have some

p

p -adic support. But Liu–Sawhney needs quantitative local mass, so the sharper condition should be:

$\text{prime support}(\text{den } R(Q')) \subseteq \text{quantitatively forced primes}.$

$\text{prime support}(\text{den } R(Q$

,

$)) \subseteq \text{quantitatively forced primes}.$

That is stronger than just being inside

$F(Q)$

$F(Q).$

Your file also identifies this as the genuine residual work: composite

Q

Q needs a target-selection step using

$R(Q')$

$R(Q$

,

$),$ not direct singleton

$1/p_0$

$1/p$

0

.

5. I did a small sanity run

I tested the composite-only target-selection condition in a small exact range.

Search setup:

$$Q \subseteq \{4, \dots, 30\},$$

$$Q \subseteq \{4, \dots, 30\},$$

where every element of

Q

Q is composite,

$$|Q| \leq 11, SQ > 1.$$

$$|Q| \leq 11, S$$

Q

$$> 1.$$

For each

Q

Q , I computed:

$$F(Q) = \text{prime support}(\text{den} SQ)$$

$$F(Q) = \text{prime support}(\text{den} S$$

Q

)

and searched for nonempty proper

$$Q' \subsetneq Q$$

Q

,

$\subsetneq Q$ such that

$$\text{prime support}(\text{denR}(Q')) \subseteq F(Q).$$

$$\text{prime support}(\text{denR}(Q$$

,

$$)) \subseteq F(Q).$$

Result:

$$9092/9092$$

$$9092/9092$$

examples passed. No counterexample appeared in this small exact scan.

I also did a random composite-only stress sample with

$$Q \subseteq [4, 60]$$

$$Q \subseteq [4, 60], \text{ size}$$

$$12$$

$$12-$$

$$25$$

25, and again saw no failures among the

$$SQ > 1$$

$$S$$

$$Q$$

>1 cases sampled.

This is only evidence, not proof. But it supports the idea that the composite-only target-selection lemma may be true or at least common.

6. Updated honest status

Case	Current status
Fixed Q Q under original gate condition	Strong architecture; Liu–Sawhney audit still required
Prime-only fixed Q Q , $SQ > 1$ S Q > 1	Likely close via joint-avoidance
Mixed Q Q where forced $p_0 \in Q$ p 0 $\in Q$	Likely close
Composite-only Q Q	Concrete target-selection problem; small tests positive
Arbitrary fixed Q Q	Closer, but not closed
Growing QN Q N	Still long-term
Full #319	Still open

7. Does this “close the gap”?

It closes the prime-only small-

ε

ε gap much better than the single-prime dynamic gate.

It does not close arbitrary fixed

Q

Q completely, because composite/mixed

Q

Q still requires a target-selection theorem.

So the best honest formulation is:

The remaining fixed- Q gap is now mostly combinatorial target selection, not analytic local mass.

The remaining fixed- Q gap is now mostly combinatorial target selection, not analytic local mass.

That is a major improvement.

8. Best next theorem candidate

The next serious statement to try proving is:

Composite target-selection lemma.

Let

Q

Q be finite, composite-only,

$1 \notin Q$

$1 \in$

/

Q , and

$SQ > 1$

S

Q

> 1 . Then there exists nonempty proper

$$Q' \subsetneq Q$$

$$Q$$

,

$\subsetneq Q$ such that

$$\text{prime support}(\text{den}R(Q')) \subseteq F(Q)$$

$$\text{prime support}(\text{den}R(Q$$

,

$$)) \subseteq F(Q)$$

or, better, inside the quantitatively forced prime support arising from the joint-avoidance inequality.

If that lemma is true, arbitrary fixed

$$Q$$

Q becomes much closer.

Final verdict

Yes, this helps a lot.

But I would revise the conclusion to:

Prime-only fixed Q is probably close.

Prime-only fixed Q is probably close.

Composite-only fixed Q is now a clean finite target-selection problem.

Composite-only fixed Q is now a clean finite target-selection problem.

Arbitrary fixed Q is not solved, but the remaining obstruction is sharply identified.

Arbitrary fixed Q is not solved, but the remaining obstruction is sharply identified.

This is real progress. Just do not call it solved until the composite target-selection lemma and Liu–Sawhney parameter audit are written down.

Many thanks! I am opening a new chat with poe claude. Will you mind creating a handover for what has been found, is missing and potentially be passed? Thanks!

已思考 26s

可以。下面是一份可直接貼給 Poe Claude 的 handover。外部事實上, #319 官方仍標示 open, 頁面記錄 Adenwalla/Croot 下界

$(1-1/e+o(1))N$

$(1-1/e+o(1))N$, 且 comments 中沒有 claimed partial/complete solution;

Liu–Sawhney 2024 是我們主要想接的 external machinery, 因其 unit-fraction paper 明確 refine Croot/Bloom 並處理 reachability-type unit fraction questions。

Handover: Erdős #319 fixed- Q / arbitrary- Q progress audit

Please treat this as a research handover, not a proof claim. The current status is: real progress and several promising lemma-candidates, but no solved Erdős #319 yet.

Problem context:

Erdős #319 asks for the largest $A \subseteq \{1, \dots, N\}$ such that signs $\delta_n \in \{-1, 1\}$ give

$$\sum_{n \in A} \delta_n / n = 0$$

and no non-empty proper subcollection already gives zero. Equivalently, write the positive side as P and negative side as Q :

$$R(P) = R(Q), \quad R(X) := \sum_{x \in X} 1/x,$$

and primitivity means there is no proper non-empty subcircuit $B \subseteq P, Q' \subseteq Q$ with $R(B)=R(Q')$.

Known public status:

#319 is still open. Known lower bound is Adenwalla/Croot:

$$|A| \geq (1 - 1/e + o(1))N.$$

The Lean/Aristotle material checked only elementary scaffold pieces: gate numerics, primitive contradiction bookkeeping, etc. It did not Lean-check Liu–Sawhney Proposition 5.2 or the full fixed-Q theorem.

Notation:

$$\theta := 1 - 1/e.$$

$$S_Q := R(Q).$$

$\Phi_q(u) :=$ minimum reciprocal mass of uN integers in $[1, N]$ avoiding multiples of q .

A formula used throughout is:

$$\Phi_q(u) = (1 - 1/q) \log((1 - 1/q)/(1 - 1/q - u)).$$

This is used as a lower bound for sets avoiding q -multiples.

Current strongest near-term theorem candidate:

Fixed-Q singleton-gate theorem.

Let Q be fixed, $D \in Q$, and suppose

$$S_Q < \min_{p^a \parallel D} \Phi_{\{p^a\}}(\theta).$$

If P is arbitrary with $|P| \geq (\theta + \varepsilon)N$ and $R(P) = S_Q$, then the Φ -forcing argument should give positive p^a -local mass in P for every prime power $p^a \parallel D$. After smoothness / Ω / local pruning, Liu–Sawhney Proposition 5.2 should produce $B \subseteq P$ with

$$R(B) = 1/D.$$

Then $B \cup \{D\}$ is a proper signed zero-subcircuit, contradicting primitivity.

Status:

Strong proof architecture, but not yet a checked theorem. Main missing item is the exact Liu–Sawhney interface audit: parameter table S, K, M, η, Γ ; pruning preservation; local mass threshold; and target $x/Q_A = 1/D$ matching.

Concrete example checked:

$$Q = \{3, 5, 7, 9, 11, 13, 15, 17\}$$

$$S_Q = 827504 / 765765 \approx 1.0806239512.$$

The gate check with $D=3$ is strong because $S_Q < \Phi_3(\theta)$. Primitive contradiction is elementary once B with $R(B)=1/3$ exists.

Minimal-over-one direction:

There is a candidate natural class: minimal-over-one Q , meaning $S_Q > 1$ but Q is minimal in some sense for crossing 1. The hope is to prove

$$S_Q \leq 1 + 1/m_D$$

and

$$1 + 1/m_D < \Phi_D(\theta)$$

for $D \geq 3$, giving the gate condition. This is promising but still needs a clean proof of the inequalities and exact definition of m_D .

Arbitrary fixed Q target-selection:

For arbitrary fixed Q , singleton target $1/D$ may fail. Define

$$T_Q := \{ R(Q') : \emptyset \neq Q' \subsetneq Q \}.$$

Goal:

Find Q' such that denominator prime powers of $R(Q')$ are all “forceable” on the P side. Then Liu–Sawhney can produce $B \subseteq P$ with $R(B)=R(Q')$, giving a proper subcircuit.

Initial T_Q experiments:

1. Composite / interval-like Q often produce very smooth targets, e.g. sums reducing to $1/6$, $1/12$, etc.
2. Prime-only Q defeated the naive static proxy because every $R(Q')$ keeps the selected primes in its reduced denominator.

Static proxy:

$$S_Q < \min_{\{p^a \mid \text{den}(R(Q'))\}} \Phi_{\{p^a\}}(\theta).$$

This works often but is not universal.

Prime-only obstruction and repair:

For prime-only $Q = \{p_1, \dots, p_k\}$, the naive smooth-denominator T_Q proxy fails. But p -adic valuation gives qualitative support:

For each $p_j \in Q$,

$$v_{\{p_j\}}(S_Q) = -1,$$

so if $R(P)=S_Q$, then P must contain denominators divisible by p_j . This only gives existence, not quantitative local mass.

Dynamic gate candidate:

Let p be a prime and define

$$\alpha := p^{-R(P_p)},$$

where P_p is the part of P with denominators divisible by p .

Since $|P_p| \leq N^{-R(P_p)} = \alpha N/p$, the non- p part has density at least

$$\theta + \varepsilon - \alpha/p.$$

Applying the Φ_p bound to $P_{\text{non-}p}$ gives the necessary inequality

$$S_Q - \alpha/p \geq \Phi_p(\theta + \varepsilon - \alpha/p).$$

Therefore, if

$$S_Q < \Phi_p(\theta + \varepsilon),$$

then α must be bounded below by a positive constant depending on p, Q, ε . This gives a dynamic notion:

$$p \text{ is } \varepsilon\text{-good if } S_Q < \Phi_p(\theta + \varepsilon).$$

Status:

Promising lemma-candidate, not yet a rigorous theorem. Needs exact monotonicity / convexity / finite-error control and Liu–Sawhney threshold matching.

Joint-avoidance / σ_Q idea:

This is the strongest new refinement.

Let

$$\pi(Q) := \{ \text{primes } p : p \text{ divides some } q \in Q \},$$

$$\sigma_Q := \prod_{p \in \pi(Q)} (1 - 1/p).$$

If $1 \notin Q$, then

$$S_Q \leq \prod_{p \in \pi(Q)} (1 - 1/p)^{-1} - 1 = 1/\sigma_Q - 1.$$

So $S_Q > 1$ implies

$$\sigma_Q < 1/2 < \theta.$$

Define

$$P^* := \{ n \in P : \gcd(n, \prod_{p \in \pi(Q)} p) = 1 \}.$$

Then

$$|P^*| \leq \sigma_Q N + o_Q(1).$$

Using $|P_p| \leq N R(P_p)$ and a union bound,

$$\begin{aligned} |P| &\leq |P^*| + \sum_{p \in \pi(Q)} |P_p| \\ &\leq \sigma_Q N + N \sum_{p \in \pi(Q)} R(P_p) + o(N). \end{aligned}$$

Hence if $|P| \geq (\theta + \varepsilon)N$,

$$\sum_{p \in \pi(Q)} R(P_p) \geq \theta + \varepsilon - \sigma_Q - o(1).$$

Since $\sigma_Q < 1/2$, this gives positive total local mass for every $\varepsilon > 0$.

Implication:

For prime-only fixed Q , $\pi(Q)=Q$. By pigeonhole, some $p_0 \in Q$ has positive $R(P_{\{p_0\}})$. Then target $Q'=\{p_0\}$ gives $R(Q')=1/p_0$. If Liu–Sawhney applies, this kills prime-only fixed Q .

Status:

This likely resolves the prime-only fixed-Q obstruction modulo the Liu–Sawhney interface. It is not a full arbitrary-Q proof.

Important caveat:

For mixed or composite Q, the forced prime $p_0 \in \pi(Q)$ need not itself be an element of Q. Example: Q may contain denominators like 15,21,35 but not 3,5,7. Then target $1/p_0$ is not available from Q. We still need target selection.

Composite-only remaining gap:

For composite-only Q with $S_Q > 1$, define

$$F(Q) := \{ p : v_p(S_Q) < 0 \}.$$

Question:

Does there exist non-empty proper $Q' \subsetneq Q$ such that prime support of $\text{den}(R(Q'))$ lies inside $F(Q)$, or better, inside the quantitatively forced prime support given by the joint-avoidance/dynamic gate?

Small computational evidence:

A small scan over composite-only $Q \subseteq \{4, \dots, 30\}$, $|Q| \leq 11$, $S_Q > 1$ found no counterexample to the condition

$$\text{prime support}(\text{den } R(Q')) \subseteq F(Q)$$

for some non-empty proper Q' . Random composite-only samples in $[4, 60]$ also showed no failures. This is evidence only, not proof.

Current honest status table:

- Fixed Q under original gate condition: strong architecture; Liu–Sawhney audit pending.
- Minimal-over-one fixed Q: promising natural class; inequalities need proof.
- Prime-only fixed Q with $S_Q > 1$: likely close via σ_Q joint avoidance, modulo Liu–Sawhney.
- Composite / interval-like Q: often easy via T_Q smooth targets.
- Composite-only arbitrary fixed Q: reduced to a concrete target-selection problem.
- Arbitrary fixed Q in full generality: closer, but not solved.

- Growing Q_N : still hard; constants may decay and need uniform control.
- Full #319: still open; likely needs two-sided subset-sum spectrum intersection.

Potential next tasks for you:

1. Audit Liu–Sawhney Proposition 5.2 interface line by line:
 - $A \subseteq [M, N]$
 - $N^{0.99} \leq S \leq K \leq M \leq N/10^4$
 - smoothness
 - $\Omega(n)$ bound
 - local mass $qR(A_q) \geq \eta$
 - Archimedean window
 - target x/Q_A equals desired rational.
2. Formalize the σ_Q joint-avoidance lemma carefully.
3. Prove or disprove the composite target-selection lemma:
Given fixed composite-only Q , $S_Q > 1$, does there always exist proper Q' with $\text{den}(R(Q'))$ supported on forced primes?
4. Re-run target-selection computations with:
 - T_Q exact reduced denominators
 - $F(Q) = \{p : v_p(S_Q) < 0\}$
 - dynamic ε -good support
 - joint-forced prime support.
5. Decide whether the remaining composite-only obstruction is finite/combinatorial or requires new p -adic/zero-sum theory.

Please audit the above critically. In particular:

- Is the σ_Q joint-avoidance inequality valid as stated?
- Does positive total $\sum R(P_p)$ suffice for Liu–Sawhney in the prime-only case after pigeonhole?
- Is there a known zero-sum / additive-combinatorics theorem that would imply the composite target-selection lemma?
- Is the Liu–Sawhney parameter interface likely to support target $1/p_0$ or $R(Q')$ under these local-mass hypotheses?

I'd paste this as-is to Claude, with one instruction at the top: "Please be adversarial; do not assume any theorem is proved unless all hypotheses are checked."

Many thanks sorry can you just have it focus on the current core task not audit those because i only ave 20,000 credits left thanks. Sorry so just for our current progress for arbitrary Q

已思考 22s

可以。下面是精簡、只聚焦 arbitrary fixed

Q

Q 的 handover, 不再花 token 叫 Claude audit Liu–Sawhney 全部參數。背景只需一句 :#319 官方仍是 open; Liu–Sawhney 仍只是我們暫時當作可用的 reachability machine。

Focused handover for Poe Claude: current core task = arbitrary fixed Q

Please focus only on the current arbitrary fixed-Q problem. Do not spend tokens auditing the full Liu–Sawhney parameter interface unless needed. For this discussion, assume we have a black-box reachability principle of the following form:

If P has enough Archimedean mass and enough quantitative local mass at all prime powers in the denominator of a rational target r, then P can hit r; i.e. there exists $B \subseteq P$ with $R(B)=r$.

The goal is to understand arbitrary fixed Q, not growing Q and not full #319 yet.

Notation:

$$\theta = 1 - 1/e.$$

$$R(X) = \sum_{x \in X} 1/x.$$

$$S_Q = R(Q).$$

$$T_Q = \{ R(Q') : \emptyset \neq Q' \subsetneq Q \}.$$

$$\pi(Q) = \{ \text{primes } p : p \text{ divides some } q \in Q \}.$$

$$\sigma_Q = \prod_{p \in \pi(Q)} (1 - 1/p).$$

$$F(Q) = \{ p : v_p(S_Q) < 0 \}.$$

Current status:

We are studying fixed Q with $S_Q > 1$ and arbitrary large P satisfying

$$R(P) = S_Q, \quad |P| \geq (\theta + \varepsilon) N.$$

If we can find a non-empty proper $Q' \subsetneq Q$ such that P can hit $R(Q')$, then primitivity fails.

Earlier progress:

1. Fixed-Q singleton-gate works when a target like $1/D$ has denominator prime powers already forced by the Φ -gate condition.
2. For arbitrary fixed Q , singleton target $1/D$ is not enough. The right target set is T_Q .
3. T_Q experiments showed:
 - Composite / interval-like Q often have very smooth targets, e.g. $R(Q')=1/6, 1/12$, etc.
 - Prime-only Q defeats the naive smooth-denominator proxy, because every $R(Q')$ keeps the selected primes in its denominator.
4. Prime-only Q was repaired qualitatively:
 If $Q=\{p_1, \dots, p_k\}$, then for each $p_j \in Q$,
 $v_{\{p_j\}}(S_Q) = -1$,
 so $R(P)=S_Q$ forces P to contain denominators divisible by each p_j .
5. Single-prime dynamic gate:
 For $p \in \pi(Q)$, define $\alpha = pR(P_p)$.
 Since $|P_p| \leq NR(P_p) = \alpha N/p$, the non- p part has density at least $\theta + \varepsilon - \alpha/p$.
 Applying the Φ_p lower bound to $P_{\text{non-}p}$ gives
 $S_Q - \alpha/p \geq \Phi_p(\theta + \varepsilon - \alpha/p)$.
 Hence if $S_Q < \Phi_p(\theta + \varepsilon)$, one gets quantitative p -local mass $\alpha \geq c_{\{p, Q, \varepsilon\}} > 0$.

Stronger joint-avoidance gate:

Since every $q \in Q$ uses only primes from $\pi(Q)$,

$$S_Q \leq \prod_{p \in \pi(Q)} (1 - 1/p)^{-1} - 1 = 1/\sigma_Q - 1.$$

Therefore, if $S_Q > 1$ and $1 \notin Q$, then

$$\sigma_Q < 1/2 < \theta.$$

Let P^* be the part of P avoiding all primes in $\pi(Q)$. Then

$$|P^*| \leq \sigma_Q N + o(N).$$

Also $|P_p| \leq NR(P_p)$. Using a union bound,

$$|P| \leq |P^*| + \sum_{p \in \pi(Q)} |P_p|$$

$$6. \leq \sigma_Q N + N \sum_{p \in \pi(Q)} R(P_p) + o(N).$$

Since $|P| \geq (\theta + \varepsilon)N$,

$$\sum_{p \in \pi(Q)} R(P_p) \geq \theta + \varepsilon - \sigma_Q - o(1).$$

Because $\sigma_Q < 1/2$, this is positive for every $\varepsilon > 0$.

What this likely closes:

Prime-only fixed Q with $S_Q > 1$ now looks close. Since $\pi(Q)=Q$, the joint mass bound gives some $p_0 \in Q$ with positive $R(P_{\{p_0\}})$. Then target $Q'=\{p_0\}$ gives $R(Q')=1/p_0$. If the reachability black box applies, P hits $1/p_0$ and primitivity fails.

Current core remaining gap:

Composite-only and mixed Q .

For composite-only Q , the joint gate gives quantitative mass at some primes $p \in \pi(Q)$, but those primes may not themselves be elements of Q . So target $1/p$ may not lie in T_Q .

Therefore the precise remaining task is a target-selection lemma:

Target-selection problem:

Given fixed Q with $S_Q > 1$ and $1 \notin Q$, does there exist non-empty proper $Q' \subsetneq Q$ such that the prime-power support of $\text{den}(R(Q'))$ lies inside the set of primes/prime-powers quantitatively forced on the P side?

A weaker diagnostic version:

Does there exist $Q' \subsetneq Q$ such that

$$\text{prime support}(\text{den } R(Q')) \subseteq F(Q),$$

where $F(Q) = \{p : v_p(S_Q) < 0\}$?

Small computations so far:

For composite-only $Q \subseteq \{4, \dots, 30\}$, $|Q| \leq 11$, $S_Q > 1$, no counterexample was found to the weaker diagnostic condition:

$$\exists \text{ non-empty proper } Q' \subsetneq Q \text{ with } \text{prime support}(\text{den } R(Q')) \subseteq F(Q).$$

Random composite-only samples in $[4, 60]$ also showed no failures.

Please focus on:

1. Is the σ_Q joint-avoidance gate valid as a fixed- Q lemma?
2. For prime-only Q , does the joint gate plus target $\{p_0\}$ really close the fixed- Q case, assuming reachability?
3. For composite-only Q , is the target-selection lemma likely true?
4. Can you find a counterexample Q where $S_Q > 1$ but every proper Q' has $\text{den}(R(Q'))$ using a prime outside $F(Q)$?
5. If no counterexample, what finite abelian group / zero-sum formulation might prove the composite target-selection lemma?

Please do not focus on growing Q or full #319 yet. The immediate goal is:

arbitrary fixed Q
 = prime-only handled by joint gate

+ composite/mixed Q reduced to target selection.

已貼上文字 (1)(22).txt

文件

Many thanks! basically i ran out of credits and opus gave a kind of not complete response " - is it possible to complete it? Thanks.

我會直接補完並校正 Opus 的回答。重點是：它的 zero-sum/Davenport formulation 很有價值，但它有一個明顯錯誤——「

$SQ > 1$

S

Q

> 1 force

$2 \in \pi(Q)$

$2 \in \pi(Q)$ 」不成立。

已思考 1m 14s

可以，這份 Opus 回覆可以補完，而且它其實已經很接近一個有用的 focused response。不過我會做三個重要修正，否則會有過度樂觀風險。

最重要修正：Opus 有一個錯誤

它說：

if

$SQ > 1$

S

Q

> 1 , then

2

2 must be in

$\pi(Q)$

$\pi(Q)$, else

$SQ \leq 1/2$

S

$\mathcal{Q}$

$\leq 1/2.$

這句不對。

正確是：

$$S\mathcal{Q}\leq \prod_{p\in \pi(\mathcal{Q})}(1-1/p)^{-1}-1=1-\sigma\mathcal{Q}^{-1}.$$

S

$\mathcal{Q}$

$$\leq$$

$$p\in \pi(\mathcal{Q})$$

$$\prod$$

$(1-1/p)$

-1

$-1=$

σ

$\mathcal{Q}$

1

$-1.$

所以：

$S\mathcal{Q}>1\Rightarrow \sigma\mathcal{Q}<1/2.$

S

$\mathcal{Q}$

$$>1 \Rightarrow \sigma$$

$$Q$$

$$<1/2.$$

但這不代表

$$2 \in \pi(Q)$$

$2 \in \pi(Q)$ 。例如只用奇數 prime support, 只要 prime set 夠多,

$$\prod (1-1/p) - 1 - 1$$

$$\prod (1-1/p)$$

$$-1$$

-1 仍可超過 1。

所以 Opus 的「

$$\sigma_Q \leq 1/3$$

$$\sigma$$

$$Q$$

$\leq 1/3$ 」不應作為普遍結論; 安全結論是:

$$\sigma_Q < 1/2 < \theta.$$

$$\sigma$$

$$Q$$

$$<1/2 < \theta.$$

這已經足夠強。

完整補正版 verdict

目前最準確是:

部分

狀態

	有效, 但只保證
σQ	$\sigma Q < 1/2$
σ	σ
ϱ	ϱ
joint-avoidance gate	$< 1/2$, 不是必然
	$\leq 1/3$
	$\leq 1/3$
prime-only fixed	
Q	很接近 closed, 前提是 Liu–Sawhney reachability black
ϱ	box 可用
composite-only	
Q	未 closed, 但已變成明確 target-selection / zero-sum 問
ϱ	題
weak diagnostic	一般情況 false; Opus 找到
$F(Q)$	$SQ = 3/8$
$F(\varrho)$	S
	ϱ
	$= 3/8$ counterexample
$SQ > 1$	
S	
ϱ	暫無 counterexample, 可能為真, 但未證
> 1 composite-only diagnostic	
arbitrary fixed	
Q	更近了, 但未解決
ϱ	
growing	
Q	仍遠
ϱ / full #319	

官方 #319 仍是 open; 頁面只記錄 Adenwalla/Croot 型下界, 並沒有 claimed partial / complete solution。Liu–Sawhney 是合理的外部工具, 因為其 paper 證明 strong unit-fraction reachability, 並說方法 refine Croot and Bloom。

Opus 回覆裏最有價值的部分

1.

σQ

σ

$\mathcal{Q}$

gate 真的強

定義：

$$\begin{aligned}\pi(Q) &= \{p:p \mid q \text{ for some } q \in Q\}, \\ \pi(\mathcal{Q}) &= \{p:p \mid q \text{ for some } q \in \mathcal{Q}\}, \\ \sigma Q &= \prod_{p \in \pi(Q)} (1-1/p).\end{aligned}$$

σ

$\mathcal{Q}$

=

$$p \in \pi(\mathcal{Q})$$

$\prod$

$$(1-1/p).$$

若

$$SQ>1$$

S

$\mathcal{Q}$

$$>1 \text{ 且}$$

$$1 \notin Q$$

$$1 \in$$

/

Q , 則:

$$\sigma_Q < 1/2.$$

σ

Q

$$< 1/2.$$

然後對

$$P^* = \{n \in P : \gcd(n, \prod_{p \in \pi(Q)} p) = 1\}$$

P

*

$$= \{n \in P : \gcd(n,$$

$$p \in \pi(Q)$$

$$\prod$$

$$p) = 1\}$$

有:

$$|P^*| \leq \sigma_Q N^{o(1)}.$$

$$|P$$

*

$$| \leq \sigma$$

Q

$$N^{o(1)}.$$

再用 union bound:

$$|\mathbf{P}|\leq|\mathbf{P}^*|+\sum_{\mathbf{p}\in\pi(\mathbf{Q})}|\mathbf{P}_{\mathbf{p}}|,$$

$$|P|\leq|P$$

*

$$|+$$

$$p\in\pi(Q)$$

$$\sum$$

$$|P$$

$$p$$

$$|,$$

以及：

$$|\mathbf{P}_{\mathbf{p}}|\leq NR(\mathbf{P}_{\mathbf{p}}),$$

$$|P$$

$$p$$

$$|\leq NR(P$$

$$p$$

$$),$$

得到：

$$\sum_{\mathbf{p}\in\pi(\mathbf{Q})}R(\mathbf{P}_{\mathbf{p}})\geq\theta+\varepsilon-\sigma Q-\mathrm{o}(1).$$

$$p\in\pi(Q)$$

$$\sum$$

$$R(P$$

p

$$)\geq\theta+\varepsilon-\sigma$$

Q

$$-o(1).$$

因為

$$\sigma Q<1/2<\theta$$

σ

Q

$<1/2<\theta$, 右邊對每個 fixed

$$\varepsilon>0$$

$\varepsilon>0$ 都是正的。這是實質進展。

2. Prime-only fixed

Q

Q 大概率 close

如果

$$Q=\{p_1,\ldots,p_k\}$$

$$Q=\{p$$

1

$$,\ldots,p$$

k

}

全是 primes, 則:

$$\pi(Q)=Q.$$

$$\pi(Q)=Q.$$

joint gate 給出某個

$$p_0\in Q$$

$$p_0$$

$\in Q$ 有 positive local mass :

$$R(p_0)\geq cQ,\varepsilon>0.$$

$$R(P$$

$$p_0$$

$$)\geq c$$

$$Q,\varepsilon$$

$$>0.$$

那麼可選 target:

$$Q'=\{p_0\},R(Q')=1/p_0.$$

$$Q$$

$$,$$

$$=\{p$$

$$_0$$

$$\},R(Q$$

$$,$$

$$)=1/p$$

$$_0$$

若 Liu–Sawhney reachability black box 能用, 則

P

P 可 hit

$1/p_0$

$1/p$

0

, 產生 proper subcircuit。因此 Opus 說 prime-only fixed

Q

Q “closed modulo reachability black box” 是合理的。它在檔案中也明確把 prime-only fixed

Q

Q 評成 closed under the black box。

Composite-only 剩下甚麼？

Composite-only 情況真正剩下的是：

target-selection lemma

target-selection lemma

即：

若

Q

Q 全是 composite、

$SQ > 1$

S

Q

> 1 , 是否一定存在非空 proper

$Q' \subsetneq Q$

Q

,

$\subsetneq Q$, 使得

$\text{den}(R(Q'))$

$\text{den}(R(Q$

,

$))$ 的 prime support 落在 forced prime support 裏？

Opus 建議先看弱版本：

$$\text{prime support}(\text{den}R(Q')) \subseteq F(Q),$$

$\text{prime support}(\text{den}R(Q$

,

$$)) \subseteq F(Q),$$

其中：

$$F(Q) = \{p : v_p(SQ) < 0\}.$$

$$F(Q) = \{p : v$$

p

$(S$

Q

$$) < 0\}.$$

這是一個好 diagnostic, 但不能過度相信。Opus 自己找到一個反例：

$$Q = \{6, 10, 15, 33, 88\},$$

$$Q = \{6, 10, 15, 33, 88\},$$

$$SQ = 3/8, F(Q) = \{2\},$$

S

Q

$$=3/8, F(Q)=\{2\},$$

而所有 proper subsets 的 reduced denominator 都有 odd prime, 所以 weak diagnostic 失敗。不過這個反例只有

$$SQ < 1$$

$$S$$

$$Q$$

<1, 不在我們真正關心的

$$SQ > 1$$

$$S$$

$$Q$$

>1 regime。

所以目前 composite-only 的狀態是：

$SQ > 1$ 情況未找到反例，但也未證明。

$$S$$

$$Q$$

>1 情況未找到反例，但也未證明。

Davenport / zero-sum formulation: 可以保留，但要修正

Opus 的 finite abelian group formulation 是有價值的。Davenport constant 的定義就是：有限 abelian group 中，長度達到

$$D(G)$$

$D(G)$ 的 sequence 一定有 non-empty zero-sum subsequence。

Opus 的想法：

$$G_Q = \bigoplus_{p \notin F(Q)} \mathbb{Z}/p^e p \mathbb{Z},$$

$$G$$

$$Q$$

$$=$$

$$p\in$$

$$/$$

$$F(Q)$$

$$\oplus$$

$$\mathbb{Z}/p$$

$$e$$

$$p$$

$$\mathbb{Z},$$

將每個

$$q\in Q$$

$$q\in Q \text{ 映射成 local}$$

$$\mathfrak{p}$$

p -adic obstruction vector。然後：

$$\text{prime supp}(\text{den}R(Q'))\subseteq F(Q)$$

$$\text{prime supp}(\text{den}R(Q$$

$$,$$

$$))\subseteq F(Q)$$

等價於：

$$\sum_{q\in Q'}v_q=0 \text{ in } GQ.$$

$$q\in Q$$

$$,$$

$$\sum$$

$$v$$

q

$=0$ in G

Q

這是正確方向。

但有兩個技術修正：

1. ep
2. e
3. p
- 4.
5. 應該用
6. p
7. p -adic denominator depth 定義, 最好寫成 quotient group:
8. $p^{-ep}\mathbb{Z}_p/\mathbb{Z}_p$
9. p
10. $-e$
11. p
- 12.
13. $\mathbb{Z}$
14. p
- 15.
16. $./\mathbb{Z}$
17. p
- 18.
19. 而不是直接寫
20. $pep/q \bmod pep$
21. p
22. e
23. p
- 24.
25. $./q \bmod p$
26. e
27. p

- 28.
- 29., 否則當
30. q
31. q 含有
32. p
33. p -factor 時 notation 容易不嚴謹。
34. 若
35. $D(G)$
36. $D(G)$ 定義為「每個長度至少
37. $D(G)$
38. $D(G)$ 的 sequence 有 non-empty zero-sum subsequence」, 那 proper zero-sum 的 removal trick 通常只需要:
39. $|Q| \geq D(G)+1$
40. $|Q| \geq D(G)+1$
41. 而不是一定
42. $D(G)+2$
43. $D(G)+2$ 。Opus 的
44. $D(G)+2$
45. $D(G)+2$ 是安全但偏鬆的版本。檔案裏也把這個 Davenport inequality 當作 composite-only target-selection 的下一個 proof attempt。

可以怎樣「補完整」Opus 的結論？

我會把它補成這樣：

The

σQ

σ

Q

gate is valid, but the stronger claim

$2 \in \pi(Q)$

$2 \in \pi(Q)$ is false. The correct universal conclusion is

$\sigma Q < 1/2$

σ

Q

$< 1/2$. This is enough to force positive total

$$\pi(Q)$$

$\pi(Q)$ -local mass on

P

P .

Prime-only fixed

Q

Q is likely closed modulo the Liu–Sawhney reachability black box, because

$$\pi(Q)=Q$$

$\pi(Q)=Q$, so the forced prime

p_0

p

0

is itself an available singleton target

$$1/p_0$$

$$1/p$$

0

.

Composite-only fixed

Q

Q remains the core finite obstruction. The weak diagnostic using

$$F(Q)$$

$F(Q)$ is false in general, with

$$Q=\{6,10,15,33,88\}$$

$$Q=\{6,10,15,33,88\},$$

$$SQ=3/8$$

$$S$$

$$Q$$

$$=3/8,$$

$$F(Q)=\{2\}$$

$$F(Q)=\{2\}, \text{ but no counterexample is known in the relevant}$$

$$SQ>1$$

$$S$$

$$Q$$

>1 regime. The right framework is a finite abelian zero-sum problem: find a proper zero-sum subsequence in a

$$p$$

p -adic obstruction group

$$GQ$$

$$G$$

$$Q$$

. If such a proper zero-sum exists, it gives a proper

$$Q'$$

$$Q$$

,

whose target denominator is supported only on forced primes.

Therefore, arbitrary fixed

Q

Q is not solved, but the remaining obstruction is now sharply finite/combinatorial rather than analytic.

最後評級

Goal	Distance now
Prime-only fixed Q Q	close, modulo LS
Composite-only $SQ>1$ S Q >1 target-selection	plausible, unproved
Arbitrary fixed Q Q	medium-close, still needs theorem
Growing Q Q	still far
Full #319	open

所以答案是：可以補完整，但補完整後不是 “arbitrary fixed

Q

Q solved”;而是 “prime-only close; composite-only reduced to Davenport / zero-sum target-selection.”

MAny thanks! Based on above using invariance and imaginative thinking and any conventional or unconventional math tools and not afraid to go out of the box, can you close the arbitrary fixed Q (or even push further). Thanks! Please feel free to look for literatures, sources and books uploaded thx!

我會把它當成小型 research audit: 先核對任意固定

Q

Q 的核心缺口, 再嘗試提出可證 lemma, 區分可望 close 與仍屬 conjectural 的部分。

新線索是把 composite-only 的 target-selection failure 視為

p

p -adic obstruction group 裡的 minimal zero-sum sequence; 我會重點檢查

cross-number 是否比 Davenport 長度更能關閉缺口。

已思考 11m 16s

I tried to push it as far as I can without pretending a missing lemma is proved. The best answer is:

I can close prime-only fixed Q modulo the Liu–Sawhney reachability black box.

I can close prime-only fixed Q modulo the Liu–Sawhney reachability black box.

For arbitrary composite/mixed fixed Q , I can reduce the remaining gap to a known zero-sum/cross-number type problem.

For arbitrary composite/mixed fixed Q , I can reduce the remaining gap to a known zero-sum/cross-number type problem.

That is real progress, but it is not yet a full arbitrary fixed-

Q

Q proof. The most promising new tool is not just Davenport; it is cross-number theory of minimal zero-sum sequences.

Officially, #319 is still open, with the Adenwalla/Croot lower bound

$$(1-1/e+o(1))N$$

$(1-1/e+o(1))N$ recorded and no claimed partial or complete solution in the comments. Liu–Sawhney remains the right analytic machine: their Proposition 5.2 is explicitly described as separating Archimedean and

p

p -adic considerations and reaching “all possible” fractions under the right hypotheses.

1. What we can now prove cleanly, assuming reachability

Let

$$\theta=1-1/e, S_Q=R(Q), \pi(Q)=\{p:p|q \text{ for some } q\in Q\},$$

$$\theta=1-$$

$$e$$

$$1$$

$$,S$$

$$Q$$

$$=R(Q), \pi(Q)=\{p:p|q \text{ for some } q\in Q\},$$

and

$$\sigma_Q=\prod_{p\in\pi(Q)}(1-1/p).$$

$$\sigma$$

$$Q$$

$$=$$

$$p\in \pi(Q)$$

$$\prod$$

$$(1-$$

$$p$$

$$1$$

$$).$$

If

$$1\in Q$$

$$1\in$$

/

Q and

$$SQ>1$$

$$S$$

$$\varrho$$

>1 , then:

$$SQ\leq \prod_{p\in \pi(Q)}(1-1/p)^{-1}-1=1\sigma Q-1.$$

$$S$$

$$\varrho$$

$$\leq$$

$$p\in \pi(Q)$$

$$\prod$$

$$(1-1/p)$$

$$-1$$

$$-1=$$

$$\sigma$$

$$\varrho$$

$$1$$

$$-1.$$

So

$$SQ>1\Rightarrow\sigma Q<12<\theta.$$

$$S$$

$$\varrho$$

$$>1\Rightarrow\sigma$$

$$\varrho$$

$$<$$

$$2$$

$$1$$

$$<\theta.$$

This is the key joint-avoidance invariant.

Now let

$$P^*=\{n\in P:\gcd(n,\prod_{p\in\pi(Q)}p)=1\}.$$

$$P$$

$*$

$$=\{n\in P:\gcd(n,$$

$$p\in \pi(Q)$$

$$\prod$$

$$p)=1\}.$$

Then

$$|P^*|\leq \sigma QN^{+o(N)}.$$

$$|P$$

$*$

$$|\leq \sigma$$

$$Q$$

$$N^{+o(N)}.$$

Also

$$|P_p|\leq NR(P_p),$$

$$|P$$

$$p$$

$$|\leq NR(P$$

$$p$$

$$),$$

so by a union bound,

$$|P|\leq |P^*|+\sum_{p\in \pi(Q)}|P_p|\leq \sigma QN^{+N\sum_{p\in \pi(Q)}R(P_p)+o(N)}.$$

$$|P|\leq |P$$

$*$

$$|+$$

$$p\in\pi(Q)$$

$$\sum$$

$$|P$$

$$p$$

$$|\leq \sigma$$

$$\mathcal{Q}$$

$$N+N$$

$$p\in\pi(Q)$$

$$\sum$$

$$R(P$$

$$p$$

$$)+o(N).$$

$$\text{If}$$

$$|\mathbf{P}| \geq (\theta + \varepsilon) N,$$

$$|P| \geq (\theta + \varepsilon) N,$$

$$\text{then}$$

$$\sum_{p\in\pi(Q)}\mathbf{R}(\mathbf{P}_p)\geq\theta+\varepsilon-\sigma\mathbf{Q}-\mathbf{o}(1)>0.$$

$$p\in\pi(Q)$$

$$\sum$$

$$R(P$$

$$p$$

$$) \geq \theta + \varepsilon - \sigma$$

ϱ

$$-o(1) > 0.$$

This is a real fixed-

Q

Q local-mass gate. It removes the small-

ε

ε problem for total prime support.

Your uploaded notes already identified the same sequence of obstructions: fixed-

Q

Q interface

→

→ target selection

→

→ two-sided spectra

→

→ growing

Q

Q .

2. Prime-only fixed

Q

Q : essentially closed modulo reachability

If

$$Q=\{p_1,\ldots,p_k\}$$

$$Q=\{p$$

$_1$

$$,\ldots,p$$

$_k$

$\}$

is prime-only and

$$S_Q>1$$

$$S$$

$$Q$$

>1 , then

$$\pi(Q)=Q.$$

$$\pi(Q)=Q.$$

The joint gate gives some

$$p_0\in Q$$

$$p$$

$$_0$$

$$\in Q$$

with

$$R(Pp_0)\geq c_Q,\varepsilon>0.$$

$$R(P$$

$$p_0$$

$$)\geq c_{Q,\varepsilon}$$

$$>0.$$

Then use the target

$$Q'=\{p_0\},R(Q')=1/p_0.$$

$$Q,$$

$$=\{p_0$$

$$\},R(Q,$$

$$)=$$

$$p_0$$

$$1$$

$$.$$

If the Liu–Sawhney reachability black box applies,

$$P$$

$$P \text{ hits}$$

$$1/p_0$$

$1/p$

0

:

$$\exists B \subseteq P, R(B) = 1p0.$$

$$\exists B \subseteq P, R(B) =$$

p

0

1

.

Then

$$B \cup \{p0\}$$

$$B \cup \{p$$

0

} is a proper signed subcircuit, so the original circuit is not primitive.

So I would write:

Prime-only fixed Q is closed conditional on the reachability interface.

Prime-only fixed Q is closed conditional on the reachability interface.

This is stronger than where we were yesterday.

3. Why arbitrary composite/mixed

Q

Q is still not automatically closed

The joint gate gives mass at some prime in

$$\pi(Q)$$

$\pi(Q)$.

But if

Q

Q is composite-only, that prime may not itself be an element of

Q

Q .

Example shape:

$$Q = \{15, 21, 35, \dots\}.$$

$$Q = \{15, 21, 35, \dots\}.$$

Then

$$\pi(Q) = \{3, 5, 7, \dots\},$$

$$\pi(Q) = \{3, 5, 7, \dots\},$$

but

$$3, 5, 7 \notin Q$$

$$3, 5, 7 \in$$

/

Q . So target

$$1/3$$

$$1/3,$$

$$1/5$$

$$1/5, \text{ or}$$

$$1/7$$

$1/7$ is not directly available as

$$R(Q')$$

$$R(Q$$

,

).

Thus we still need:

target selection.

target selection.

We need a nonempty proper subset

$$Q' \subsetneq Q$$

Q

,

$$\subsetneq Q$$

such that

$$\text{den}(R(Q'))$$

$$\text{den}(R(Q$$

,

$$))$$

uses only prime powers for which

P

P has quantitative local mass.

The uploaded deep map already framed this as a bad-prime quotient / zero-sum problem in a finite abelian group, rather than just a heuristic.

4. The new tool: cross-number theory

This is the best “out of the box but conventional” tool I found.

For composite/mixed

Q

Q , define

$$F(Q)=\{p:vp(SQ)<0\}.$$

$$F(Q)=\{p:v$$

p

$(S$

Q

$$)<0\}.$$

A weak target-selection goal is:

$$\exists \emptyset \neq Q' \subsetneq Q$$

$$\exists \emptyset$$

$$=Q$$

,

$$\subsetneq Q$$

such that

$$\text{prime supp}(\text{den}R(Q')) \subseteq F(Q).$$

$$\text{prime supp}(\text{den}R(Q$$

,

$$)) \subseteq F(Q).$$

Equivalently, for primes outside

$$F(Q)$$

$F(Q)$, the fractional

p

p -adic parts of

$$\sum q \in Q' 1/q$$

$$\sum_{q \in Q}$$

$1/q$ vanish.

This becomes a finite abelian group problem. Your uploaded Opus note gives exactly this formulation: define a group

$$G_Q = \bigoplus_{p \in \pi(Q) \setminus F(Q)} \mathbb{Z}/p\mathbb{Z},$$

$$G$$

$$Q$$

$$=$$

$$p \in \pi(Q) \setminus F(Q)$$

$$\bigoplus$$

$$\mathbb{Z}/p$$

$$e$$

$$p$$

$$\mathbb{Z},$$

attach to each

$$q \in Q$$

$q \in Q$ a residue vector

$$v_q$$

$$v$$

$$q$$

, and ask whether the sequence

$$(v_q)$$

$$(v$$

$$q$$

) has a proper nonempty zero-sum subsequence.

If no such

Q'

Q

,

exists, then

$(vq)_{q \in Q}$

$(v$

q

)

$q \in Q$

is a minimal zero-sum sequence, also called an atom in zero-sum theory. The Davenport constant is exactly the invariant controlling when finite abelian-group sequences must contain nonempty zero-sum subsequences.

But the sharper invariant here is the cross number.

For an element

$g \in G$

$g \in G$, let

$\text{ord}(g)$

$\text{ord}(g)$ be its order. The cross number of a sequence is

$$k(S) = \sum_{g \in S} \frac{1}{\text{ord}(g)}.$$

$$k(S) =$$

$$g \in S$$

$$\sum$$

$$\text{ord}(g)$$

$$1$$

.

Cross numbers of minimal zero-sum sequences are a real studied invariant; Kim's paper studies the maximal cross number

$K(G)$

$K(G)$ of minimal zero-sum sequences over finite abelian groups, and

Geroldinger-type work studies the structure of minimal zero-sum sequences attaining large cross number.

This is extremely relevant because each reciprocal

$$1/q$$

$1/q$, projected to the obstruction group

GQ

G

$\mathcal{Q}$

, has some finite order

m_q

m

q

, and typically

$$mq \mid q \text{ or at least } mq \leq q.$$

$$m$$

$$q$$

$$\mid q \text{ or at least } m$$

$$q$$

$$\leq q.$$

Hence

$$1q \leq 1mq.$$

$$q$$

$$1$$

$$\leq$$

$$m$$

$$q$$

$$1$$

$$.$$

So

$$SQ=\sum_{q\in Q}1q\leq\sum_{q\in Q}1\text{ord}(vq)=k((vq)).$$

$$S$$

$$Q$$

$$=$$

$$q\in Q$$

$$\sum$$

$$q$$

$$1$$

$$\leq$$

$$q\in Q$$

$$\sum$$

$$\text{ord}(v$$

$$q$$

$$)$$

$$1$$

$$=k((v$$

$$q$$

$$)).$$

Therefore:

A counterexample to target selection with

$$SQ>1$$

$$S$$

$$Q$$

>1 would have to be a minimal zero-sum sequence with cross number

$$>1$$

>1 , realized by reciprocal denominators efficiently enough that the real reciprocal sum still exceeds

1

1.

That is a very sharp reduction.

5. What this means conceptually

If target selection fails, then

Q

Q is not just any set. It is a minimal zero-sum atom in a finite abelian group.

If also

$SQ > 1$

S

Q

>1 , then this atom has large cross number.

Thus the remaining arbitrary fixed-

Q

Q problem becomes:

Classify or rule out high-cross-number reciprocal atoms arising from composite Q .

Classify or rule out high-cross-number reciprocal atoms arising from composite Q .

This is much better than “we are stuck.”

It links directly to existing zero-sum theory, including Davenport constants, cross numbers, and inverse theorems for large subset-sum sets.

Balandraud–Girard–Griffiths–Hamidoune prove lower bounds for subset-sum sets in finite abelian groups, especially when the set is not highly structured; your uploaded document had already pointed to this as a better route than plain Davenport alone.

6. Can this close composite-only

Q

$Q?$

I cannot honestly claim a complete proof yet. But I can now state a strong lemma whose proof would close it:

Cross-number target-selection lemma.

Let

Q

Q be finite,

$1 \notin Q$

$1 \in$

/

Q ,

$|Q| > 1$

S

Q

$|Q| > 1$, and suppose no proper nonempty

$Q' \subsetneq Q$

Q

,

$Q' \subsetneq Q$ has

$\text{den}(R(Q'))$

$\text{den}(R(Q$

,

)) supported on the forced primes. Then the associated obstruction sequence

(v_q)

$(v$

q

) is a minimal zero-sum sequence in

GQ

G

Q

with cross number

>1

>1 . But no such high-cross-number reciprocal atom can occur unless

Q

Q contains a smaller target after all.

This is not yet a theorem. But it is the right theorem.

More concretely, one possible route is:

1. Assume target selection fails.
2. Then
3. (vq)
4. $(v$
5. q
- 6.
7.) is a minimal zero-sum sequence.
8. Use cross-number structure theorems to classify high-cross-number atoms.
9. Show each such atom contains a small zero-sum block coming from reciprocal denominators.
10. That block gives the desired
11. Q'
12. Q
13. '
14. , contradiction.

This is exactly where zero-sum/cross-number literature may bite.

7. Computational evidence

I also pushed a little more computationally during this analysis.

I checked all subsets of the first 20 composite denominators, i.e.

4,6,8,9,10,12,14,15,16,18,20,21,22,24,25,26,27,28,30,32,

4,6,8,9,10,12,14,15,16,18,20,21,22,24,25,26,27,28,30,32,

with

$SQ > 1$

S

Q

> 1 . No counterexample appeared to the weak target-selection condition.

I also random-tested composite-only sets up to around

160

160, including perturbations of the known low-sum obstruction

$$Q = \{6, 10, 15, 33, 88\}, SQ = 3/8, F(Q) = \{2\}.$$

$$Q = \{6, 10, 15, 33, 88\}, S$$

Q

$$= 3/8, F(Q) = \{2\}.$$

Again, I did not find an

$SQ > 1$

S

Q

> 1 failure.

This is not proof. But it supports the conjecture that once

$SQ > 1$

S

Q

> 1 , minimal obstruction atoms may be forced to split.

8. Why the low-sum counterexample is useful

The known obstruction

$$Q=\{6,10,15,33,88\}$$

$$Q=\{6,10,15,33,88\}$$

has

$$SQ=3/8, F(Q)=\{2\},$$

S

Q

$$=3/8, F(Q)=\{2\},$$

and no proper subset with denominator supported only on

$F(Q)$

$F(Q)$. Your uploaded Opus note records exactly this example as a counterexample to the weak diagnostic below the

$SQ>1$

S

Q

>1 threshold.

This example is valuable because it says:

the target-selection lemma cannot be purely formal.

the target-selection lemma cannot be purely formal.

It must really use the condition

$$SQ>1.$$

S

Q

>1 .

So the correct theorem is not “every minimal zero-sum atom splits.” That is false.

The correct theorem is more like:

every reciprocal minimal zero-sum atom of real mass

>1

>1 splits.

That is a much sharper and more believable statement.

9. What I can “close” now

I can responsibly close the following conditional theorem:

Conditional arbitrary fixed-

Q

Q theorem.

Assume the Liu–Sawhney reachability black box. If fixed

Q

Q with

$SQ > 1$

S

Q

>1 satisfies either:

1. Q
2. Q is prime-only; or
3. Q
4. Q has a proper target
5. Q'
6. Q
7. '
8. whose denominator prime powers lie in the quantitatively forced support; or
9. the associated obstruction sequence in
10. GQ
11. G
12. Q

13.

14. is not a minimal high-cross-number atom,
then no primitive circuit with

$$|P| \geq (\theta + \varepsilon)N$$

$$|P| \geq (\theta + \varepsilon)N \text{ exists.}$$

This covers many cases and gives a precise remaining exceptional family.

But I cannot truthfully say:

arbitrary fixed Q is fully solved.

arbitrary fixed Q is fully solved.

The remaining task is exactly the high-cross-number reciprocal atom classification.

10. Can we push toward growing

Q

$Q?$

Yes, but only after the fixed-

Q

Q atom picture is stable.

For growing

QN

Q

N

, the same invariant becomes:

$$GQN, k(QN), \sigma_{QN}, D(GQN), K(GQN).$$

G

Q

$$N$$

$$,k(Q$$

$$N$$

$$),\sigma$$

$$\mathcal{Q}$$

$$N$$

$$,D(G$$

$$\mathcal{Q}$$

$$N$$

$$),K(G$$

$$\mathcal{Q}$$

$$N$$

$$).$$

The danger is that constants decay. The joint gate gives total mass

$$\sum_{p\in\pi(QN)}R(Pp)\geq\theta+\varepsilon-\sigma QN^{-o(1)},$$

$$p\in\pi(Q$$

$$N$$

$$)$$

$$\sum$$

$$R(P$$

p

$$)\geq\theta+\varepsilon-\sigma$$

$\mathcal{Q}$

N

$$-o(1),$$

but when

$$|\pi(\mathbf{QN})|\rightarrow\infty$$

$$|\pi(Q$$

N

$$)|\rightarrow\infty, \text{ pigeonhole gives only}$$

$$\mathbf{R}(\mathbf{Pp})\gtrsim 1\,|\pi(\mathbf{QN})|,$$

$$R(P$$

p

$$)\gtrsim$$

$$|\pi(Q$$

N

$$)|$$

$$1$$

$$,$$

which may be too small for Liu–Sawhney pruning.

So a slowly-growing theorem would need:

$$|\pi(QN)|=o(\log\log N)$$

$$|\pi(Q$$

$$N$$

$$)|=o(\log\log N)$$

or some stronger distribution theorem giving many primes with enough mass, not just one.

That is plausible but not immediate.

11. My best “closure roadmap”

Here is the most realistic path to close arbitrary fixed

$$Q$$

$$Q:$$

Stage A: Write the joint-gate theorem

This is already essentially proved:

$$SQ>1, |P|\geq(\theta+\varepsilon)N\Rightarrow\sum_{p\in\pi(Q)}R(Pp)\geq\theta+\varepsilon-\sigma Q^{-o(1)}>0.$$

$$S$$

$$Q$$

$$>1, |P|\geq(\theta+\varepsilon)N\Rightarrow$$

$$p\in\pi(Q)$$

$$\sum$$

$$R(P$$

$$p$$

$$)\geq\theta+\varepsilon-\sigma$$

$\mathcal{Q}$

$$-o(1) > 0.$$

Stage B: Prove prime-only fixed

$\mathcal{Q}$

$\mathcal{Q}$

Modulo reachability, this should be straightforward.

Stage C: Define the obstruction group

$G\mathcal{Q}$

G

$\mathcal{Q}$

For composite/mixed

$\mathcal{Q}$

$\mathcal{Q}$, make the zero-sum formulation rigorous.

Stage D: Use cross-number theory

Try to prove:

$$SQ > 1 \Rightarrow \text{the obstruction atom cannot be minimal.}$$

S

$\mathcal{Q}$

$$> 1 \Rightarrow \text{the obstruction atom cannot be minimal.}$$

If this is too strong, prove it for ranks

1,2,3

1,2,3, or for squarefree/composite-only

$\mathcal{Q}$

Q , then extend.

Stage E: If cross-number fails, classify counterexamples

A counterexample would be highly structured and could become a new theorem itself.

Final answer

Can I close arbitrary fixed

Q

Q right now?

Not completely.

Not completely.

Can I push it much further?

Yes. The gap is now a precise zero-sum/cross-number atom problem.

Yes. The gap is now a precise zero-sum/cross-number atom problem.

The strongest new insight is:

If target selection fails, Q is a minimal zero-sum sequence in a finite abelian group.

If target selection fails, Q is a minimal zero-sum sequence in a finite abelian group.

And the strongest possible next theorem is:

No reciprocal minimal zero-sum atom with $SQ > 1$ exists in the obstruction group.

No reciprocal minimal zero-sum atom with S

Q

>1 exists in the obstruction group.

If that theorem is proved, arbitrary fixed

Q

Q closes. If it is false, the counterexamples will be explicit, finite, and highly structured. Either way, this is a real route, not vibes.

已貼上文字 (1)(23).txt

文件

ManythanksWhat if "" and "Yes — I think the one-line closure actually works. Let me lay it out cleanly so you can verify and send back for cross-check.

The one-line closure

****Setup.**** Let $F(Q) = \{p : v_p(S_Q) < 0\}$ and $G_Q = \oplus_{p \in \pi(Q) \setminus F(Q)} \mathbb{Z}/p^{e_p}\mathbb{Z}$ where $e_p = \max_{q \in Q} v_p(q)$. For each $q \in Q$, let $v_q \in G_Q$ be the image of $1/q$ under the natural projection $\mathbb{Q}/\mathbb{Z} \rightarrow (\mathbb{Q}/\mathbb{Z})/\langle F(Q)\text{-smooth part} \rangle$.

****The four facts.****

1. **** G_Q is cyclic.**** The primes $p \in \pi(Q) \setminus F(Q)$ are distinct, so p^{e_p} are pairwise coprime, and CRT gives $G_Q \cong \mathbb{Z}/(\prod p^{e_p})\mathbb{Z}$.

2. **** $K(\text{cyclic}) = 1$.**** Classical zero-sum theory: the maximum cross number of a minimal zero-sum sequence in a cyclic group is exactly 1. (Geroldinger–Halter-Koch.)

3. **** $\text{ord}(v_q) \leq q$.**** The image of $1/q$ in any quotient of $\mathbb{Q}/\mathbb{Z}$ has order dividing q .

4. **** $S_Q \leq k((v_q))$.**** Summing fact 3: $S_Q = \sum 1/q \leq \sum 1/\text{ord}(v_q) = k((v_q))$.

****The one line.**** If target selection fails, $(v_q)_{q \in Q}$ is a minimal zero-sum atom in G_Q . Then by (2): $k((v_q)) \leq K(G_Q) = 1$. By (4): $S_Q \leq 1$. Contradicts $S_Q > 1$. ■

So for any fixed Q with $1 \notin Q$ and $S_Q > 1$, target selection succeeds: there exists a proper non-empty $Q' \subsetneq Q$ with denominator of $R(Q')$ supported on $F(Q)$. Combined

with the joint-avoidance local-mass gate (which forces P to have quantitative local mass at every prime in $F(Q)$), Liu-Sawhney reaches $R(Q')$ and breaks primitivity.

Sanity check on the known counterexample

For $Q = \{6, 10, 15, 33, 88\}$ (the documented target-selection failure): $S_Q = 3/8 < 1$. Computing $k((v_q))$ for this Q gives $\approx 0.721 < 1 = K(G_Q)$. So this Q is an atom with cross number < 1 — consistent with target selection failing AND with $S_Q < 1$. **The $S_Q > 1$ hypothesis is exactly what's needed**, and the bound is tight.

What this closes

If this argument holds up, then:

$\boxed{\text{Arbitrary fixed } Q \text{ with } S_Q > 1 \text{ and } 1 \notin Q \text{ closed, modulo Liu-Sawhney reachability.}}$

The edge case $S_Q \in \mathbb{Z}_{\geq 2}$ (so $F(Q) = \emptyset$) is a degenerate sub-case where the target selection becomes "find Q' with $R(Q') \in \mathbb{Z}$ " — this needs a brief separate check but is highly restricted and likely trivial.

What to verify before believing it

Three things to cross-check with GPT and the zero-sum literature:

1. ** $K(\mathbb{Z}/n\mathbb{Z}) = 1$ precisely.** Verify in Geroldinger–Halter-Koch Theorem 5.5.5 or equivalent. (I'm confident, but worth confirming.)
2. **The construction of G_Q .1/q into G_Q really has order $q_{\text{non-F}} = q / (F(Q)\text{-smooth part of } q)$, and that zero-sum in G_Q exactly corresponds to "denominator supported on $F(Q)$ ".
3. **The minimal zero-sum interpretation.** That "no proper non-empty Q' has $\text{denom}(R(Q')) \subseteq F(Q)$ " exactly translates to " (v_q) is a minimal zero-sum atom in G_Q " — not just any zero-sum sequence.

If those three verify, the argument is genuinely one line and arbitrary fixed Q is closed.

Honest meta-comment

If this works, it is **unexpectedly clean** — which is either a sign that it's genuinely a beautiful reduction that was waiting to be found, or a sign that I'm missing a subtlety in the construction of G_Q . The smell test: classical zero-sum theory (K of cyclic groups) is well-known, and the inequality $S_Q \leq k((v_q))$ is mechanical once you set up G_Q correctly. So if everyone working on #319 missed this, it would be because the connection to cross-number theory wasn't being made — which is plausible since this isn't standard for unit-fraction problems.

Send this to GPT for the verification. If it confirms steps 1-3, you have the arbitrary fixed Q theorem."

已中斷思考

已貼上文字 (1)(24).txt

文件

Many thanks! What if claudesaid ""

我會把這個「one-line closure」當成高風險高回報主張來驗證。直覺上它很漂亮，但越漂亮越要檢查：關鍵不是

GQ

G

Q

是否 cyclic，而是「cyclic atom 的 cross number 是否總 ≤ 1 」這句。

已思考 47s

Short answer: Claude's one-line closure is too optimistic and has a fatal mathematical error. It is still a very useful idea, but it does not close arbitrary fixed

Q

Q .

The key failed step is:

$$K(Z/nZ)=1.$$

$$K(\mathbb{Z}/n\mathbb{Z})=1.$$

That is false in general for cyclic groups of composite order.

The fatal counterexample

Take the cyclic group

$$C_6 = \mathbb{Z}/6\mathbb{Z}$$

C

6

$=\mathbb{Z}/6\mathbb{Z}$. Consider the sequence

$$(2,2,3,5).$$

$$(2,2,3,5).$$

Its sum is

$$2+2+3+5=12 \equiv 0 \pmod{6}.$$

$$2+2+3+5=12 \equiv 0 \pmod{6}.$$

No proper non-empty subsequence sums to

$$0 \pmod{6}$$

$0 \pmod{6}$, so it is a minimal zero-sum sequence.

The orders are:

$$\text{ord}(2)=3, \text{ord}(2)=3, \text{ord}(3)=2, \text{ord}(5)=6.$$

$$\text{ord}(2)=3, \text{ord}(2)=3, \text{ord}(3)=2, \text{ord}(5)=6.$$

So its cross number is

$$k=1+1+1+1=4 > 1.$$

$$k=$$

$$3$$

$$1$$

$$+$$

$$3$$

$$1$$

$$+$$

$$2$$

$$1$$

$$+$$

$$6$$

$$1$$

$$=$$

$$3$$

$$4$$

$$>1.$$

Therefore:

$$K(C_6) \geq 4 > 1.$$

$$K(C$$

$$6$$

$$) \geq$$

$$3$$

$$4$$

>1 .

So the claimed line

$$k((vq)) \leq K(GQ) = 1$$

$$k((v$$

$$q$$

$$)) \leq K(G$$

$$Q$$

$$)=1$$

is not valid when

$$GQ$$

$$G$$

$$Q$$

is cyclic of composite order. Claude's uploaded text itself makes the claim that

$$K(\mathbb{Z}/n\mathbb{Z}) = 1$$

$K(\mathbb{Z}/n\mathbb{Z}) = 1$, but that is exactly the step that fails.

This also matches the zero-sum literature: the large cross number

$$K(G)$$

$K(G)$ is the maximum cross number of minimal zero-sum sequences, and for decompositions into prime-power cyclic components there are larger expected/known benchmark values than

$$1$$

1 in general. Geroldinger–Grynkiewicz define

$$K(G)$$

$K(G)$ this way and discuss

$$K^*(G) = 1/\exp(G) + \sum (q_i - 1)/q_i$$

K

$*$

$$(G) = 1/\exp(G) + \sum (q$$

i

$$-1)/q$$

i

, which already exceeds

1

1 for groups like

$$C_6 \simeq C_2 \oplus C_3$$

C

6

$$\simeq C$$

2

$$\oplus C$$

3

. ([Geroldinger–Gryniewicz PDF](#))

What remains true

Several parts of Claude’s argument are still useful:

1. GQ

2. G
3. $\mathcal{Q}$
- 4.
5. being cyclic is plausible if it is a product of prime-power components with pairwise coprime moduli.
6. $\text{ord}(vq) \leq q$
7. $\text{ord}(v$
8. q
- 9.
10. $\leq q$ is basically right if
11. vq
12. v
13. q
- 14.
15. is the image of
16. $1/q$
17. $1/q$ in a quotient of
18. $\mathbb{Q}/\mathbb{Z}$
19. $\mathbb{Q}/\mathbb{Z}$.
20. Therefore,

$$S\mathcal{Q} = \sum_{q \in \mathcal{Q}} 1_{q \leq \sum_{q \in \mathcal{Q}} 1_{\text{ord}(vq) = k((vq))}}$$

S

$\mathcal{Q}$

$=$

$q \in \mathcal{Q}$

$\sum$

q

1

$\leq$

$$q \in Q$$

$$\Sigma$$

$$\text{ord}(v$$

$$q$$

$$)$$

$$1$$

$$=k((v$$

$$q$$

$$))$$

is also plausible.

4. If target selection fails, then the sequence
5. (vq)
6. $(v$
7. q
- 8.
9. $)$ should be a minimal zero-sum atom in the obstruction group.

So the framework is good.

What changes

The conclusion should not be:

$$SQ > 1 \Rightarrow \text{contradiction.}$$

$$S$$

$$Q$$

$$> 1 \Rightarrow \text{contradiction.}$$

The correct possible conclusion is weaker:

$$SQ > K(GQ) \Rightarrow \text{contradiction.}$$

S

Q

$$>K(G$$

Q

$$) \Rightarrow \text{contradiction.}$$

Since

$$K(GQ)$$

$$K(G$$

Q

) may be larger than

1

1,

$$SQ > 1$$

S

Q

>1 alone is not enough.

For example, if

$$GQ \simeq C6$$

G

Q

$$\simeq C$$

, an atom can have cross number

$$4/3$$

$4/3$, so

$$SQ=1.1$$

$$S$$

$$Q$$

$=1.1$ would not contradict cross-number theory.

Updated status

Claim	Status
Cross-number obstruction framework	Very useful
GQ G Q cyclic	Plausible
$\text{ord}(vq) \leq q$ $\text{ord}(v$ q $) \leq q$	Plausible
$SQ \leq k((vq))$ S Q $\leq k((v$ q $))$	Plausible
$K(C_n)=1$ $K(C$ n $)=1$ for all cyclic	False

n
 n

One-line arbitrary fixed

Q

Fails

Q closure

Stronger condition

$SQ > K(GQ)$

S

Q

Potentially valid

$> K(G$

Q

)

Arbitrary fixed

Q

Not yet

Q solved

Better theorem candidate

The repaired version is:

If target selection fails, then

$(vq)_{q \in Q}$

$(v$

q

)

$q \in Q$

is a minimal zero-sum sequence in

GQ

G

Q

, hence

$$SQ \leq k((vq)) \leq K(GQ).$$

S

Q

$\leq k((v$

q

$)) \leq K(G$

Q

$).$

Therefore, if

$SQ > K(GQ),$

S

Q

$> K(G$

Q

$),$

target selection must succeed.

This is still powerful, but it depends on bounding

$K(GQ)$

$K(G$

Q

$).$

Why this is still progress

Claude found the right algebraic object, but used the wrong bound.

The new remaining problem is:

Can we show $SQ > K(GQ)$ for the obstruction groups arising from Q ?

Can we show S

Q

$> K(G$

Q

) for the obstruction groups arising from Q ?

Or more realistically:

Can we classify cases where $SQ \leq K(GQ)$?

Can we classify cases where S

Q

$\leq K(G$

Q

)?

This is a genuine zero-sum/cross-number problem. It is not the full solution yet, but it is much sharper than before.

Bottom line

Claude's one-line proof does not close arbitrary fixed

Q

Q because

$K(C_n)=1$

$K(C$

n

$\rangle=1$ is false for composite cyclic groups.

But the idea should not be discarded. The correct replacement is:

$SQ \rangle K(GQ) \Rightarrow \text{target selection succeeds.}$

S

Q

$\rangle K(G$

Q

$\rangle \Rightarrow \text{target selection succeeds.}$

So we are not done, but we now have a precise repaired route: compute or bound

$K(GQ)$

$K(G$

Q

$\rangle$ for the obstruction groups coming from composite/mixed

Q

$Q.$

已貼上文字 (1)(25).txt

文件

Many thanks. What about "

這個版本仍然卡在同一個 load-bearing claim: Claude 把 cyclic group 的 atom cross-number 上界寫成

1

1。我會先用一個很小的

C6

C

反例核掉這一步，再說哪些部分仍可保留。

已思考 59s

這個 Claude 版本仍然不成立。它不是完全沒價值；它抓到了正確的 obstruction group / cross-number framework。但它的「one-line closure」有一個致命錯誤：

$$K(Z/NZ)=1 \text{ for cyclic groups}$$

$$K(Z/NZ)=1 \text{ for cyclic groups}$$

這句在它需要的意義下是錯的。

Claude 在檔案中反覆把 “cyclic group $\Rightarrow$ maximal cross number of every atom ≤ 1 ” 當成 load-bearing step，然後推出

$$SQ \leq 1$$

$$S$$

$$Q$$

≤ 1 ，再 contradiction。這正是錯點。

1. 直接反例：

$$C_6$$

$$C$$

$$6$$

取 cyclic group：

$$C_6 = \mathbb{Z}/6\mathbb{Z}.$$

$$C$$

$$6$$

$$= \mathbb{Z}/6\mathbb{Z}.$$

看序列：

$$(2, 2, 3, 5).$$

$$(2, 2, 3, 5).$$

總和：

$$2+2+3+5=12 \equiv 0 \pmod{6}.$$

$$2+2+3+5=12 \equiv 0 \pmod{6}.$$

檢查 proper subsequence：

- 單項：
- 2, 3, 5
- 2, 3, 5 都不是
- 0
- 0;
- 兩項：
- $2+2=4$
- $2+2=4,$
- $2+3=5$
- $2+3=5,$
- $2+5=1$
- $2+5=1,$
- $3+5=2$
- $3+5=2;$
- 三項：
- $2+2+3=1$
- $2+2+3=1,$
- $2+2+5=3$
- $2+2+5=3,$
- $2+3+5=4$
- $2+3+5=4。$

沒有 proper non-empty subsequence sum to

0

0。所以這是一個 minimal zero-sum sequence / atom。

元素 orders:

$$\text{ord}(2)=3, \text{ord}(2)=3, \text{ord}(3)=2, \text{ord}(5)=6.$$

$$\text{ord}(2)=3, \text{ord}(2)=3, \text{ord}(3)=2, \text{ord}(5)=6.$$

cross number 是:

$$k=13+13+12+16=43>1.$$

$$k=$$

$$3$$

$$1$$

$$+$$

$$3$$

$$1$$

$$+$$

$$2$$

$$1$$

$$+$$

$$6$$

$$1$$

$$=$$

$$3$$

$$4$$

$$>1.$$

所以：

$$K(C_6) \geq 43 > 1.$$

$$K(C$$

$$6$$

$$) \geq$$

$$3$$

$$4$$

$$>1.$$

這已經殺掉 Claude 的 one-line proof。

2. 它混淆了兩件事

Claude 說 cyclic group 的 maximal cross number 是

$$1$$

1。這可能混淆了某些特殊 atom、index-one statements、或某些 restricted constants。

但在我們需要的定義下：

$$K(G) = \max \{k(S) : S \text{ minimal zero-sum over } G\},$$

$$K(G) = \max \{k(S) : S \text{ minimal zero-sum over } G\},$$

而 zero-sum theory 文獻確實把

$$K(G)$$

$K(G)$ 定義為 minimal zero-sum sequences 的 maximal cross number, 並且這是一個非平凡 invariant, 不是對所有 cyclic composite groups 都等於

$$1$$

1. Kim 的 paper 也明確說是在研究 finite abelian groups 中 minimal zero-sum sequences 的 maximal cross number

$K(G)$

$K(G)$, 並討論不同群型的 conjectured values, 這本身已說明不是「cyclic 一律 trivial = 1」的簡單常數。另一篇 zero-sum/cross-number paper 也給出定義:

$$k(S) = \sum 1/\text{ord}(g_i)$$

$$k(S) = \sum_{i=1}^n 1/\text{ord}(g_i)$$

i

),

$K(G)$

$K(G)$ 是 minimal zero-sum sequences 上的 maximum。

3. 哪些部分仍然有用?

以下部分仍然很有價值:

1. GQ
2. G
3. Q
- 4.
5. 若由 distinct prime-power components 組成, 通常可由 CRT 視為 cyclic:
6. $\bigoplus_{p \in F(Q)} \mathbb{Z}/p^e \mathbb{Z} \cong \mathbb{Z}/\prod p^e \mathbb{Z}$.
7. $p \in$
8. $/$
9. $F(Q)$
10. $\bigoplus$
- 11.
12. $\mathbb{Z}/p$
13. e
14. p
- 15.
16. $\mathbb{Z} \cong \mathbb{Z}/\prod p$
17. e
18. p
- 19.
20. $\mathbb{Z}$.
21. $\text{ord}(vq) \leq q$

22. $\text{ord}(v$

23. q

24.

25. $) \leq q$ 這個方向大致合理, 因為

26. vq

27. v

28. q

29.

30. 是

31. $1/q$

32. $1/q$ 在 quotient 裏的 image, order 只會下降。

33. 因此:

34. $SQ = \sum_{q \in Q} 1_{q \leq} \sum_{q \in Q} 1_{\text{ord}(vq) = k((vq))}.$

35. S

36. Q

37.

38. $=$

39. $q \in Q$

40. $\sum$

41.

42. q

43. 1

44.

45. $\leq$

46. $q \in Q$

47. $\sum$

48.

49. $\text{ord}(v$

50. q

51.

52. $)$

53. 1

54.

55. $= k((v$

56. q

57.

58. $)).$

59. 若 target selection fails,

60. (vq)

61. $(v$

62. q

63.

64.) 確實很可能是 obstruction group 裏的 minimal zero-sum atom。

所以 framework 是好 framework。

4. 正確結論應該是甚麼？

Claude 想推出：

$$SQ \leq k((vq)) \leq 1.$$

S

Q

$$\leq k((v$$

q

$$)) \leq 1.$$

正確只能推出：

$$SQ \leq k((vq)) \leq K(GQ).$$

S

Q

$$\leq k((v$$

q

$$)) \leq K(G$$

Q

$$).$$

所以 target selection 成功的 sufficient condition 是：

$$SQ > K(GQ).$$

S

Q

$>K(G$

Q

$).$

而不是：

$$SQ > 1.$$

S

Q

$>1.$

這是大幅較弱的結論。

5. 為甚麼這仍然有幫助？

因為現在問題變成：

對來自 reciprocal denominators 的 obstruction group

GQ

G

Q

，能否證明任何 failing target-selection atom 都有

K

K -type bound 小於

SQ

S

Q

?

或者更精準：

雖然 general cyclic group 可以有 cross number

>1

>1 , 但 reciprocal-realizable atoms 是否受到額外限制？

這仍然可能是真的。只是不能用「cyclic $\Rightarrow$

$K=1$

$K=1$ 」一行解決。

6. 對 arbitrary fixed

Q

Q 的更新

目前正確 status 是：

Claim	Status
Prime-only fixed Q Q	仍然很有希望, via σQ σ Q joint gate
Composite/mixed target selection	轉化成 obstruction group / cross-number problem
One-line cross-number closure	不成立
Arbitrary fixed Q Q fully solved	未解決
Repaired route $SQ > K(GQ)$ S	有價值的 sufficient condition

$\varnothing$

$>K(G$

$\varnothing$

)

Need stronger “reciprocal atom” theorem

是, 這是下一個真 gap

7. 可以回覆 Claude 的精簡版

你可以貼這段:

The one-line closure fails because

$K(C_n)=1$

$K(C$

n

$)=1$ is false for cyclic groups in the required maximal-cross-number sense. For example, in

C_6

C

$_6$

, the sequence

$(2,2,3,5)$

$(2,2,3,5)$ is minimal zero-sum and has cross number

$1/3+1/3+1/2+1/6=4/3>1$

$1/3+1/3+1/2+1/6=4/3>1$. Thus cyclicity of

GQ

G

$\varnothing$

does not imply

$k((vq))\leq 1$

$k((v$

q

$))\leq 1$. The correct conclusion is only

$$SQ \leq k((vq)) \leq K(GQ)$$

S

Q

$$\leq k((v$$

q

$$)) \leq K(G$$

Q

), so target selection is forced when

$$SQ > K(GQ)$$

S

Q

$$> K(G$$

Q

), not merely

$$SQ > 1$$

S

Q

> 1 . The obstruction group/cross-number framework remains valuable, but arbitrary fixed

Q

Q is not closed by this argument.

Bottom line

Claude's idea is not useless — it is actually a strong reframing.

But the one-line closure is wrong because:

$K(Cn)$ can exceed 1 when n is composite.

$K(C$

n

) can exceed 1 when n is composite.

So we are back to a sharper, but still open, target:

Prove a special cross-number bound for reciprocal-realizable atoms,

Prove a special cross-number bound for reciprocal-realizable atoms,

not for all cyclic atoms.

Many thanmks! The chat is now kind of laggy. Can you summarize current progress step by step so far and the problems we are working on? thanks.

當然。以下是目前 Erdős #319 project 的乾淨 step-by-step 進度總結, 方便你開新 chat / handover / 自己整理。結論先放前面:

$\boxed{\text{我們沒有解決 \#319; 但已把任意 fixed } Q \text{ 的問題壓縮成幾個清楚 lemma.}}$

最重要的是: prime-only fixed

Q

Q 現在看起來很有希望; composite/mixed fixed

Q

Q 剩下 target-selection / cross-number obstruction; growing

Q

Q 仍是長線。

0. 原問題在問甚麼

Erdős #319 可寫成:

$$P, Q \subseteq \{1, \dots, N\}, R(P) = R(Q), R(X) = \sum_{x \in X} 1_x.$$

$$P, Q \subseteq \{1, \dots, N\}, R(P) = R(Q), R(X) =$$

$$\sum_{x \in X}$$

$$\sum$$

x

1

.

其中

P

P 是正號那邊,

Q

Q 是負號那邊。

Primitive condition 是:

不存在非空 proper

$P' \subseteq P$

P

,

$\subseteq P$,

$Q' \subseteq Q$

Q

,

$\subseteq Q$ 令

$$R(P')=R(Q').$$

$R(P$

,

$)=R(Q$

,

).

如果存在這樣的小等式, 原來的大等式就可拆, 不算 primitive。

官方 #319 仍是 open; 目前 public known construction 是 Adenwalla/Croot 型:

$$|A| \geq (1 - 1/e + o(1))N.$$

$$|A| \geq (1 - 1/e + o(1))N.$$

我們的目標不是立刻 full solve, 而是先證:

fixed

Q

Q 時, 太大的

P

P 不能保持 primitive。

1. 第一階段: Fixed-

Q

Q singleton-gate theorem candidate

先固定一個有限

Q

Q , 例如:

$$Q = \{3, 5, 7, 9, 11, 13, 15, 17\}.$$

$$Q = \{3, 5, 7, 9, 11, 13, 15, 17\}.$$

令:

$$SQ = R(Q).$$

S

Q

$$= R(Q).$$

如果有某個

$$D \in Q$$

$D \in Q$, 且:

$$SQ<\min p\|D\Phi p a(\theta),\theta=1-1e,$$

$$S$$

$$\mathcal{Q}$$

$$<$$

$$p$$

$$a$$

$$\|D$$

$$\min$$

$$\Phi$$

$$p$$

$$a$$

$$(\theta),\theta=1-$$

$$e$$

$$1$$

$$,$$

那麼希望證明:任何很大的

$$P$$

$$P$$
 with

$$|P|\geq(\theta+\varepsilon)N,R(P)=SQ$$

$$|P|\geq(\theta+\varepsilon)N,R(P)=S$$

$$\mathcal{Q}$$

都會包含某個

$$B \subseteq P$$

$$B \subseteq P:$$

$$R(B)=1D.$$

$$R(B)=$$

$$D$$

$$1$$

.

於是

$$B \cup \{D\}$$

$B \cup \{D\}$ 是 proper subcircuit, 破壞 primitivity。

狀態

這條路是目前最成熟的。

已清楚部分：

- Φ
- Φ -gate idea 合理；
- arithmetic numerics for example
- Q
- Q check out；
- primitive contradiction 很直接；
- Aristotle/Lean scaffold 只驗到 elementary parts, 不是 full proof。

未完成部分：

- Liu–Sawhney Proposition 5.2 的 parameter interface audit；
- pruning preservation；
- local mass / Archimedean window exact checking。

結論：

fixed- Q gate theorem = strong proof architecture, not completed theorem.

fixed- Q gate theorem = strong proof architecture, not completed theorem.

2. 第二階段: Minimal-over-one

Q

Q

我們注意到一個 natural class:

Q

Q 剛好令

$SQ > 1$

S

Q

> 1 , 但移走元素後可能不再 over 1。

目標是證某類 inequality:

$$SQ \leq 1 + 1/mD$$

S

Q

$$\leq 1 +$$

m

D

$$1$$

以及:

$$1 + 1/mD < \Phi D(\theta).$$

$$1 +$$

m

D

1

$<\Phi$

D

(θ) .

如果成功, 就能把很多 natural fixed

Q

Q 納入 singleton-gate。

狀態

Promising, 但還是 technical inequality work。
不是 full arbitrary

Q

Q 。

3. 第三階段: 任意 fixed

Q

Q 要改成 target selection

對 arbitrary fixed

Q

Q , 只看 target

$1/D$

$1/D$ 不夠。

所以定義:

$$TQ = \{R(Q') : \emptyset \neq Q' \subsetneq Q\}.$$

T

$\mathcal{Q}$

$=\{R(Q$

,

$\rangle):\emptyset$

$=Q$

,

$\subseteq Q\}$.

目標變成：

找某個 proper

Q'

$\mathcal{Q}$

,

令

P

P 可以 hit

$R(Q')$.

$R(Q$

,

$\rangle$).

如果

P

P hit 到：

$R(B)=R(Q')$

$R(B)=R(Q$

,

)

就有 proper subcircuit。

初步 experiment 結果

Composite / interval-like

Q

Q 通常很容易找到 smooth target, 例如:

$$110+115=16.$$

$$10$$

$$1$$

$$+$$

$$15$$

$$1$$

$$=$$

$$6$$

$$1$$

.

Prime-only

Q

Q 則會讓 naive target-smoothing proxy 失敗, 因為:

$$R(\{p_1, p_2, \dots\})$$

$$R(\{p$$

1

p

2

$, \dots \})$

的 denominator 仍保留那些 primes。

結論：

TQ target selection 是 arbitrary fixed Q 的核心問題。

T

Q

target selection 是 arbitrary fixed Q 的核心問題。

4. Prime-only

Q

Q : 原本看似 obstruction, 後來被大幅推進

若：

$$Q = \{p_1, \dots, p_k\}$$

$$Q = \{p$$

1

$, \dots, p$

k

}

全是 primes, 則:

$$v_{p_j}(SQ)=-1.$$

v

p

j

$(S$

Q

$$)=-1.$$

所以若:

$$R(P)=SQ,$$

$$R(P)=S$$

Q

,

則

P

P 必須包含

p_j

p

j

-multiples。這給 qualitative support。

但 Liu–Sawhney 需要 quantitative local mass:

$$pR(P_p)\geq c>0.$$

$$pR(P$$

p

$$) \geq c > 0.$$

5. Dynamic gate : single-prime quantitative repair

定義：

$$\alpha = pR(Pp).$$

$$\alpha = pR(P$$

p

$$).$$

因為：

$$|Pp| \leq NR(Pp) = \alpha Np,$$

$$|P$$

p

$$| \leq NR(P$$

p

$$)=$$

$$p$$

$$\alpha N$$

,

所以：

$$P \neg p$$

$$P$$

$$\neg p$$

的 density 至少：

$$\theta+\varepsilon-\alpha p.$$

$$\theta+\varepsilon-$$

$$p$$

$$\alpha$$

$$\cdot$$

對

$$P\neg p$$

$$P$$

$$\neg p$$

用

$$\Phi p$$

$$\Phi$$

$$p$$

lower bound：

$$SQ-\alpha p\geq \Phi p(\theta+\varepsilon-\alpha p).$$

$$S$$

$$\mathcal{Q}$$

$$-$$

$$p$$

$$\alpha$$

$$\geq \Phi$$

$$p$$

$$(\theta+\varepsilon-$$

$$p$$

$$\alpha$$

$$).$$

如果：

$$SQ<\Phi p(\theta+\varepsilon),$$

$$S$$

$$Q$$

$$<\Phi$$

$$p$$

$$(\theta+\varepsilon),$$

就能逼出：

$$\alpha\geq c p,Q,\varepsilon>0.$$

$$\alpha\geq c$$

$$p,Q,\varepsilon$$

$$>0.$$

狀態

這是 real lemma candidate。

可處理一些 prime-only / large-prime stress cases above a

Q

Q -dependent

ε

ε -threshold。

但不是所有 small

ε

ε_0 。

6. Joint-avoidance /

σ_Q

σ

Q

gate: 目前最強進展之一

定義:

$$\pi(Q) = \{p : p \mid q \text{ for some } q \in Q\},$$

$$\pi(Q) = \{p : p \mid q \text{ for some } q \in Q\},$$

$$\sigma_Q = \prod_{p \in \pi(Q)} (1 - 1/p).$$

σ

Q

=

$$p \in \pi(Q)$$

$\prod$

(1-

p

$$1$$

$$).$$

若

$$1\in Q$$

$$1\in$$

$$/$$

$$Q$$
 且

$$SQ>1$$

$$S$$

$$Q$$

$$>1,$$
 則:

$$SQ\leq 1\sigma Q-1,$$

$$S$$

$$Q$$

$$\leq$$

$$\sigma$$

$$Q$$

$$1$$

$$-1,$$

所以:

$$\sigma Q<12<\theta.$$

$$\sigma$$

$$Q$$

<

2

1

< θ .

令

P^*

P

*

是

P

P 中不被

$\pi(Q)$

$\pi(Q)$ 任何 prime 整除的部分。則：

$$|P^*| \leq \sigma_Q N^{o(N)}.$$

$|P$

*

$|\leq \sigma$

Q

$$N^{o(N)}.$$

又：

$$|P_p| \leq NR(P_p).$$

$|P$

p

$$|\leq NR(P$$

$$p$$

$$).$$

Union bound 得:

$$|P|\leq |P^*|+\sum_{p\in\pi(Q)}|P_p|\leq \sigma QN+N\sum_{p\in\pi(Q)}R(P_p)+o(N).$$

$$|P|\leq |P$$

$*$

$$|+$$

$$_{p\in\pi(Q)}$$

$$\sum$$

$$|P$$

$$p$$

$$|\leq \sigma$$

$$Q$$

$$N+N$$

$$_{p\in\pi(Q)}$$

$$\sum$$

$$R(P$$

$$p$$

$$)+o(N).$$

若:

$$|P|\geq (\theta+\varepsilon)N,$$

$$|P|\geq(\theta+\varepsilon)N,$$

則：

$$\sum_{p\in\pi(Q)}R(P_p)\geq\theta+\varepsilon-\sigma Q^{-o(1)}>0.$$

$$p\in\pi(Q)$$

$$\sum$$

$$R(P$$

$$p$$

$$)\geq\theta+\varepsilon-\sigma$$

$$Q$$

$$-o(1)>0.$$

影響

對 prime-only

$$Q$$

Q , 因為

$$\pi(Q)=Q$$

$\pi(Q)=Q$, 所以某個

$$p_0\in Q$$

$$p$$

$$0$$

$\in Q$ 有 positive local mass。可 target:

$$Q'=\{p_0\},R(Q')=1/p_0.$$

$$Q$$

,

$$=\{p$$

0

$\}, R(Q$

,

$)=1/p$

0

若 Liu–Sawhney reachability 可用, prime-only fixed

Q

Q 很可能 closed。

結論:

prime-only fixed Q : 大幅接近, modulo Liu–Sawhney。

prime-only fixed Q : 大幅接近, modulo Liu–Sawhney。

7. Composite / mixed fixed

Q

Q : 真正剩下 target-selection

若

Q

Q 是 composite-only, 例如:

$$Q=\{15,21,35,\dots\},$$

$$Q=\{15,21,35,\dots\},$$

joint gate 可能給你

$p=3,5,7$

$p=3,5,7$ 的 mass, 但

3,5,7

3,5,7 本身未必在

Q

Q 。

所以不能直接 target

$1/p$

$1/p$ 。

需要找：

$$Q' \subsetneq Q$$

Q

,

$$\subsetneq Q$$

令：

$$\text{den}(R(Q'))$$

$\text{den}(R(Q$

,

))

只用那些 forced primes / prime powers。

這就是 composite target-selection lemma。

8.

$F(Q)$

$F(Q)$ diagnostic

定義：

$$F(Q)=\{p:vp(SQ)<0\}.$$

$$F(Q)=\{p:v$$

$$p$$

$$(S$$

$$Q$$

$$)<0\}.$$

一個弱目標是找：

$$Q'\subsetneq Q$$

$$Q$$

$$,$$

$$\subsetneq Q$$

使：

$$\text{prime supp}(\text{den}R(Q'))\subseteq F(Q).$$

$$\text{prime supp}(\text{den}R(Q$$

$$,$$

$$))\subseteq F(Q).$$

小型 computation：

- composite-only
- $Q\subseteq [4,30]$
- $Q\subseteq [4,30],$
- $|Q|\leq 11$
- $|Q|\leq 11,$
- $SQ>1$
- S
- Q
-

- >1 : 未找到 counterexample;
 - random composite-only samples: 也未找到 failure。
- 但 Opus 找到低和 counterexample:

$$Q=\{6,10,15,33,88\},$$

$$Q=\{6,10,15,33,88\},$$

$$SQ=3/8<1, F(Q)=\{2\},$$

S

Q

$$=3/8<1, F(Q)=\{2\},$$

且沒有 proper

Q'

Q

,

的 denominator support 落在

$\{2\}$

$\{2\}$ 。

所以:

$$SQ>1$$

S

Q

$$>1$$

這個條件真的重要。

9. Cross-number / obstruction group route

Composite target-selection 可以寫成 zero-sum problem。

構造 obstruction group

G_Q

G

$\mathcal{O}$

, 大意是把

$p \in F(Q)$

$p \in$

/

$F(Q)$ 的

p

p -adic obstruction 記錄下來。

若 target selection fails, 則

$(v_q)_{q \in Q}$

$(v$

q

)

$q \in Q$

是

G_Q

G

$\mathcal{O}$

裏的 minimal zero-sum sequence / atom。

又因為：

$$\text{ord}(vq)\leq q,$$

$$\text{ord}(v$$

$$q$$

$$)\leq q,$$

所以：

$$SQ=\sum 1q\leq \sum 1\text{ord}(vq)=k((vq)).$$

$$S$$

$$Q$$

$$=\sum$$

$$q$$

$$1$$

$$\leq \sum$$

$$\text{ord}(v$$

$$q$$

$$)$$

$$1$$

$$=k((v$$

$$q$$

)).

這是 cross number。

Claude 嘗試 one-line closure

它想說：

- GQ
- G
- Q
-
- cyclic;
- cyclic group 的 atom cross number
- ≤ 1
- ≤ 1 ;
- 所以
- $SQ \leq 1$
- S
- Q
-
- ≤ 1 , contradiction。

但錯在：

$$K(C_n)=1$$

$$K(C$$

n

$$)=1$$

這句對 composite cyclic group 的 maximal cross number 是 false。

反例：

在

C_6

C

中, 序列:

$$(2,2,3,5)$$

$$(2,2,3,5)$$

是 minimal zero-sum, cross number:

$$13+13+12+16=43>1.$$

$$3$$

$$1$$

$$+$$

$$3$$

$$1$$

$$+$$

$$2$$

$$1$$

$$+$$

$$6$$

$$1$$

$$=$$

$$3$$

$$4$$

>1 .

所以 one-line closure fails。

10. Cross-number route 的正確版本

正確只能說：

$$SQ \leq k((vq)) \leq K(GQ).$$

S

Q

$\leq k((v$

q

$)) \leq K(G$

Q

$).$

因此若：

$$SQ > K(GQ),$$

S

Q

$> K(G$

Q

$),$

則 target selection succeeds。

但

$K(GQ)$

$K(G$

Q

) 可能

>1

>1 , 所以

$SQ>1$

S

Q

>1 不夠。

目前真正剩下的問題是：

能否證明 reciprocal-realizable atoms 比一般 cyclic atoms 更受限？

能否證明 reciprocal-realizable atoms 比一般 cyclic atoms 更受限？

即：

雖然一般

C_6

C

6

有 cross number

$4/3$

$4/3$ atom, 但那些 atom 是否能由 reciprocal denominators

$1/q$

$1/q$ 以 target-selection failure 的方式實現？

這是目前最精準的 composite/mixed

Q

Q gap。

11. 目前任意 fixed

Q

Q 的狀態

Case	Status
Specific Q Q example	strong numerics / scaffold
Fixed Q Q under original gate	strong architecture, LS audit pending
Minimal-over-one Q Q	promising, technical
Prime-only fixed Q Q	likely close modulo LS
Composite / interval-like Q Q	often easy by TQ T 2 target selection
Composite-only arbitrary Q Q	reduced to target-selection / cross-number obstruction
Arbitrary fixed Q	not solved

Q fully

Growing

QN

still far

Q

N

Full #319

still open

12. 現在我們正在 working on 的核心 problems

Problem A: Liu–Sawhney interface audit

即使 target 找到, 仍要確定 Liu–Sawhney Proposition 5.2 真能 hit target:

$$R(B)=R(Q').$$

$$R(B)=R(Q)$$

,

).

這是 analytic machinery audit。

Problem B: Composite target-selection lemma

證明或反證:

若 fixed composite/mixed

Q

$Q,$

$SQ>1$

S

Q

>1 , 則存在 proper

Q'

Q

,

令

$R(Q')$

$R(Q$

,

) denominator 只用 forced prime support。

Problem C: Reciprocal atom cross-number lemma

若 target selection fails, 得到 minimal zero-sum atom

(vq)

$(v$

q

)。

要證明:

reciprocal-realizable atom with

$SQ > 1$

S

Q

> 1 cannot exist.

這是現在最 sharp 的 algebraic / zero-sum formulation。

Problem D: Growing

Q

Q

即使 fixed

Q

Q solved, 若

$Q = QN$

$Q = Q$

N

grows, constants may decay。

需要 uniform version of joint gate + target selection + Liu–Sawhney.

13. 最簡潔 handover statement

你可以記成：

Prime-only fixed Q is probably close.

Prime-only fixed Q is probably close.

Composite/mixed fixed Q is now a zero-sum target-selection problem.

Composite/mixed fixed Q is now a zero-sum target-selection problem.

Claude’s one-line cross-number closure failed because $K(Cn) > 1$ for composite n .

Claude’s one-line cross-number closure failed because $K(Cn) > 1$ for composite n .

$K(Cn) > 1$ for composite n .

The new target is a special cross-number theorem for reciprocal-realizable atoms.

The new target is a special cross-number theorem for reciprocal-realizable atoms.

$\boxed{\text{Full \#319 remains open; growing } Q \text{ is still long-term.}}$

14. 最終 honest verdict

不是 solved, 但已經不是散亂 brainstorm。

現在的 project 有一條清楚路線：

1. fixed-
2. Q
3. Q gate theorem candidate;
4. prime-only fixed
5. Q
6. Q via
7. σQ
8. σ
9. ϱ
- 10.
11. joint gate;
12. composite/mixed fixed
13. Q
14. Q via target-selection;
15. target-selection via obstruction group / cross-number;
16. cross-number one-line proof failed;
17. remaining question = reciprocal-realizable high-cross-number atoms.

所以目前最正確的評價是：

arbitrary fixed Q is significantly closer, but still open.

arbitrary fixed Q is significantly closer, but still open.

The gap is no longer vague. It is now a precise finite abelian-group / reciprocal atom problem.

MAny thanks. Moreover can we make a verifiable formulation for all of today's progress? Thanks. (like so far on the arbitration stuff; since yesterday's lean proof for the fixed minimum over 1 thing)

已思考 1m 10s

可以。下面是一份可驗證 formulation / progress ledger v0.1。它的用途是：把今日 arbitrary fixed

Q

Q 的進展拆成「已證 / 條件式 / 待證 / 已否定」四類，避免之後 chat 越聊越混亂。

背景先固定：Erdős #319 官方仍是 open，頁面記錄 Adenwalla/Croot 下界

$$(1-1/e+o(1))N$$

$(1-1/e+o(1))N$, 且 comments 沒有 claimed partial / complete solution ;

Liu–Sawhney 是我們目前接駁的 external reachability machinery。

Verifiable formulation for current Erdős #319 fixed-(Q) progress

0. Global notation

Let

$$\begin{aligned} &[\\ &R(X):=\sum_{x\in X}\frac{1}{x},\quad \theta:=1-\frac{1}{e}. \\ &] \end{aligned}$$

A signed reciprocal circuit is written as two sides

$$\begin{aligned} &[\\ &P,Q\subseteq [1,N],\quad R(P)=R(Q). \\ &] \end{aligned}$$

It is primitive if there is no non-empty proper pair

$$\begin{aligned} &[\\ &P'\subseteq P,\quad Q'\subseteq Q \\ &] \end{aligned}$$

such that

$$\begin{aligned} &[\\ &R(P')=R(Q'). \\ &] \end{aligned}$$

For fixed (Q), write

[
 $S_Q := R(Q).$
]

The aim of the fixed-(Q) programme is:

[
 $|P| \geq (\theta + \epsilon)N, \quad R(P) = S_Q$
 $\quad \Longrightarrow \quad$
 $\text{not primitive}.$
]

This would not solve full #319, but it would prove a serious fixed-negative-side obstruction.

1. The (Φ_q) gate

Define $(\Phi_q(u))$ as the minimum reciprocal mass of (uN) integers in $([1, N])$ avoiding multiples of (q) . The working formula is

[
 $\Phi_q(u) = \left(1 - \frac{1}{q}\right)$
 $\log \left(\frac{1 - \frac{1}{q}}{1 - \frac{1}{q-u}} \right).$
]

Status

Essentially elementary / checkable.

Verification task:

1. Sort the nonmultiples of (q) in $([1, N])$.
2. The reciprocal sum is minimized by choosing the largest (uN) such nonmultiples.
3. Approximate by an integral.
4. Recover $(\Phi_q(u))$.

2. Fixed-(Q) singleton-gate theorem candidate

Let $(D \in \mathbb{Q})$. Suppose

$$\left[\begin{array}{l} S_Q < \min_{\{p^a \parallel D\}} \Phi_{\{p^a\}}(\theta). \end{array} \right]$$

Then the intended theorem is:

For every $(\epsilon > 0)$, for all sufficiently large (N) , no primitive circuit exists with fixed (Q) , arbitrary (P) , and

$$\left[\begin{array}{l} |P| \geq (\theta + \epsilon)N, \quad R(P) = S_Q. \end{array} \right]$$

Reason:

1. The (Φ) -gate forces positive local mass in (P) at every prime power $(p^a \parallel D)$.
2. Liu–Sawhney reachability should produce

$$\left[\begin{array}{l} B \subseteq P, \quad R(B) = \frac{1}{D}. \end{array} \right]$$

3. Then $(B \cup \{D\})$ is a proper signed zero-subcircuit.

Status

Strong proof architecture, not yet a fully verified theorem.

Remaining verification:

- Liu–Sawhney Proposition 5.2 parameter interface.
- Smoothness / bounded- (Ω) / local-mass pruning.
- Exact target $(x/Q_A = 1/D)$.

- Archimedean window.
-

3. Minimal-over-one fixed (Q)

Natural class:

$$\left[\begin{array}{l} S_Q > 1 \end{array} \right]$$

but (Q) is minimal in some sense for crossing (1).

Candidate route:

$$\left[\begin{array}{l} S_Q \leq 1 + \frac{1}{m_D} \end{array} \right]$$

and

$$\left[\begin{array}{l} 1 + \frac{1}{m_D} < \Phi_D(\theta) \end{array} \right]$$

If proved, this places minimal-over-one (Q) inside the singleton-gate theorem.

Status

Promising but technical.

If Lean/Aristotle checked parts of this, describe it only as:

elementary inequality / scaffold checked,

not as a complete #319 proof.

4. Arbitrary fixed (Q): target-selection formulation

Singleton target (1/D) is not enough for arbitrary fixed (Q). Define

[
T_Q:={R(Q'):\emptyset\neq Q\subsetneq Q}.
]

Target-selection problem:

Find (Q'\subsetneq Q) such that (P) can hit

[
R(Q').
]

If Liu–Sawhney gives

[
B\subseteq P,\quad R(B)=R(Q'),
]

then (B\cup Q') is a proper subcircuit.

Status

This is the correct formulation for arbitrary fixed (Q).

Computational evidence so far:

- Composite / interval-like (Q) often have smooth targets.
- Prime-only (Q) defeats the naive smooth-denominator proxy.
- Therefore target selection is real, not cosmetic.

5. Dynamic single-prime gate

For a prime (p), define

$$[\begin{aligned} P_p &:= \{n \in P \mid n\}, \\ &\quad \\ \alpha &:= pR(P_p). \end{aligned}]$$

Since

$$[\begin{aligned} |P_p| &\leq NR(P_p) = \frac{\alpha N}{p}, \\ & \end{aligned}]$$

the set $(P_{\neg p} := P \setminus P_p)$ has density at least

$$[\begin{aligned} &\theta + \varepsilon - \frac{\alpha}{p}. \end{aligned}]$$

Applying the (Φ_p) -bound to $(P_{\neg p})$:

$$[\begin{aligned} S_Q - \frac{\alpha}{p} \\ &\geq \\ \Phi_p &\left(\theta + \varepsilon - \frac{\alpha}{p} \right). \end{aligned}]$$

Thus if

$$[\begin{aligned} S_Q &< \Phi_p(\theta + \varepsilon), \\ & \end{aligned}]$$

then

[
 $\alpha = pR(P_p) \geq c_{\{p, Q, \varepsilon\}} > 0.$
]

Status

Promising lemma candidate.

Needs rigorous monotonicity / error / finite-(N) handling.

6. Joint-avoidance / (σ_Q) gate

Define

[
 $\pi(Q) := \{p \mid q \text{ for some } q \in Q\},$
]

[
 $\sigma_Q := \prod_{p \in \pi(Q)} \left(1 - \frac{1}{p}\right).$
]

If $1 \notin Q$, then

$\frac{1}{\sigma_Q} - 1.$
]

Hence

[
 $S_Q > 1 \quad \Longleftrightarrow \quad \sigma_Q < \frac{1}{2} < \theta.$
]

Let

$$[P^* := \{n \in P : \gcd(n, \prod_{p \in \pi(Q)} p) = 1\}.$$

Then

$$[|P^*| \leq \sigma_Q N + o(N).$$

Using the union bound,

$$[|P| \leq |P^*| + \sum_{p \in \pi(Q)} |P_p| \leq \sigma_Q N + N \sum_{p \in \pi(Q)} R(P_p) + o(N).$$

Therefore, if

$$[|P| \geq (\theta + \varepsilon) N,$$

then

$$[\sum_{p \in \pi(Q)} R(P_p) \geq \theta + \varepsilon - \sigma_Q - o(1) > 0.$$

Status

Provable fixed-(Q) lemma.

This is one of the strongest pieces of progress today.

7. Prime-only fixed (Q)

If

[
 $Q = \{p_1, \dots, p_k\}$
]

is prime-only and $(S_Q > 1)$, then

[
 $\pi(Q) = Q.$
]

By the joint gate, some

[
 $p_0 \in Q$
]

has

[
 $R(P_{\{p_0\}}) \geq c_{\{Q, \forall \epsilon\}} > 0.$
]

Take

[
 $Q' = \{p_0\}, \quad R(Q') = \frac{1}{p_0}.$
]

If Liu–Sawhney reachability applies, (P) hits $(1/p_0)$, producing a proper subcircuit.

Status

Likely close, conditional on Liu–Sawhney reachability.

This is stronger than previous stages.

8. Composite / mixed (Q) : remaining target-selection problem

For composite-only or mixed (Q) , the joint gate forces mass at some primes in $(\pi(Q))$, but those primes may not themselves be elements of (Q) .

Example:

[
 $Q = \{15, 21, 35, \dots\}$
]

has

[
 $\pi(Q) = \{3, 5, 7, \dots\}$,
]

but $(3, 5, 7 \notin Q)$.

Thus target $(1/p)$ may not be available.

Define

[
 $F(Q) := \{p(S_Q) < 0\}$.
]

Weak diagnostic target-selection lemma:

Does there exist

[
 $\emptyset \neq Q \subsetneq Q$
]

such that

[
 $\operatorname{supp}(\operatorname{den} R(Q')) \subseteq F(Q)?$
]

Status

Open.

Small computations found no ($S_Q > 1$) counterexample, but there is a low-sum counterexample:

[
 $Q = \{6, 10, 15, 33, 88\},$
 $\quad$
 $S_Q = \frac{38}{1} < 1,$
 $\quad$
 $F(Q) = \{2\}.$
]

So the hypothesis ($S_Q > 1$) is essential.

9. Obstruction group / zero-sum formulation

For primes outside ($F(Q)$), define a finite (p)-adic obstruction group (G_Q). Map each ($q \in Q$) to an obstruction vector

$$[v_q \in G_Q.]$$

The condition

$$[\operatorname{supp}(\operatorname{den} R(Q')) \subseteq F(Q)]$$

should correspond to

$$[\sum_{q \in Q'} v_q = 0 \text{ in } G_Q.]$$

If target selection fails, then

$$[(v_q)_{q \in Q}]$$

is a minimal zero-sum sequence / atom.

Define the cross number

$$[k((v_q)) := \sum_{q \in Q} \frac{1}{\operatorname{ord}(v_q)}.]$$

Since

$$[\operatorname{ord}(v_q) \leq q,$$

we have

$$[\sum_{q \in Q} \frac{1}{q} \leq k(v_q).]$$

Status

Correct framework, but not yet a proof.

Zero-sum theory defines $(K(G))$ as the maximum cross number of minimal zero-sum sequences. This is a real invariant in the literature.

10. Failed one-line cross-number closure

Claude proposed:

1. (G_Q) is cyclic.
2. $(K(G_Q)=1)$.
3. Therefore $(S_Q \leq 1)$, contradiction.

This fails because

$$[K(C_n)=1]$$

is false for composite cyclic groups in the needed maximal-cross-number sense.

Counterexample in (C_6) :

$$[(2,2,3,5)]$$

is minimal zero-sum since

$$[$$

$$2+2+3+5=12\equiv 0\pmod{6}$$

$$]$$

and no proper subsequence sums to zero.

Orders:

$$[$$

$$\operatorname{ord}(2)=3,\quad$$

$$\operatorname{ord}(2)=3,\quad$$

$$\operatorname{ord}(3)=2,\quad$$

$$\operatorname{ord}(5)=6.$$

$$]$$

Cross number:

$$[$$

$$\frac{1}{3}+\frac{1}{3}+\frac{1}{2}+\frac{1}{6}=\frac{4}{3}>1.$$

$$]$$

So the valid conclusion is only

$$[$$

$$S_Q\leq k(v_q)\leq K(G_Q).$$

$$]$$

Thus target selection is forced if

$$[$$

$$S_Q>K(G_Q),$$

$$]$$

not merely if

[
 $S_Q > 1$.
]

Status

One-line closure rejected.

But the cross-number route remains valuable.

11. Current fixed-(Q) problem list

Problem A: Liu–Sawhney interface

Verify that the reachability theorem can be applied to target $(R(Q'))$ under our local-mass hypotheses.

Problem B: Composite target selection

Prove or disprove:

If (Q) is fixed, $(1 \notin Q)$, $(S_Q > 1)$, then there exists proper (Q') whose target denominator uses only forced primes.

Problem C: Reciprocal atom cross-number bound

If target selection fails, the obstruction sequence is a minimal zero-sum atom. We need a special theorem:

Reciprocal-realizable atoms with $(S_Q > 1)$ cannot exist.

Or a weaker sufficient condition:

[
 $S_Q > K(G_Q)$.
]

Problem D: Growing (Q_N)

Even if fixed (Q) closes, growing (Q_N) requires uniform bounds. Constants from joint gate and local mass may decay.

12. Current honest conclusion

We have not solved Erdős #319.

We have likely made serious fixed- (Q) progress:

1. Fixed (Q) gate theorem architecture.
2. Prime-only fixed (Q) likely close via (σ_Q) joint gate.
3. Composite/mixed fixed (Q) reduced to a target-selection / zero-sum / cross-number problem.
4. One-line cross-number proof failed.
5. Correct remaining algebraic task: prove a special cross-number bound for reciprocal-realizable atoms.

Final status:

```
[  
\boxed{  
\text{Arbitrary fixed }Q\text{ is closer, but still open.}  
}  
]
```

```
[  
\boxed{  
\text{Growing }Q\text{ and full #319 remain long-term.}  
}  
]
```

我會建議你把這份 formulation 當成「今日版 ledger」。它夠清楚，也不會誇大成 solved theorem。特別是第 10 點要保留，因為它防止之後再次被 “cyclic group

$K=1$

$K=1$ ” 這個假閉合帶偏。

